CHAPTER 3

Kantorovich Relaxation

Kantorovich's relaxation is the decisive move that turns transport into convex optimization. This chapter replaces deterministic maps by couplings, first for discrete histograms and then for general measures; it develops the resulting linear-programming structure, existence theory and characterization of optimizers by cyclical monotonicity. The metric geometry induced when the cost is a power of a ground distance is the subject of the next chapter. Historically, this viewpoint grew from Kantorovich's economic planning work [296] and is now the standard foundation of OT [530, 531, 442].

3.1 Discrete Relaxation

The discrete relaxation is the cleanest place to see mass splitting. It replaces permutations by a transportation polytope and reveals the linear-programming structure that algorithms exploit.

Mass splitting and couplings. Monge's discrete matching problem is problematic because it cannot be applied when $n \neq m$. A faithful formulation must keep track of masses $(\mathrm{a}_i, \mathrm{b}_j)$. The continuous Monge problem (2.5), based on push-forward maps, has the same obstruction in another form: there may be no map T such that $T_\sharp \alpha = \beta$. This happens, for instance, when a single Dirac mass should be sent to several Dirac masses.

This lack of mass splitting also makes the Monge formulation asymmetric in α and β: one can map two Dirac masses to a single one, but not the other way around. Even when a feasible map exists, the resulting optimization problem is non-convex and therefore difficult to solve numerically.

Kantorovich's key idea [296] is to relax the deterministic nature of transportation. Instead of requiring each source point x_i to be sent to exactly one target, the mass at x_i may be dispatched across several locations. This moves from deterministic transport maps to probabilistic, or fuzzy, transport plans. The relaxation is encoded, in place of a permutation σ or a map T, by a coupling matrix $\mathrm{P} \in \mathbb{R}_+^{n \times m}$, where $\mathrm{P}_{i,j}$ describes the amount of mass flowing from x_i to y_j in the formalism of discrete measures

$$\alpha = \sum_i \mathrm{a}_i \delta_{x_i}, \qquad \beta = \sum_j \mathrm{b}_j \delta_{y_j}.$$

Definition 3.1 (Discrete couplings and mass conservation)**.** Admissible couplings are only constrained to satisfy conservation of mass:

$$\mathrm{U}(\mathrm{a}, \mathrm{b}) := \left\{ \mathrm{P} \in \mathbb{R}_+^{n \times m} \; ; \; \mathrm{P}\mathbb{1}_m = \mathrm{a} \quad \text{and} \quad \mathrm{P}^\top \mathbb{1}_n = \mathrm{b} \right\}. \tag{3.1}$$

Equivalently,

$$\mathrm{P}\mathbb{1}_m = \Big(\sum_j \mathrm{P}_{i,j}\Big)_i \in \mathbb{R}^n, \qquad \mathrm{P}^\top \mathbb{1}_n = \Big(\sum_i \mathrm{P}_{i,j}\Big)_j \in \mathbb{R}^m.$$

Remark 3.2 (Small transportation polytopes). The definition is already informative in the smallest dimensions. If $n = m = 1$, mass conservation fixes the only entry, so the feasible set is a singleton. The same happens for $(n, m) = (2, 1)$, and by symmetry for $(n, m) = (1, 2)$: the unique coupling is forced by its only column, or by its only row.
The first nontrivial case is $(n, m) = (2, 2)$. Let $\mathrm{a} = (p, 1-p)$ and $\mathrm{b} = (q, 1-q)$ with $p, q \in [0, 1]$. Once $s := \mathrm{P}_{1,1}$ is chosen, the marginal constraints force all other entries, hence every coupling has the form

$$\mathrm{P}(s) = \begin{pmatrix} s & p-s \\ q-s & 1-p-q+s \end{pmatrix}.$$

The nonnegativity constraints are exactly $s \in [\max(0, p+q-1), \min(p, q)]$, so $\mathrm{U}(\mathrm{a}, \mathrm{b})$ is a segment, possibly reduced to a point at the boundary. In general, when all marginal entries are positive, the transportation polytope has affine dimension $(n-1)(m-1)$ before the nonnegativity inequalities cut out its faces.

The first useful consequence of this relaxation is feasibility: there is always at least one admissible plan, obtained by making source and target independent.

Definition 3.3 (Discrete product coupling)**.** Given weights $\mathrm{a} \in \Sigma_n$ and $\mathrm{b} \in \Sigma_m$, the discrete product, or trivial, coupling is

$$(\mathrm{a} \otimes \mathrm{b})_{i,j} := \mathrm{a}_i \mathrm{b}_j.$$

It belongs to $\mathrm{U}(\mathrm{a}, \mathrm{b})$ and corresponds to choosing the source and target labels independently.

This feasible set is the bounded intersection of an affine space with the nonnegative orthant, hence a convex polytope. In one dimension, an ordered coupling can be read as a matrix: rows index source bins, columns index target bins, and the marginal constraints appear as prescribed row and column sums.

Proposition 3.4 (Discrete product optimality is degenerate)**.** *Assume that the zero-mass rows and columns have been removed, so that* $\mathrm{a}_i > 0$ *and* $\mathrm{b}_j > 0$*, and let* C *be a finite cost matrix. The product plan* $\mathrm{a} \otimes \mathrm{b}$ *minimizes* $\mathrm{P} \mapsto \langle \mathrm{C}, \mathrm{P} \rangle$ *over* $\mathrm{U}(\mathrm{a}, \mathrm{b})$ *if and only if every coupling* $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ *minimizes it.*

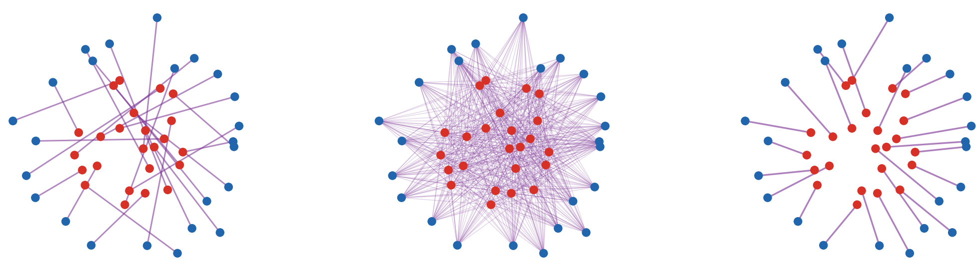
Proof. The reverse implication is immediate. Assume conversely that $\mathrm{a} \otimes \mathrm{b}$ is optimal and let $\mathrm{Q} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ be arbitrary. Since all entries of $\mathrm{a} \otimes \mathrm{b}$ are positive, there exists $t > 0$ small enough that $\mathrm{R} := (1+t)(\mathrm{a} \otimes \mathrm{b}) - t\mathrm{Q}$ has nonnegative entries. Its row and column sums are $\mathrm{R}\mathbb{1}_m = (1+t)\mathrm{a} - t\mathrm{a} = \mathrm{a}$ and $\mathrm{R}^\top \mathbb{1}_n = (1+t)\mathrm{b} - t\mathrm{b} = \mathrm{b}$, so $\mathrm{R} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$. Moreover, $\mathrm{a} \otimes \mathrm{b} = \frac{1}{1+t}\mathrm{R} + \frac{t}{1+t}\mathrm{Q}$. By optimality of $\mathrm{a} \otimes \mathrm{b}$, both $\langle \mathrm{C}, \mathrm{R} \rangle$ and $\langle \mathrm{C}, \mathrm{Q} \rangle$ are at least $\langle \mathrm{C}, \mathrm{a} \otimes \mathrm{b} \rangle$. Taking the scalar product of the convex-combination identity with C gives

$$\langle \mathrm{C}, \mathrm{a} \otimes \mathrm{b} \rangle = \frac{1}{1+t} \langle \mathrm{C}, \mathrm{R} \rangle + \frac{t}{1+t} \langle \mathrm{C}, \mathrm{Q} \rangle.$$

A convex average of two numbers not smaller than $\langle \mathrm{C}, \mathrm{a} \otimes \mathrm{b} \rangle$ can equal $\langle \mathrm{C}, \mathrm{a} \otimes \mathrm{b} \rangle$ only if both numbers are equal to it. Hence $\langle \mathrm{C}, \mathrm{Q} \rangle = \langle \mathrm{C}, \mathrm{a} \otimes \mathrm{b} \rangle$, and Q is optimal. Since Q was arbitrary, all couplings are optimal. □

Thus the product plan is mainly a feasibility witness. Except in the degenerate situation where the linear cost is constant on the whole transportation polytope, it is not expected to solve optimal transport. It is also maximally diffuse: when all masses are positive, it has nm positive entries, whereas Proposition 3.8 shows below that sparse optimal plans with at most $n + m - 1$ positive entries always exist.

Figure 3.1 contrasts deterministic, product and optimal couplings through weighted transport segments.



deterministic graph　　　product plan $\mathrm{a} \otimes \mathrm{b}$　　　optimal plan

Figure 3.1: Discrete couplings represented as straight transport segments on the canonical point clouds used in the matching section. The deterministic graph is a feasible Monge-type plan, the product plan spreads every source mass over all targets, and the optimal Kantorovich plan minimizes the quadratic transport cost. Line width and opacity encode the transported mass.

Figure 3.2 gives the complementary matrix representation and attaches the prescribed marginals directly to each plan.

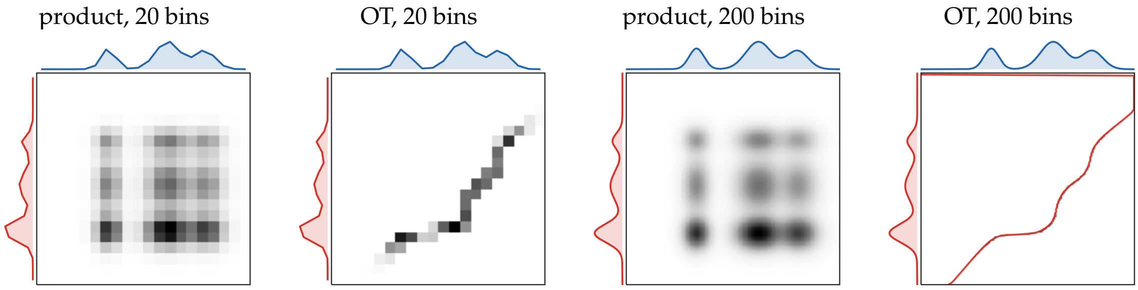


Figure 3.2: Coupling matrices with their prescribed marginals. The central grayscale image displays $\mathrm{P}_{i,j}$; the red curve on the left is the source marginal a, and the blue curve on top is the target marginal b. The independent product plan is diffuse, whereas the one-dimensional optimal plan concentrates near the monotone quantile correspondence. Only the dense 200-bin optimal panel overlays the barycentric projection $i \mapsto \sum_j \mathrm{P}_{i,j} j/\mathrm{a}_i$, because that curve is meaningful visually only at sufficient resolution.

Whereas the Monge formulation is intrinsically asymmetric, Kantorovich's relaxed formulation is symmetric at the level of feasible sets: a coupling P belongs to $\mathrm{U}(\mathrm{a},\mathrm{b})$ if and only if $\mathrm{P}^\top$ belongs to $\mathrm{U}(\mathrm{b},\mathrm{a})$.

Linear-programming structure. Kantorovich, aiming for economic planning, made a strong simplifying assumption: the cost of transportation should be linear in the amount of transported mass.

Definition 3.5 (Discrete Kantorovich problem)**.** For $\mathrm{a} \in \Sigma_n$, $\mathrm{b} \in \Sigma_m$, and $\mathrm{C} \in \mathbb{R}^{n\times m}$, define

$$\mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b}) := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{C},\, \mathrm{P}\rangle = \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \sum_{i,j} \mathrm{C}_{i,j}\mathrm{P}_{i,j}. \tag{3.2}$$

A minimizer P is an optimal coupling, or optimal transport plan.

Here $\mathrm{C}_{i,j}$ is the cost of moving one unit of mass from x_i to y_j. This is a linear program, and its solutions need not be unique.

The discrete transport value has opposite curvature in its two types of data: mixing marginals gives convexity, whereas minimizing a linear objective over couplings gives concavity in the cost matrix.

Proposition 3.6 (Convexity in the marginals and concavity in the cost)**.** *For every fixed cost matrix* $\mathrm{C} \in \mathbb{R}^{n\times m}$*, the map*

$$(\mathrm{a},\mathrm{b}) \longmapsto \mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b})$$

is jointly convex on $\Sigma_n \times \Sigma_m$*. For every fixed* $(\mathrm{a},\mathrm{b}) \in \Sigma_n \times \Sigma_m$*, the map*

$$\mathrm{C} \longmapsto \mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b})$$

is concave on $\mathbb{R}^{n\times m}$.

Proof. For $r \in \{0,1\}$, let P_r be optimal between a_r and b_r. For $t \in [0,1]$, the matrix $\mathrm{P}_t = (1-t)\mathrm{P}_0 + t\mathrm{P}_1$ belongs to $\mathrm{U}((1-t)\mathrm{a}_0 + t\mathrm{a}_1, (1-t)\mathrm{b}_0 + t\mathrm{b}_1)$. Therefore

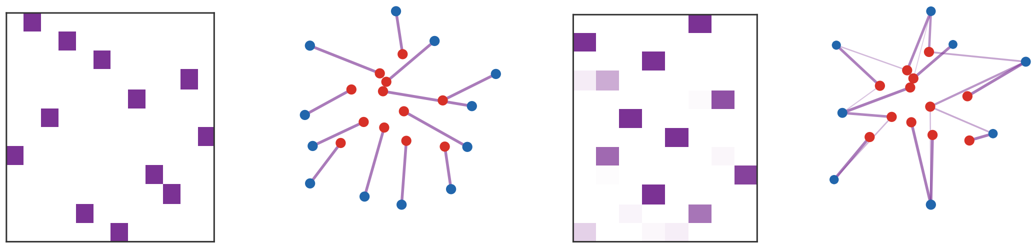
$$\mathrm{L}_\mathrm{C}\big((1-t)\mathrm{a}_0 + t\mathrm{a}_1, (1-t)\mathrm{b}_0 + t\mathrm{b}_1\big) \leqslant \langle \mathrm{C},\, \mathrm{P}_t\rangle = (1-t)\mathrm{L}_\mathrm{C}(\mathrm{a}_0,\mathrm{b}_0) + t\mathrm{L}_\mathrm{C}(\mathrm{a}_1,\mathrm{b}_1).$$

For fixed marginals, every $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ satisfies

$$\langle (1-t)\mathrm{C}_0 + t\mathrm{C}_1,\, \mathrm{P}\rangle \geqslant (1-t)\mathrm{L}_{\mathrm{C}_0}(\mathrm{a},\mathrm{b}) + t\mathrm{L}_{\mathrm{C}_1}(\mathrm{a},\mathrm{b}).$$

Minimizing the left-hand side over P proves concavity in C. □

Figure 3.3 contrasts the permutation plans inherited from equal-weight assignments with the genuinely splitting couplings permitted by nonuniform marginals.



permutation matrix P_σ | assignment segments | splitting matrix P | splitting segments

Figure 3.3: From permutation matrices to splitting couplings. When the two empirical measures have the same number of atoms and uniform weights, an optimal plan can be a permutation matrix. Once target masses are nonuniform, the admissible set $\mathrm{U}(\mathrm{a},\mathrm{b})$ also contains plans in which one source sends mass to several targets and several sources merge into the same target.

Sparsity here is not peculiar to transport. For nonnegative variables, the relevant quantity is the rank of the constraint operator—not the raw number of listed constraints, which may contain redundancies. The following standard linear-programming principle makes this precise; see, for instance, [52].

Proposition 3.7 (Rank-controlled sparse minimizers)**.** *Let $\mathcal{A} : \mathbb{R}^{n\times m} \to \mathbb{R}^q$ be linear, let $\mathbf{d} \in \mathbb{R}^q$ and $\mathrm{C} \in \mathbb{R}^{n\times m}$, and consider*

$$\inf_{\mathrm{P}\geqslant 0,\ \mathcal{A}(\mathrm{P})\leqslant \mathbf{d}} \langle \mathrm{C},\, \mathrm{P}\rangle, \tag{3.3}$$

where the inequality is componentwise. If the feasible set is nonempty and the objective is bounded below on it, then (3.3) admits a minimizer $\mathrm{P}^\star$ satisfying

$$\#\left\{(i,j)\ ;\ \mathrm{P}^\star_{i,j} > 0\right\} \leqslant \operatorname{rank}(\mathcal{A}). \tag{3.4}$$

Here $\operatorname{rank}(\mathcal{A}) = \dim(\operatorname{Im}\mathcal{A})$*. The same conclusion holds when $\mathcal{A}(\mathrm{P}) \leqslant \mathbf{d}$ is replaced by $\mathcal{A}(\mathrm{P}) = \mathbf{d}$.*

Proof. The feasible region is a nonempty polyhedron, and the objective is bounded below on it. The attainment theorem for linear programs therefore provides a minimizer. Among all minimizers, choose $\mathrm{P}^\star$ with the smallest number of positive entries, and denote its support by S.

Suppose that $\#S > \operatorname{rank}(\mathcal{A})$. The restriction of $\mathcal{A}$ to the $\#S$-dimensional space of matrices supported on S then has a nontrivial kernel. Hence there is a nonzero matrix H, supported on S, such that $\mathcal{A}(H) = 0$. Since $\mathrm{P}^\star$ is strictly positive on S, both $\mathrm{P}^\star + tH$ and $\mathrm{P}^\star - tH$ remain nonnegative for all sufficiently small $t > 0$. Moreover, $\mathcal{A}(\mathrm{P}^\star \pm tH) = \mathcal{A}(\mathrm{P}^\star)$, so both perturbations satisfy exactly the same linear inequalities, or equalities, as $\mathrm{P}^\star$. Optimality forces $\langle \mathrm{C},\, H\rangle = 0$; otherwise one of the two signs would strictly decrease the objective.

Choose $\sigma \in \{-1, 1\}$ such that σH has a negative entry, and set $t_\star := \min_{(i,j):\,\sigma H_{i,j}<0} \mathrm{P}^\star_{i,j}/(-\sigma H_{i,j}) > 0$. Then $\mathrm{P}^\star + t_\star\sigma H$ is nonnegative, feasible, and has the same cost as $\mathrm{P}^\star$. At least one of its positive entries has become zero, while no new entry has appeared outside S. This contradicts the minimality of S and proves (3.4). □

Proposition 3.8 (Sparse optimal plans)**.** *Assume $\mathrm{a}_i > 0$, $\mathrm{b}_j > 0$ and $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j = 1$. The linear program (3.2) admits an optimal coupling with at most $n + m - 1$ nonzero entries.*

Proof. The transportation polytope is nonempty and compact, so the minimum is finite. Apply the equality case of Proposition 3.7 to the marginal operator $\mathcal{A}_{\mathrm{marg}}(\mathrm{P}) := (\mathrm{P}\mathbb{1}_m, \mathrm{P}^\top\mathbb{1}_n) \in \mathbb{R}^{n+m}$. Its rank is $n + m - 1$. Indeed, every matrix satisfies $\sum_i(\mathrm{P}\mathbb{1}_m)_i = \sum_j(\mathrm{P}^\top\mathbb{1}_n)_j$, so there is one linear dependence among the $n + m$ marginal coordinates. It is the only one: if $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^m$ satisfy

$$\langle u,\, \mathrm{P}\mathbb{1}_m\rangle + \langle v,\, \mathrm{P}^\top\mathbb{1}_n\rangle = 0 \qquad \text{for every P},$$

then testing this identity on each elementary matrix gives $u_i + v_j = 0$ for every (i,j). Thus all u_i equal one scalar and all v_j equal its opposite, so the annihilator of $\operatorname{Im}(\mathcal{A}_{\mathrm{marg}})$ is one-dimensional. It follows that $\operatorname{rank}(\mathcal{A}_{\mathrm{marg}}) = n + m - 1$, and Proposition 3.7 supplies the claimed optimal coupling. □

For transportation polytopes, this rank argument also has a graph interpretation. A cycle in the bipartite support produces an alternating nonzero perturbation with zero row and column sums, exactly the kernel direction used in the general proof. Consequently, the positive support of an extreme coupling is a forest.

Proposition 3.9 (North-west corner feasible plan)**.** *Let* $\mathrm{a} \in \mathbb{R}_+^n$ *and* $\mathrm{b} \in \mathbb{R}_+^m$ *have the same positive total mass. The following greedy sweep constructs a coupling* $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ *with at most* $n + m - 1$ *positive entries and an acyclic positive support. Starting from* $(i, j) = (1, 1)$ *with residual masses* $r_i = \mathrm{a}_i$ *and* $s_j = \mathrm{b}_j$*, skip zero residuals, set* $\mathrm{P}_{i,j} = \min(r_i, s_j)$*, subtract this value from both residuals, and advance every index whose residual has become zero. Repeat until all residual masses are exhausted.*

Proof. All assignments are nonnegative. At each step, the mass placed in entry (i, j) is subtracted from exactly one current row residual and one current column residual, so no row or column can receive more mass than prescribed. Conversely, an index is advanced only when its residual has been fully filled. When the algorithm stops, the total assigned mass is $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j$, hence all row and column sums are exactly a and b.

Each positive assignment exhausts at least one current row or one current column. Before the final assignment, at most $n - 1$ row advances and $m - 1$ column advances can occur without terminating the construction. Hence the number of positive entries is at most $(n - 1) + (m - 1) + 1 = n + m - 1$. For acyclicity, view the positive support as a bipartite graph. Once a row or column index is advanced, it never appears again, so each new positive edge either starts a new component or attaches at least one new vertex to the component currently being swept. No edge is ever added between two old vertices of the same component, so no cycle can be created. □

Algorithm 3.1: North-west corner coupling

Input: Source weights $\mathrm{a} \in \Sigma_n$ and target weights $\mathrm{b} \in \Sigma_m$.
Output: Sparse feasible coupling $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$.
Initialize: Set $\mathrm{P} = 0$, $r = \mathrm{a}$, $s = \mathrm{b}$, and $(i, j) = (1, 1)$.
While $i \leqslant n$ and $j \leqslant m$ **do**:
  $\eta = \min(r_i, s_j)$, $\mathrm{P}_{i,j} \leftarrow \eta$.
  Update residuals: $r_i \leftarrow r_i - \eta$, $s_j \leftarrow s_j - \eta$.
  If $r_i = 0$ **then**:
    Set $i \leftarrow i + 1$.
  If $s_j = 0$ **then**:
    Set $j \leftarrow j + 1$.
Return P.

The north-west corner rule, summarized in Algorithm 3.1, does not use the cost matrix and is therefore not meant to solve (3.2). Its role is algorithmic: an acyclic support corresponds to linearly independent marginal constraints. When the support has fewer than $n + m - 1$ positive entries, transportation simplex implementations complete it with zero-mass basic variables to obtain a degenerate basic feasible solution. This gives a cheap initialization for the pivoting methods discussed in Section 3.2.

One-dimensional cases. In one dimension, the transportation polytope has a canonical monotone optimizer. This is the weighted version of the sorting rule from Section 1.1.

Proposition 3.10 (One-dimensional weighted sweep)**.** *Let* $x_1 \leqslant \cdots \leqslant x_n$ *and* $y_1 \leqslant \cdots \leqslant y_m$ *be points on the line, and let* $c(x, y) = h(x - y)$ *with* h *convex. The north-west corner plan between the sorted weighted atoms is optimal for* (3.2)*. Consequently, for unsorted one-dimensional inputs, an optimal plan is obtained in* $O(n \log n + m \log m)$ *time by sorting and then sweeping the masses once from left to right.*

Proof. The north-west plan is monotone: if $i < i'$ and $j > j'$, it cannot put positive mass on both (i, j) and (i', j'), because the sweep exhausts rows and columns in increasing order. To prove optimality, start from an arbitrary feasible plan P. If $\mathrm{P}_{1,1} < \min(\mathrm{a}_1, \mathrm{b}_1)$, then row 1 sends positive mass to some column $j > 1$, while column 1 receives positive mass from some row $i > 1$. Moving $\eta = \min(\mathrm{P}_{1,j}, \mathrm{P}_{i,1})$ from $(1, j)$ and $(i, 1)$ to $(1, 1)$ and (i, j) preserves both marginals. Convexity of h gives the Monge inequality

$$h(x_1 - y_j) + h(x_i - y_1) \geqslant h(x_1 - y_1) + h(x_i - y_j),$$

so this exchange does not increase the cost. Each exchange either removes the chosen entry in row 1 or removes the chosen entry in column 1. Repeating finitely many times forces $P_{1,1} = \min(a_1, b_1)$, exactly the first north-west assignment. It exhausts row 1, column 1, or both. Removing every exhausted index reduces the problem to a smaller ordered transportation problem. Induction on $n + m$ produces the complete north-west plan without increasing cost. Sorting costs $O(n \log n + m \log m)$ and the sweep uses at most $n + m - 1$ positive assignments. □

Permutation matrices as couplings. We now restrict attention to the special case $n = m$ and uniform weights $a = b = \mathbb{1}_n/n$. In this case a matching can be encoded as a matrix with exactly one active entry per row and per column.

Definition 3.11 (Permutation matrices)**.** For a permutation $\sigma \in \mathrm{Perm}(n)$, its permutation matrix P_σ is

$$(P_\sigma)_{i,j} = \begin{cases} 1 & \text{if} \quad j = \sigma(i), \\ 0 & \text{otherwise.} \end{cases}$$

The set of all permutation matrices is

$$\mathcal{P}_n^{\text{perm}} := \{P_\sigma \; ; \; \sigma \in \mathrm{Perm}(n)\}.$$

The corresponding probability coupling is P_σ/n. If the matching cost matrix is C, then

$$\langle C, P_\sigma/n \rangle = \frac{1}{n} \sum_{i=1}^{n} C_{i,\sigma(i)}.$$

Thus the assignment problem is the minimization of a linear function over the discrete, non-convex set of permutation matrices.

The convex relaxation replaces this finite set by all bistochastic matrices.

Definition 3.12 (Birkhoff polytope)**.** The Birkhoff polytope is the convex set of bistochastic matrices

$$\mathcal{B}_n := \left\{P \in \mathbb{R}_+^{n\times n} \; ; \; P\mathbb{1}_n = \mathbb{1}_n \quad \text{and} \quad P^\top \mathbb{1}_n = \mathbb{1}_n\right\}.$$

Then $U(\mathbb{1}_n/n, \mathbb{1}_n/n) = \mathcal{B}_n/n$, and permutation couplings are included in this convex relaxation. More precisely,

$$\mathcal{P}_n^{\text{perm}} = \mathcal{B}_n \cap \{0,1\}^{n\times n} \subset \mathcal{B}_n,$$

so before using the structure of $\mathcal{B}_n$ one first obtains only the relaxed inequality

$$\min_{P\in\mathcal{B}_n} \langle C, P \rangle \leqslant \min_{P\in\mathcal{P}_n^{\text{perm}}} \langle C, P \rangle.$$

The next elementary facts lead to the Birkhoff–von Neumann theorem [61, 534], which explains why this relaxation is tight for uniform matching.

Definition 3.13 (Extreme points)**.** For a convex set C in a finite-dimensional vector space,

$$\mathrm{Extr}(C) := \left\{x \in C \; ; \; x = (y+z)/2, \; y, z \in C \Rightarrow y = z = x\right\}.$$

Proposition 3.14 (Existence of extreme points)**.** *If C is a non-empty compact convex subset of a finite-dimensional vector space, then* $\mathrm{Extr}(C)$ *is non-empty.*

Proof. Among all non-empty faces of C, choose one of minimal affine dimension. If this face contained two distinct points, maximizing a linear functional that is not constant on the face would produce a non-empty proper exposed subface, contradicting minimality. Hence the minimal face is a singleton, and its point is extreme. □

Example 3.15 (An unbounded convex set without extreme points). Compactness cannot be dropped from Proposition 3.14. For instance, the affine line $C = \mathbb{R} \times \{0\} \subset \mathbb{R}^2$ is closed, convex and unbounded, but it has no extreme point: every $(x, 0)$ is the midpoint of the distinct points $(x - 1, 0)$ and $(x + 1, 0)$.

Proposition 3.16 (Linear programs have extreme minimizers)**.** *Let C be a non-empty compact convex set. For every linear form ℓ,*

$$\mathrm{Extr}(C) \cap \operatorname*{argmin}_{x \in C} \ell(x) \neq \emptyset.$$

Proof. The set $S = \operatorname{argmin}_{x\in C} \ell(x)$ is non-empty, compact and convex. By Proposition 3.14, it has an extreme point x. If $x = (y+z)/2$ with $y, z \in C$, then by linearity and optimality of x, both y and z also minimize ℓ on C, hence $y, z \in S$. Since x is extreme in S, $y = z = x$. Thus x is extreme in C. □

Theorem 3.17 (Birkhoff–von Neumann)**.** *The extreme points of $\mathcal{B}_n$ are exactly the permutation matrices.*

Figure 3.4 shows the non-extreme mechanism used in the proof below. The displayed matrix is bistochastic but not a permutation matrix: the three unit entries already behave like isolated matching edges, while the scattered fractional support contains a longer alternating cycle.

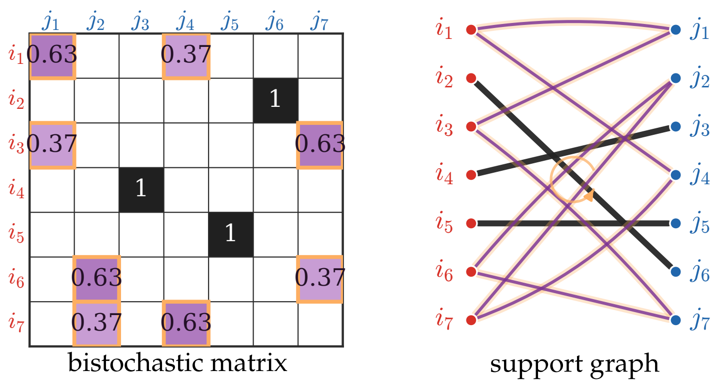


Figure 3.4: Cycle certificate in the Birkhoff–von Neumann proof. The left panel is a 7×7 bistochastic matrix which is not a permutation matrix. It has only three isolated entries equal to one, while the non-binary entries are scattered across four rows and columns. The right panel shows the bipartite positive-support graph, with the column nodes sorted as $j_1, \dots, j_7$ from top to bottom to match the matrix order: red nodes are rows, blue nodes are columns, thin purple edges correspond to $0 < \mathrm{P}_{i,j} < 1$, and bold black edges correspond to isolated entries $\mathrm{P}_{i,j} = 1$. The orange halo marks the longer alternating fractional cycle along which one can add and subtract mass while preserving all row and column sums.

Proof. We first prove that permutation matrices are extreme. Let $\mathrm{P}_\sigma \in \mathcal{P}_n^{\mathrm{perm}}$ and assume that $\mathrm{P}_\sigma = (\mathrm{Q}+\mathrm{R})/2$ with $\mathrm{Q}, \mathrm{R} \in \mathcal{B}_n$. Every bistochastic matrix has entries in $[0,1]$. Since the only extreme points of $[0,1]$ are 0 and 1, each entry of P_σ fixes the corresponding entries of Q and R: if $(\mathrm{P}_\sigma)_{i,j} = 0$, then $\mathrm{Q}_{i,j} = \mathrm{R}_{i,j} = 0$, while if $(\mathrm{P}_\sigma)_{i,j} = 1$, then $\mathrm{Q}_{i,j} = \mathrm{R}_{i,j} = 1$. Hence $\mathrm{Q} = \mathrm{R} = \mathrm{P}_\sigma$, so P_σ is extreme.

We now prove the converse by contrapositive. Pick $\mathrm{P} \in \mathcal{B}_n \setminus \mathcal{P}_n^{\mathrm{perm}}$. Since an integral bistochastic matrix is necessarily a permutation matrix, P has at least one fractional entry. We shall split $\mathrm{P} = \frac{\mathrm{Q}+\mathrm{R}}{2}$ with $\mathrm{Q}, \mathrm{R} \in \mathcal{B}_n$ and $\mathrm{Q} \neq \mathrm{R}$, proving that P is not extreme.

Associate with P the bipartite graph whose left vertices are the rows, whose right vertices are the columns, and whose edges are the fractional entries $0 < \mathrm{P}_{i,j} < 1$. An entry equal to 1 uses the whole mass of its row and column, so it is isolated in the positive support and does not appear in this fractional graph. If a left vertex i is incident to a fractional edge (i, j_1), then it must be incident to at least one other fractional edge. Indeed, the row sum is one; after the contribution $\mathrm{P}_{i,j_1} \in (0,1)$, a positive amount $1 - \mathrm{P}_{i,j_1}$ remains in the same row, and it cannot be carried by an entry equal to 1. The same argument applies to right vertices, using the column sums. Thus every non-isolated vertex of the fractional graph has degree at least two.

Starting from any fractional edge, one may therefore walk through adjacent fractional edges without immediately backtracking and without getting stuck. Since the graph is finite, some vertex is eventually visited twice; the portion of the walk between the two visits contains a cycle. Choose a shortest such cycle and write it in alternating form $(i_1, j_1, i_2, j_2, \dots, i_p, j_p)$, with $i_{p+1} = i_1$, where both (i_s, j_s) and (i_{s+1}, j_s) are fractional for every s. The minimality of the cycle implies that the vertices i_s are all distinct and that the vertices j_s are all distinct. In particular,

$$0 < \mathrm{P}_{i_s, j_s} < 1 \quad \text{and} \quad 0 < \mathrm{P}_{i_{s+1}, j_s} < 1.$$

Define

$$\varepsilon := \min_{1 \leqslant s \leqslant p} \left\{ \mathrm{P}_{i_s,j_s},\ \mathrm{P}_{i_{s+1},j_s},\ 1 - \mathrm{P}_{i_s,j_s},\ 1 - \mathrm{P}_{i_{s+1},j_s} \right\}.$$

All these numbers are positive, so $\varepsilon > 0$. Split the cycle edges into the two alternating families

$$A := \{(i_s, j_s)\}_{s=1}^{p}, \qquad B := \{(i_{s+1}, j_s)\}_{s=1}^{p}.$$

We now perform the standard alternating-cycle perturbation:

$$\mathrm{Q}_{i,j} := \begin{cases} \mathrm{P}_{i,j}, & (i,j) \notin A \cup B, \\ \mathrm{P}_{i,j} + \varepsilon/2, & (i,j) \in A, \\ \mathrm{P}_{i,j} - \varepsilon/2, & (i,j) \in B, \end{cases} \qquad \mathrm{R}_{i,j} := \begin{cases} \mathrm{P}_{i,j}, & (i,j) \notin A \cup B, \\ \mathrm{P}_{i,j} - \varepsilon/2, & (i,j) \in A, \\ \mathrm{P}_{i,j} + \varepsilon/2, & (i,j) \in B. \end{cases}$$

By the definition of ε, all modified entries stay in $[0,1]$, so Q and R are nonnegative. Each row vertex i_s of the cycle is incident to exactly one edge of A and one edge of B; the $+\varepsilon/2$ and $-\varepsilon/2$ perturbations therefore cancel in that row. The same cancellation holds in each column vertex j_s, and all other rows and columns are unchanged. Consequently, $\mathrm{Q}\mathbb{1}_n = \mathrm{R}\mathbb{1}_n = \mathbb{1}_n$ and $\mathrm{Q}^\top \mathbb{1}_n = \mathrm{R}^\top \mathbb{1}_n = \mathbb{1}_n$, so $\mathrm{Q}, \mathrm{R} \in \mathcal{B}_n$. Finally, $\mathrm{Q} \neq \mathrm{R}$ because $\varepsilon > 0$ and the cycle is non-empty, while by construction $\mathrm{P} = (\mathrm{Q} + \mathrm{R})/2$. Thus P is not extreme. Consequently every extreme point of $\mathcal{B}_n$ is integral, and every integral bistochastic matrix is a permutation matrix. ◻

Corollary 3.18 (Kantorovich for matching)**.** *If $m = n$ and $\mathrm{a} = \mathrm{b} = \mathbb{1}_n/n$, then the discrete Kantorovich problem (3.2) admits an optimal solution of the form P_σ/n. The associated permutation σ solves the assignment problem of Section 1.1.*

Proof. The feasible set is $\mathcal{B}_n/n$. By Proposition 3.16, the linear objective has an optimal extreme point. Since scaling preserves extreme points and Theorem 3.17 identifies the extreme points of $\mathcal{B}_n$, this optimizer is P_σ/n for some permutation σ. Its cost is exactly $n^{-1} \sum_i \mathrm{C}_{i,\sigma(i)}$, so σ is an optimal assignment. ◻

Equivalently, for uniform empirical measures, one can always choose a permutation matrix among the minimizers of the relaxed Kantorovich problem: the relaxation is tight for assignment problems.

> **Remark 3.19 (General discrete case).** For general input measures, one does not have equivalence between Monge and Kantorovich problems, since the Monge constraint can be empty. In finite dimension, however, the support of an optimal coupling still enjoys strong sparsity: one can choose an optimal basic feasible plan whose bipartite support is cycle-free, hence with at most $n + m - 1$ nonzero entries. Figure 3.3 illustrates the difference between the tight uniform matching case and the genuinely splitting nonuniform case.

Constructive Birkhoff–von Neumann decomposition. The same combinatorial idea gives both a finite convex decomposition of every bistochastic matrix and an effective procedure for extracting its permutation matrices. The existence statement follows immediately from the extreme-point theorem above and Carathéodory's theorem; the algorithmic statement makes the matching argument constructive.

Corollary 3.20 (Birkhoff–von Neumann decomposition)**.** *Every $\mathrm{P} \in \mathcal{B}_n$ admits a representation*

$$\mathrm{P} = \sum_{r=1}^{L} \lambda_r \mathrm{P}_{\sigma_r}, \qquad \lambda_r > 0, \qquad \sum_{r=1}^{L} \lambda_r = 1, \qquad L \leqslant (n-1)^2 + 1,$$

where each P_{σ_r} is a permutation matrix.

Proof. Let $\mathcal{K}$ be the convex hull of the permutation matrices. Clearly $\mathcal{K} \subset \mathcal{B}_n$. Conversely, if some $\mathrm{P} \in \mathcal{B}_n$ did not belong to the compact convex set $\mathcal{K}$, strict separation would provide a linear form ℓ such that $\ell(\mathrm{P}) < \min_{\mathrm{Q} \in \mathcal{K}} \ell(\mathrm{Q})$. Proposition 3.16 and Theorem 3.17 give

$$\min_{\mathrm{Q} \in \mathcal{B}_n} \ell(\mathrm{Q}) = \min_{\sigma \in \mathrm{Perm}(n)} \ell(\mathrm{P}_\sigma) = \min_{\mathrm{Q} \in \mathcal{K}} \ell(\mathrm{Q}),$$

contradicting $\mathrm{P} \in \mathcal{B}_n$. Hence $\mathcal{B}_n = \mathcal{K}$.

The $2n$ row- and column-sum equations have rank $2n - 1$, and the strictly positive matrix $\mathbb{1}_n \mathbb{1}_n{}^\top/n$ lies in their affine space. Therefore $\mathcal{B}_n$ has affine dimension $n^2 - (2n - 1) = (n-1)^2$. Carathéodory's theorem in this affine hull now reduces any convex decomposition to at most $(n-1)^2 + 1$ permutation matrices; zero coefficients can be discarded. ◻

The decomposition can be computed without first enumerating the extreme points. Hall's theorem states that a finite bipartite graph admits a matching covering all row vertices if and only if every row subset I has at least $|I|$ neighbors [267]. To apply it, let $\mathrm{R} \geqslant 0$ have every row and column sum equal to the same number $s > 0$, and let G_{R} be its support graph. If $N(I)$ denotes the neighbor set of I, then

$$s|I| = \sum_{i \in I} \sum_{j \in N(I)} \mathrm{R}_{i,j} \leqslant \sum_{j \in N(I)} \sum_{i} \mathrm{R}_{i,j} = s|N(I)|.$$

Thus $|N(I)| \geqslant |I|$, so G_{R} contains a perfect matching. The following algorithm extracts that matching by successive augmenting paths and then removes the largest possible multiple of its permutation matrix.

Algorithm 3.2: Birkhoff–von Neumann decomposition

Input: $\mathrm{P} \in \mathcal{B}_n$.
Initialize: $\mathrm{R} \leftarrow \mathrm{P}$, $s \leftarrow 1$, and an empty list $\mathcal{D}$.
While $s > 0$:
- Form the support graph G_{R} with edge (i, j) whenever $\mathrm{R}_{i,j} > 0$.
- **Initialize matching:** $M \leftarrow \varnothing$.
- **For** $q = 1, \ldots, n$ **do**:
  - Choose an unmatched row i and find by breadth-first search an M-augmenting path A from i to an unmatched column, alternating between edges outside and inside M.
  - Set $M \leftarrow M \triangle A$.
- Set $\sigma(i) = j$ for every $(i, j) \in M$ and $\lambda \leftarrow \min_i \mathrm{R}_{i,\sigma(i)}$.
- Append (λ, σ) to $\mathcal{D}$, set $\mathrm{R} \leftarrow \mathrm{R} - \lambda \mathrm{P}_\sigma$, and set $s \leftarrow s - \lambda$.

Return $\mathcal{D}$, so that $\mathrm{P} = \sum_{(\lambda,\sigma)\in\mathcal{D}} \lambda \mathrm{P}_\sigma$.

Hall's condition proves that each augmenting-path search succeeds. Indeed, if a search rooted at an unmatched row were to fail, its alternating tree would reach a row set I and its full neighbor set $N(I)$. Every reached column would be matched to a distinct reached row other than the root, giving $|N(I)| = |I| - 1$ and contradicting Hall's condition. This augmenting-path argument is the standard computational proof of Hall's theorem; bistochasticity supplies its neighborhood hypothesis here.

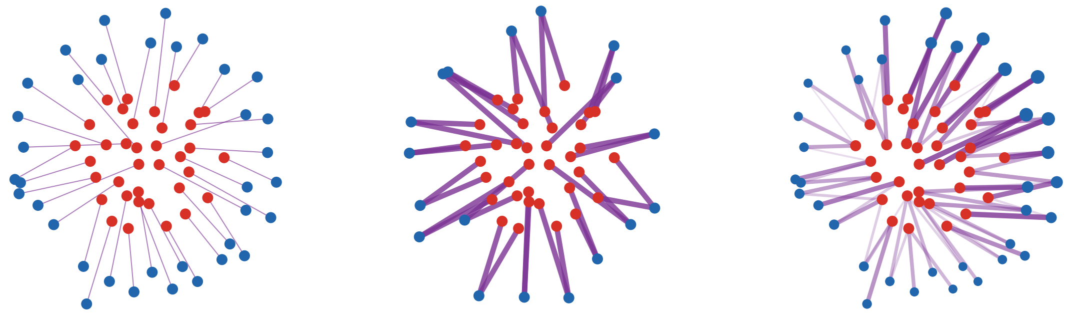
The residual update preserves nonnegativity and changes every row and column sum from s to $s - \lambda$. To verify both termination and the sharper term bound, normalize the current residual as $\widehat{\mathrm{P}} = \mathrm{R}/s \in \mathcal{B}_n$ and let F be its minimal face. The chosen permutation matrix belongs to F because its support is contained in that of $\widehat{\mathrm{P}}$. If $\lambda < s$, then

$$\widehat{\mathrm{P}}' := \frac{\mathrm{R} - \lambda \mathrm{P}_\sigma}{s - \lambda}$$

is the next normalized residual. Maximality of $\lambda = \min_i \mathrm{R}_{i,\sigma(i)}$ makes at least one formerly positive coordinate vanish, so $\widehat{\mathrm{P}}'$ lies in a proper face of F. Each nonterminal iteration therefore lowers the face dimension by at least one; if $\lambda = s$, the residual vanishes. Since $\dim(\mathcal{B}_n) = (n-1)^2$, Algorithm 3.2 terminates after at most $(n-1)^2 + 1$ extracted permutations. Summing its residual updates gives the asserted decomposition and $\sum_{(\lambda,\sigma)\in\mathcal{D}} \lambda = 1$.

If the current support graph has E edges, the elementary matching subroutine performs at most n breadth-first searches and costs $O(nE)$. Thus a dense implementation costs $O(n^3)$ per extracted permutation and, using the term bound above, at most $O(n^5)$ arithmetic and graph operations overall, with $O(n^2)$ memory. This is a conservative worst-case estimate rather than a bit-complexity bound; faster bipartite-matching routines improve it, and sparse residual supports reduce E substantially [316, 84].

Rational weights. The strict assignment model is tied to equal cardinalities and equal weights, whereas the coupling set $\mathrm{U}(\mathrm{a}, \mathrm{b})$ of Definition 3.1 accommodates arbitrary discrete masses and different support sizes. Figure 3.5 contrasts these regimes. For rational weights, the relaxed problem can in fact be reduced back to a larger uniform assignment problem.



$n = m$ $\qquad$ $m < n$ $\qquad$ $n = m$ but $b_j \neq 1/n$

Figure 3.5: From assignments to transport plans, using the same disk-to-annulus geometry as Figure 1.5. In the balanced equal-weight case, each source atom is matched to one target atom. With a target cloud that has half as many atoms, or with strongly nonuniform target weights, the coupling matrix can merge or split mass; segment thickness and opacity encode its nonzero entries, and blue marker areas encode the prescribed target masses.

Proposition 3.21 (Rational weights as duplicated uniform matching)**.** *Let*

$$\alpha = \sum_{i=1}^{n} \frac{k_i}{N}\delta_{x_i}, \qquad \beta = \sum_{j=1}^{m} \frac{\ell_j}{N}\delta_{y_j}, \qquad \sum_i k_i = \sum_j \ell_j = N,$$

with positive integers k_i, ℓ_j. Replace each x_i by k_i identical copies and each y_j by ℓ_j identical copies, producing two uniform N-point clouds. Then the duplicated assignment problem and the discrete Kantorovich problem between α and β have the same optimal value. Moreover, there exists an optimal coupling of the form

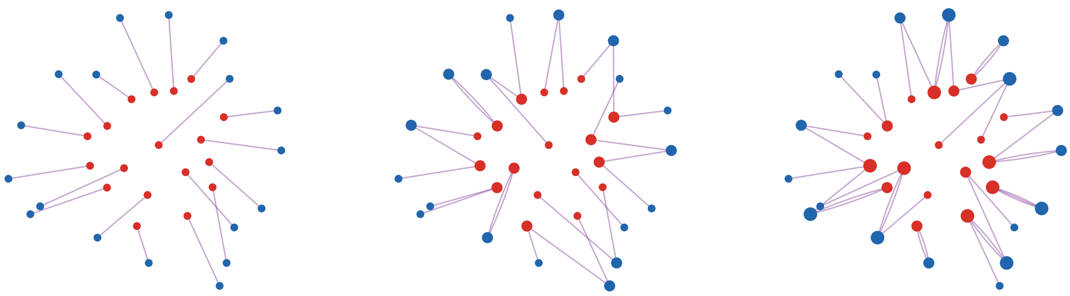
$$\mathrm{P}_{i,j} = \frac{n_{i,j}}{N}, \qquad n_{i,j} \in \mathbb{N}, \qquad \sum_j n_{i,j} = k_i, \quad \sum_i n_{i,j} = \ell_j,$$

and couplings whose scaled entries $N\mathrm{P}_{i,j}$ are integral are exactly the collapsed assignments between duplicated clouds. Fractional optimal couplings may nevertheless coexist when the optimum is degenerate.

Proof. Any assignment between the duplicated source and target clouds defines integers $n_{i,j}$ counting how many copied particles of type x_i are matched to copied particles of type y_j. These counts satisfy $\sum_j n_{i,j} = k_i$ and $\sum_i n_{i,j} = \ell_j$, and the associated coupling $\mathrm{P}_{i,j} = n_{i,j}/N$ has marginals k_i/N and ℓ_j/N. The assignment cost is $\frac{1}{N}\sum_{i,j} n_{i,j} c(x_i, y_j) = \sum_{i,j} \mathrm{P}_{i,j} c(x_i, y_j)$. Conversely, any nonnegative integer count matrix with those row and column sums can be realized by allocating the k_i copies of each x_i among the target copies according to $(n_{i,j})_j$. It remains to show that restricting to integral counts does not increase the optimum. Scale a feasible coupling by N and write $\mathrm{Q} = N\mathrm{P}$. The constraints on Q have integer right-hand sides $(k_1, \ldots, k_n, \ell_1, \ldots, \ell_m)$. After multiplying the target rows by -1, their coefficient matrix is the oriented node-edge incidence matrix of a bipartite graph and is therefore totally unimodular. Every vertex of this transportation polytope is integral. Since a linear objective attains its minimum at a vertex, an integral optimal Q exists, proving equality of the two optimal values. The argument establishes existence, not integrality of every optimizer; convex combinations of distinct integral optima give fractional optima. □

This network-flow integrality mechanism complements the Birkhoff–von Neumann theorem above (Theorem 3.17): in both cases, a linear transport relaxation has an optimizer represented by integer edge flows after scaling. The equal-unit-margin case specializes these flows to permutation matrices, whereas a degenerate optimal face may also contain fractional couplings.

Figure 3.6 makes this reduction explicit by replacing each rational mass with identical unit copies and then regrouping the resulting uniform assignment.



$k_i = \ell_j = 1$ $\quad$ $k_i, \ell_j \in \{1,2\}$ $\quad$ $k_i, \ell_j \in \{1,2,3\}$

Figure 3.6: Rational weights as duplicated uniform matchings, using the same disk-to-annulus geometry as Figure 3.5 with fewer displayed atoms. The red and blue locations are kept fixed, while disk areas encode the integer multiplicities k_i and ℓ_j. Solving the assignment problem after duplicating particles produces several collapsed segments attached to high-multiplicity atoms; this is the integer count matrix of Proposition 3.21.

Applications of discrete transport. Discrete Kantorovich transport applies whenever weighted data points or histograms must be compared through a geometry-aware correspondence. The following examples illustrate information transfer between datasets, comparison of visual distributions, and inference of temporal relations between cell populations.

Example 3.22 (Application to domain adaptation). In unsupervised domain adaptation, labeled source samples $(x_i^s, y_i^s)_i$ and unlabeled target samples $(x_j^t)_j$ define empirical laws $\alpha_s = \sum_i a_i \delta_{x_i^s}$ and $\alpha_t = \sum_j b_j \delta_{x_j^t}$. Writing $\mathrm{a} = (a_i)_i$ and $\mathrm{b} = (b_j)_j$, a Kantorovich coupling $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ gives soft correspondences between the two clouds. For $b_j > 0$, disintegrating P with respect to its target marginal and averaging the source labels gives the backward barycentric rule $\widetilde{y}_j = b_j^{-1} \sum_i \mathrm{P}_{i,j} y_i^s$, for either real-valued labels or one-hot class vectors. This is the label-space analogue of the barycentric projection introduced later in Definition 12.31. Alternatively, the transport can be optimized jointly with a classifier by adding label-prediction terms to the feature cost. This is the mechanism behind OT domain adaptation and JDOT: the plan is not only a distance certificate, but an explicit cross-domain alignment [161, 160]. Learning or adapting the ground cost is the inverse viewpoint developed later in Section 12.5.

Example 3.23 (Application to visual distributions). Images, color histograms, texture descriptors and shape samples can all be represented as discrete measures. The ground cost then encodes color differences, image-plane displacements or surface distances. This viewpoint underlies the Earth Mover's Distance for image retrieval [452], regularized transport for imaging [544], convolutional transport on geometric domains [489], and Wasserstein barycenters for texture mixing or Radon-domain image processing [76, 75]. The practical gain is that the discrepancy respects geometry: moving a small amount of mass to a nearby color or pixel is cheaper than moving it far away.

Example 3.24 (Application to single-cell population dynamics). Single-cell sequencing observes different cells at each collection time. If $x_i^{[k]} \in \mathcal{X}$ denotes the state of cell i at time t_k, write

$$\alpha_{t_k} = \sum_{i=1}^{n_k} a_i^{[k]} \delta_{x_i^{[k]}} \quad \text{and} \quad \mathrm{a}^{[k]} = (a_i^{[k]})_{i=1}^{n_k} \in \Sigma_{n_k}.$$

Thus a cell is one atom of an empirical population measure, rather than itself a measure over genes as in Example 4.10. A balanced transport between two consecutive snapshots is represented by a matrix $\mathrm{P}^{[k]} \in \mathbb{R}_+^{n_k \times n_{k+1}}$ satisfying

$$\mathrm{P}^{[k]} \mathbb{1}_{n_{k+1}} = \mathrm{a}^{[k]} \quad \text{and} \quad (\mathrm{P}^{[k]})^\top \mathbb{1}_{n_k} = \mathrm{a}^{[k+1]}.$$

Its entry $\mathrm{P}_{i,j}^{[k]}$ is a soft ancestor–descendant mass, not evidence that the same cell was observed twice.

For every row with positive sum, the normalization

$$K_{i,j}^{[k]} := \frac{P_{i,j}^{[k]}}{\sum_{j'=1}^{n_{k+1}} P_{i,j'}^{[k]}}$$

defines a conditional transition probability; multiplying successive matrices $K^{[k]}$ yields population-level lineage probabilities across several collection times.
Waddington-OT modifies this balanced baseline by incorporating growth estimates and relaxing the marginal constraints through unbalanced transport, as discussed in Example 11.4 [470]. Related continuous and entropic constructions include trajectory inference and generative transport models [514, 326, 308]. Figure 3.7 displays three observed population snapshots in the common force-layout embedding used by Waddington-OT; this visualization need not coincide with the ground geometry used to compute $P^{[k]}$.

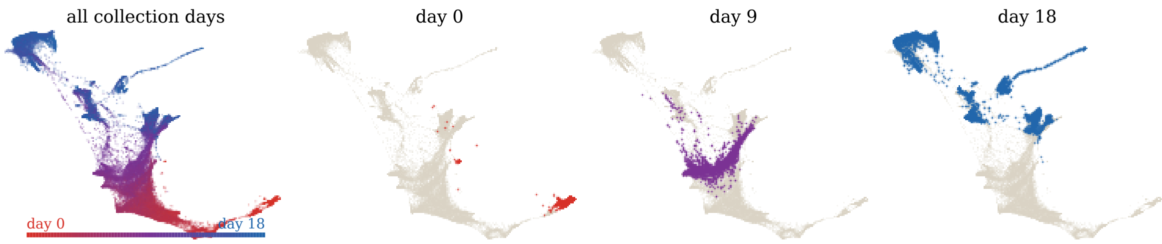


Figure 3.7: Three snapshots of the Waddington-OT single-cell reprogramming time course. The first panel aggregates a representative sample from all collection times and colors cells from red to blue according to time. The remaining panels highlight the populations observed at days 0, 9, and 18, while cells from other times remain in gray. Every panel uses the same official force-layout embedding and viewport; no transport coupling or interpolated trajectory is displayed.

3.2 Linear-Programming Algorithms

The discrete Kantorovich problem is a linear program with much more structure than a generic dense LP. Its variables are the arcs of a complete bipartite network, its equality constraints are flow-conservation constraints, and its extreme points are sparse tree-like couplings. This is why classical transportation algorithms remain important even though the formulation is finite-dimensional and convex.

Transportation simplex and network simplex. The transportation simplex goes back to Dantzig's formulation of the transportation problem [178]. It works on basic feasible couplings, whose positive support is completed into a spanning tree of the bipartite supply-demand graph. Starting from a basis, for instance one produced by the north-west corner rule, reduced costs identify whether an unused arc can decrease the objective. Adding such an arc creates a unique cycle in the tree; one then pushes as much mass as possible around this cycle and removes the exhausted arc. This is exactly the simplex method specialized to the transportation polytope, with pivots that can be implemented using graph operations instead of a generic basis inverse.

The network simplex is the corresponding pivoting method for general minimum-cost-flow problems [51]. It keeps node potentials, reduced costs and a spanning-tree basis, and it has become one of the most effective exact solvers for medium-scale discrete OT. Like the ordinary simplex method, its worst-case number of pivots can be exponential for adversarial instances, but the per-pivot operations exploit sparsity and are very efficient in practice. For theoretical polynomial guarantees, one can instead use strongly polynomial minimum-cost-flow algorithms, such as Orlin's algorithm [407]. In a dense balanced transportation problem with n sources and n targets, the graph has $O(n)$ vertices and $O(n^2)$ arcs, so these general bounds are polynomial but still much heavier than the nearly matrix-vector structure that Sinkhorn will exploit.

Interior-point methods. Generic interior-point methods approach the same LP through a smooth central path. Assume here that all entries of a and b are positive; zero-mass rows and columns must first be removed. The logarithmic-barrier problem on the resulting transport polytope is

$$\mathrm{P}_\varepsilon := \underset{\substack{\mathrm{P}\mathbb{1}_m=\mathrm{a},\ \mathrm{P}^\top\mathbb{1}_n=\mathrm{b} \\ \mathrm{P}_{i,j}>0}}{\operatorname{argmin}} \ \langle \mathrm{C}, \mathrm{P}\rangle - \varepsilon \sum_{i,j} \log \mathrm{P}_{i,j}, \tag{3.5}$$

where $\varepsilon > 0$ is decreased along the algorithm. The barrier is singular at the boundary, so each iterate stays strictly inside the transportation polytope; as $\varepsilon \downarrow 0$, the central path approaches the set of LP minimizers. Each Newton step solves a linear system involving the marginal constraints and the current diagonal Hessian $\operatorname{diag}(\varepsilon/\mathrm{P}_{i,j}^2)$, which gives robust polynomial complexity but can be expensive for dense couplings [397].

Figure 3.8 isolates this mechanism on a two-dimensional polytope: decreasing the barrier parameter moves the minimizer along the central path toward the optimal face.

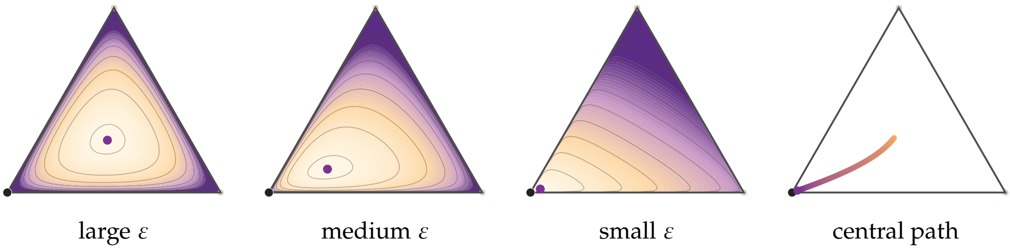


Figure 3.8: Logarithmic-barrier central path for a two-dimensional equilateral triangular slice of a linear program. The first three panels display level sets of $\ell^\top z - \varepsilon \sum_i \log(b_i - \langle a_i, z\rangle)$ for decreasing barrier strengths. The last panel removes the objective background and shows only the central path inside the feasible triangle. Large ε selects a central interior point; decreasing ε moves the minimizer toward the optimal vertex while never touching the boundary. This should be contrasted with entropic OT, where the entropy temperature ε is usually fixed and defines the regularized problem itself.

The comparison with Sinkhorn is therefore subtle. Both methods keep iterates positive, but they use positivity in different ways. Interior-point algorithms solve the original LP by decreasing the barrier parameter ε and following a central path. Sinkhorn fixes an entropic temperature ε and solves a different, KL-regularized OT problem by alternating diagonal scalings. The parameter ε may be decreased in continuation strategies, but for a fixed run it is part of the objective rather than only an algorithmic barrier.

3.3 Relaxation for Arbitrary Measures

This section lifts the finite-dimensional coupling matrix to a joint probability measure. The payoff is that existence, duality and metric properties can be stated for arbitrary laws, including discrete, singular and continuous distributions.

Continuous couplings. The first step is to formalize what it means for a joint law to have prescribed marginals.

Definition 3.25 (Marginals of a joint measure)**.** Let $\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})$ and let $P_\mathcal{X}(x,y) = x$, $P_\mathcal{Y}(x,y) = y$ be the coordinate projections. The marginals of π are

$$\pi_1 := (P_\mathcal{X})_\sharp \pi \in \mathcal{M}_+^1(\mathcal{X}), \qquad \pi_2 := (P_\mathcal{Y})_\sharp \pi \in \mathcal{M}_+^1(\mathcal{Y}).$$

Equivalently, for all bounded continuous test functions f on $\mathcal{X}$ and g on $\mathcal{Y}$,

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\pi(x,y) = \int_\mathcal{X} f\mathrm{d}\pi_1, \qquad \int_{\mathcal{X}\times\mathcal{Y}} g(y)\mathrm{d}\pi(x,y) = \int_\mathcal{Y} g\mathrm{d}\pi_2.$$

A useful mnemonic for the marginal constraint $\pi_1 = \alpha$ and $\pi_2 = \beta$ is the formal row-and-column notation

$$\int_\mathcal{Y} \mathrm{d}\pi(x,y) = \mathrm{d}\alpha(x), \qquad \int_\mathcal{X} \mathrm{d}\pi(x,y) = \mathrm{d}\beta(y),$$

which is made rigorous by Definition 3.25, or equivalently by the identities $\pi(A \times \mathcal{Y}) = \alpha(A)$ and $\pi(\mathcal{X} \times B) = \beta(B)$ for measurable sets $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$.

Definition 3.26 (Couplings)**.** Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between α and β is

$$\mathcal{U}(\alpha, \beta) := \left\{\pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y})\,;\; \pi_1 = \alpha \quad \text{and} \quad \pi_2 = \beta\right\}. \tag{3.6}$$

This is the continuous analogue of the transportation polytope (3.1).

Remark 3.27 (Probabilistic interpretation of couplings). If $X \sim \alpha$ and $Y \sim \beta$, then $\pi \in \mathcal{U}(\alpha, \beta)$ means that π is the law of a pair (X, Y) whose coordinates have laws α and β. The coupling encodes the dependence between X and Y. The tensor product $\alpha \otimes \beta$ corresponds to independence, whereas a graph coupling $(\mathrm{Id}, T)_\sharp \alpha$ corresponds to the deterministic relation $Y = T(X)$.
In the discrete case, when $\alpha = \sum_i \mathrm{a}_i \delta_{x_i}$ and $\beta = \sum_j \mathrm{b}_j \delta_{y_j}$, the constraint $\pi_1 = \alpha$ and $\pi_2 = \beta$ forces every coupling to have the form $\pi = \sum_{i,j} \mathrm{P}_{i,j} \delta_{(x_i, y_j)}$ with $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$. The discrete formulation is therefore a special case of the continuous one, not merely an approximation.

Unlike the Monge constraint, the coupling constraint is never empty. The continuous feasibility witness is the tensor product coupling, the measure-theoretic version of the discrete product plan above.

Definition 3.28 (Tensor product and trivial coupling)**.** Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the tensor product coupling $\alpha \otimes \beta$ is the probability measure on $\mathcal{X} \times \mathcal{Y}$ defined by

$$\int_{\mathcal{X}\times\mathcal{Y}} h(x, y) \mathrm{d}(\alpha \otimes \beta)(x, y) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} h(x, y) \mathrm{d}\beta(y) \right) \mathrm{d}\alpha(x)$$

for every bounded measurable h. It is also called the trivial coupling because it makes the two coordinates independent.

Indeed, for every $f \in C_b(\mathcal{X})$,

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x) \mathrm{d}(\alpha \otimes \beta)(x, y) = \left(\int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) \right) \left(\int_{\mathcal{Y}} \mathrm{d}\beta(y) \right) = \int f \mathrm{d}\alpha,$$

and similarly for the second marginal, so $\alpha \otimes \beta \in \mathcal{U}(\alpha, \beta)$.

The next result echoes Proposition 3.4 in the continuous setting. In both cases, the independent coupling is optimal precisely when the objective is flat over the whole admissible set; for continuous costs, this flatness is equivalently the additive form $c(x, y) = u(x) + v(y)$ on the product support.

Proposition 3.29 (Product optimality is degenerate)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in C(\mathcal{X} \times \mathcal{Y})$. The following statements are equivalent:*

$$\alpha \otimes \beta \in \operatorname*{argmin}_{\pi \in \mathcal{U}(\alpha, \beta)} \int c \,\mathrm{d}\pi, \qquad \textit{every coupling in } \mathcal{U}(\alpha, \beta) \textit{ is optimal.}$$

They are also equivalent to the additive decomposition of the cost on the product support,

$$c(x, y) = u(x) + v(y), \qquad u \in C(\operatorname{supp}(\alpha)), \quad v \in C(\operatorname{supp}(\beta)).$$

Proof. If every coupling is optimal, then $\alpha \otimes \beta$ is optimal. Conversely, assume that $\alpha \otimes \beta$ is optimal. We first show that, for every $x_0, x_1 \in \operatorname{supp}(\alpha)$ and $y_0, y_1 \in \operatorname{supp}(\beta)$,

$$c(x_0, y_0) + c(x_1, y_1) = c(x_0, y_1) + c(x_1, y_0).$$

Indeed, if this equality failed, after exchanging y_0 and y_1 if necessary one would have a strict inequality

$$c(x_0, y_0) + c(x_1, y_1) > c(x_0, y_1) + c(x_1, y_0).$$

Failure of the equality forces $x_0 \neq x_1$ and $y_0 \neq y_1$. By continuity, the strict inequality persists with a uniform margin on small neighborhoods U_0, U_1 of x_0, x_1 and V_0, V_1 of y_0, y_1, chosen disjoint within each pair. Since the four points lie in the supports, these neighborhoods have positive marginal mass. Denote by α_i and β_i the normalized restrictions of α to U_i and of β to V_i, and choose

$$0 < \lambda \leqslant \min\{\alpha(U_0)\beta(V_0), \alpha(U_1)\beta(V_1)\}.$$

The exchanged measure

$$\tilde{\pi} = \alpha \otimes \beta - \lambda\, \alpha_0 \otimes \beta_0 - \lambda\, \alpha_1 \otimes \beta_1 + \lambda\, \alpha_0 \otimes \beta_1 + \lambda\, \alpha_1 \otimes \beta_0$$

is nonnegative and has the same two marginals as $\alpha \otimes \beta$. The uniform strict inequality on the neighborhoods implies that $\int c\, \mathrm{d}\tilde{\pi} < \int c\, \mathrm{d}(\alpha \otimes \beta)$, contradicting optimality.

Fixing any $x_\star \in \mathrm{supp}(\alpha)$ and $y_\star \in \mathrm{supp}(\beta)$, the equality of cross differences gives, for all $(x, y) \in \mathrm{supp}(\alpha) \times \mathrm{supp}(\beta)$,

$$c(x, y) = c(x, y_\star) + c(x_\star, y) - c(x_\star, y_\star).$$

Thus $c = u + v$ on the product support, with $u(x) = c(x, y_\star) - c(x_\star, y_\star)$ and $v(y) = c(x_\star, y)$. These functions are continuous on the corresponding supports. Every coupling is concentrated on this product support, so for any $\pi \in \mathcal{U}(\alpha, \beta)$, $\int c\, \mathrm{d}\pi = \int u\, \mathrm{d}\alpha + \int v\, \mathrm{d}\beta$, which depends only on the marginals. Hence all couplings are optimal. □

The tensor product is therefore a trivial feasible coupling, not a typical optimizer. Product optimality means that the cost cannot distinguish between dependences once the marginals are fixed. The continuity assumption is important: if $\alpha = \beta$ is the uniform law on $[0, 1]$ and $c(x, y) = \mathbb{1}_{\{x=y\}}$, then $\alpha \otimes \beta$ has zero cost and is optimal, whereas the identity coupling has cost one. Thus, for arbitrary merely measurable costs, changing the cost on an $\alpha \otimes \beta$-negligible set may affect singular couplings without changing the product cost.

If there exists a map $T : \mathcal{X} \to \mathcal{Y}$ such that $T_\sharp \alpha = \beta$, then the Monge map induces the graph coupling $\pi = (\mathrm{Id}, T)_\sharp \alpha \in \mathcal{U}(\alpha, \beta)$, characterized by

$$\int_{\mathcal{X} \times \mathcal{Y}} h(x, y) \mathrm{d}\pi(x, y) = \int_{\mathcal{X}} h(x, T(x)) \mathrm{d}\alpha(x).$$

Applying this identity to $h(x, y) = f(x)$ or $h(x, y) = g(y)$ gives respectively $\pi_1 = \alpha$ and $\pi_2 = \beta$. Thus graph couplings are precisely the Kantorovich representation of deterministic Monge maps. A last important class consists of semi-discrete problems, where α has a density and β is discrete. Every coupling is then supported on the union of the slices $\mathcal{X} \times \{y_j\}$. When an optimal coupling is induced by a map, these slices are selected by a partition of $\mathcal{X}$ into transport cells, as developed in Chapter 6.

Remark 3.30 (Book-shifting as a flat Kantorovich face). The Monge book-shifting example in Example 2.21 has a transparent coupling interpretation. Let α be uniform on $[0, 2]$ and β uniform on $[1, 3]$. For every $\pi \in \mathcal{U}(\alpha, \beta)$,

$$\int |y - x| \mathrm{d}\pi(x, y) \geqslant \int (y - x) \mathrm{d}\pi(x, y) = \int y \mathrm{d}\beta(y) - \int x \mathrm{d}\alpha(x) = 1.$$

Equality holds exactly for couplings concentrated on the half-plane $\{(x, y) : y \geqslant x\}$, where $|y - x| = y - x$. Hence the optimal set is a whole face of the coupling polytope, not a single graph. The translation and book-shifting maps give two graph couplings inside this face, but many non-deterministic couplings are optimal as well.

Continuous Kantorovich problem. The discrete Kantorovich relaxation of Section 3.1, formalized in Definition 3.5, extends from nonnegative coupling matrices to coupling measures between arbitrary probability measures.

Definition 3.31 (Continuous Kantorovich problem). For $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, and a nonnegative Borel cost $c : \mathcal{X} \times \mathcal{Y} \to [0, +\infty]$, define

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y) \mathrm{d}\pi(x, y). \tag{3.7}$$

A minimizer is an optimal coupling, or optimal transport plan. The value is understood in $[0, +\infty]$.

This is an infinite-dimensional linear program over a space of measures.

The following result generalizes Proposition 3.6 from coupling matrices to coupling measures: the Kantorovich value is convex in the marginals but concave in the ground cost.

Proposition 3.32 (Convexity in the marginals and concavity in the cost). *Fix a nonnegative Borel cost c. The extended-valued map*

$$(\alpha, \beta) \longmapsto \mathcal{L}_c(\alpha, \beta)$$

is jointly convex on pairs of probability measures. Conversely, for fixed marginals (α, β), the map $c \mapsto \mathcal{L}_c(\alpha, \beta)$ is concave on the cone of nonnegative Borel costs: for any such c_0, c_1 and $t \in [0, 1]$,

$$\mathcal{L}_{(1-t)c_0+tc_1}(\alpha, \beta) \geqslant (1 - t)\mathcal{L}_{c_0}(\alpha, \beta) + t\mathcal{L}_{c_1}(\alpha, \beta).$$

The inequality is understood in $[0, +\infty]$; in particular, the map is concave on every convex family of costs on which it is finite.

Proof. For joint convexity, let (α_0, β_0) and (α_1, β_1) be two pairs of probability measures. The claim is immediate at $t \in \{0, 1\}$ or if one of the two values on the right-hand side is infinite. Otherwise, for $\eta > 0$, choose $\pi_i \in \mathcal{U}(\alpha_i, \beta_i)$ such that $\int c \mathrm{d}\pi_i \leqslant \mathcal{L}_c(\alpha_i, \beta_i) + \eta$ for $i \in \{0, 1\}$. Then $(1 - t)\pi_0 + t\pi_1$ couples $(1 - t)\alpha_0 + t\alpha_1$ and $(1 - t)\beta_0 + t\beta_1$, and hence

$$\mathcal{L}_c\big((1 - t)\alpha_0 + t\alpha_1, (1 - t)\beta_0 + t\beta_1\big) \leqslant (1 - t)\mathcal{L}_c(\alpha_0, \beta_0) + t\mathcal{L}_c(\alpha_1, \beta_1) + \eta.$$

Letting $\eta \to 0$ proves joint convexity.

For concavity, the endpoint cases are immediate, so assume $t \in (0, 1)$ and fix $\pi \in \mathcal{U}(\alpha, \beta)$. Linearity of integration for nonnegative functions gives

$$\int \big((1 - t)c_0 + tc_1\big)\mathrm{d}\pi \geqslant (1 - t)\mathcal{L}_{c_0}(\alpha, \beta) + t\mathcal{L}_{c_1}(\alpha, \beta).$$

Taking the infimum over π proves the stated inequality. □

Remark 3.33 (Probabilistic interpretation of Kantorovich's problem). The continuous Kantorovich problem of Definition 3.31 can equivalently be written as

$$\mathcal{L}_c(\alpha, \beta) = \inf_{X\sim\alpha, Y\sim\beta} \mathbb{E}(c(X, Y)).$$

The minimization is not over the marginal laws, which are fixed, but over all possible dependences between the two random variables. OT therefore chooses the cheapest joint law among all couplings.

Existence is one of the decisive gains of Kantorovich relaxation. The Monge problem of Definition 2.14 may have no admissible map, and even when maps exist its infimum need not be attained because graph couplings are not closed under weak limits. By contrast, $\mathcal{U}(\alpha, \beta)$ always contains the product coupling $\alpha \otimes \beta$; tightness and weak closedness then make the direct method available.

Proposition 3.34 (Existence for lower-semicontinuous costs). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces and that $c : \mathcal{X} \times \mathcal{Y} \to [0, +\infty]$ is lower semicontinuous. Then the Kantorovich problem (3.7) admits at least one minimizer. In particular, this applies to every continuous nonnegative cost on compact metric spaces.*

Proof. The constraint set is non-empty because it contains the product coupling $\alpha\otimes\beta$. It is uniformly tight: given $\varepsilon > 0$, choose compact sets $K_\mathcal{X} \subset \mathcal{X}$ and $K_\mathcal{Y} \subset \mathcal{Y}$ such that $\alpha(K_\mathcal{X}) \geqslant 1 - \varepsilon/2$ and $\beta(K_\mathcal{Y}) \geqslant 1 - \varepsilon/2$. Every $\pi \in \mathcal{U}(\alpha, \beta)$ then satisfies $\pi(K_\mathcal{X} \times K_\mathcal{Y}) \geqslant 1 - \varepsilon$. By Prokhorov's theorem [69, 531], the set of couplings is relatively compact for weak convergence; the weak topology is reviewed in Section 4.2. It is also weakly closed, because the coordinate projections are continuous and therefore preserve the marginal identities in the limit. Hence $\mathcal{U}(\alpha, \beta)$ is weakly compact. Finally, Portmanteau's theorem makes $\pi \mapsto \int c\mathrm{d}\pi$ weakly lower semicontinuous. The direct method gives a minimizer. □

Axiomatic characterization of the relaxation. Kantorovich relaxation is not only a convenient enlargement of Monge's problem: convexity and weak stability single it out as the maximal extension of the pointwise ground cost from pairs of points to pairs of probability measures.

Theorem 3.35 (Kantorovich cost as the maximal convex relaxation)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c : \mathcal{X} \times \mathcal{Y} \to [0, +\infty)$ is continuous. Then $\mathcal{L}_c$ is the pointwise largest proper functional*

$$\mathsf{F} : \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \longrightarrow (-\infty, +\infty]$$

that is jointly convex, weakly lower semicontinuous, and satisfies

$$\mathsf{F}(\delta_x, \delta_y) \leqslant c(x, y) \qquad \text{for every } (x, y) \in \mathcal{X} \times \mathcal{Y}.$$

More precisely, every such F satisfies $\mathsf{F}(\alpha, \beta) \leqslant \mathcal{L}_c(\alpha, \beta)$ for all (α, β), while $\mathcal{L}_c$ itself has these properties and obeys $\mathcal{L}_c(\delta_x, \delta_y) = c(x, y)$.

Proof. Joint convexity follows from Proposition 3.32. For weak lower semicontinuity, consider converging marginals and optimal plans along a subsequence realizing the lower limit. Compactness gives a further weakly convergent subsequence; its limit has the limiting marginals, and lower semicontinuity of the integral gives the claim, exactly as in Proposition 3.34. The only coupling of (δ_x, δ_y) is $\delta_{(x,y)}$, which proves the identity on Dirac pairs.

Fix $\pi \in \mathcal{U}(\alpha, \beta)$. Its marginal constraints are exactly the barycentric identity $(\alpha, \beta) = \int (\delta_x, \delta_y) \mathrm{d}\pi$. Jensen's inequality for lower-semicontinuous convex functionals therefore gives

$$\mathsf{F}(\alpha, \beta) \leqslant \int \mathsf{F}(\delta_x, \delta_y) \mathrm{d}\pi(x, y) \leqslant \int c(x, y) \mathrm{d}\pi(x, y).$$

Taking the infimum over $\pi \in \mathcal{U}(\alpha, \beta)$ proves maximality. □

This is the probability-measure specialization of the convex-envelope theorem of Savaré and Sodini [466, Theorem 4.4], which treats finite measures on completely regular spaces. The word *largest* is essential: joint convexity, weak lower semicontinuity and equality on Dirac pairs only imply that another extension lies below $\mathcal{L}_c$; they do not force it to coincide with $\mathcal{L}_c$.

Monge–Kantorovich equivalence. The proof of Brenier's theorem 2.23 relies on Kantorovich relaxation and duality. It proves that, under its hypotheses, the relaxation is tight: it has the same cost as the Monge problem and its optimal coupling is induced by a map.

Corollary 3.36 (Monge–Kantorovich equivalence under Brenier)**.** *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, assume that α is absolutely continuous with respect to Lebesgue measure, and let $c(x, y) = |x - y|^2$. If T is the Brenier map solving Monge's problem, then $\pi = (\mathrm{Id}, T)_\sharp \alpha$ is the unique optimal coupling solving the Kantorovich problem. In particular, Monge and Kantorovich costs are the same.*

Proof. The proof of Brenier's theorem shows that the support of any optimal Kantorovich plan is contained in the subdifferential $\partial\varphi$ of a convex function φ. When α has a density, φ is differentiable α-almost everywhere, so $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for α-almost every x. Thus every optimal coupling is concentrated on the graph of $T = \nabla\varphi$ and must equal $(\mathrm{Id}, T)_\sharp \alpha$. The graph coupling is feasible and optimal, and the two formulations have the same value. □

The density assumption is exactly what prevents the relaxed plan from using several destinations at a nonsmooth point.

> **Remark 3.37 (Nonsmooth potentials and splitting).** If α does not have a density, then φ may be non-smooth on a set charged by α, and non-smooth points can lead to mass splitting. For instance, moving δ_0 to $(\delta_{-1} + \delta_{+1})/2$ can be represented by a plan concentrated on the set-valued subdifferential of $\varphi(x) = |x|$, but not by a deterministic map. This is the continuous counterpart of the gap between the uniform matching case of Corollary 3.18 and the general splitting case.

> **Remark 3.38 (Probabilistic form of tightness).** If (X, Y) has the optimal Kantorovich law under the assumptions of Corollary 3.36, then $Y = T(X)$ almost surely with $X \sim \alpha$ and $T(X) \sim \beta$. This is analogous to the Birkhoff–von Neumann result in the fully discrete uniform case: in both settings, the convex relaxation admits an optimizer satisfying the original deterministic constraint. The hypotheses are quite different, however: Birkhoff–von Neumann is finite-dimensional and need

not give uniqueness, whereas Brenier's theorem uses absolute continuity of the source and gives uniqueness of the optimal map almost everywhere.

Kantorovich solution in 1D. The atomless assumption in the Monge statement of Section 2.6 is a limitation of maps, not of one-dimensional optimality. Once couplings are allowed, atoms can be split by assigning subintervals of quantile levels to different target points. The common quantile parameter therefore defines an optimal relaxed coupling for arbitrary probability measures.

Theorem 3.39 (One-dimensional Kantorovich solution)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ and let $h : \mathbb{R} \to [0, +\infty)$ be convex. Consider the cost $c(x, y) = h(x - y)$. Write $q_\alpha := C_\alpha^{-1}$ and $q_\beta := C_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0, 1)$. Then the quantile coupling*

$$\pi^\star = (q_\alpha, q_\beta)_\sharp \mathrm{Leb}_{[0,1]} = (C_\alpha^{-1}, C_\beta^{-1})_\sharp \mathrm{Leb}_{[0,1]}$$

solves the Kantorovich problem (3.7) for $c(x, y) = h(x - y)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional optimal coupling for every $p \geqslant 1$.

Proof. The push-forward statement $\pi^\star \in \mathcal{U}(\alpha, \beta)$ follows from Proposition 2.40. It remains to prove optimality.

The key point is the one-dimensional uncrossing inequality. If $x < x'$ and $y > y'$, set $a = x - y$, $\delta = x' - x > 0$ and $\eta = y - y' > 0$. Convexity of h implies that increments are monotone, hence

$$h(a + \eta) - h(a) \leqslant h(a + \delta + \eta) - h(a + \delta),$$

which is exactly $h(x - y) + h(x' - y') \geqslant h(x - y') + h(x' - y)$. Thus removing a crossing never increases the cost. For a finite transport matrix on two ordered grids, if $i < i'$ and $j > j'$ carry crossed masses $\mathrm{P}_{i,j}$ and $\mathrm{P}_{i',j'}$, one may move $\theta = \min(\mathrm{P}_{i,j}, \mathrm{P}_{i',j'})$ units of mass from the crossed entries (i, j), (i', j') to the uncrossed entries (i, j'), (i', j). The row and column sums are unchanged, and the preceding inequality shows that the cost does not increase. Repeating this elementary uncrossing step yields an ordered plan; on an ordered uniform quantile grid, this is the diagonal plan.

For general measures, apply the same argument in quantile coordinates. Let $\pi \in \mathcal{U}(\alpha, \beta)$. Let U_α and U_β be uniform random variables on $(0, 1)$, and denote by $K_\alpha(x, \mathrm{d}r)$ and $K_\beta(y, \mathrm{d}s)$ regular conditional laws of U_α given $q_\alpha(U_\alpha) = x$ and of U_β given $q_\beta(U_\beta) = y$. Given $(X, Y) \sim \pi$, draw conditionally $R \sim K_\alpha(X, \cdot)$ and $S \sim K_\beta(Y, \cdot)$. The law γ of (R, S) has uniform marginals and satisfies $\pi = (q_\alpha, q_\beta)_\sharp \gamma$.

We now justify the approximation. Let $\kappa_M(r) = \max(-M, \min(r, M))$ and set $q_{\alpha,M} = \kappa_M \circ q_\alpha$ and $q_{\beta,M} = \kappa_M \circ q_\beta$. These functions are bounded and nondecreasing. Approximate them almost everywhere by nondecreasing step functions $q_{\alpha,M}^N$ and $q_{\beta,M}^N$ that are constant on the uniform intervals $I_k = ((k - 1)/N, k/N]$. The matrix $G_{k,\ell}^N = \gamma(I_k \times I_\ell)$ is a coupling of the two uniform histograms. Proposition 3.10, applied to the ordered step values, therefore gives

$$\int_0^1 h(q_{\alpha,M}^N(r) - q_{\beta,M}^N(r))\mathrm{d}r \leqslant \int_{(0,1)^2} h(q_{\alpha,M}^N(r) - q_{\beta,M}^N(s))\mathrm{d}\gamma(r, s).$$

For fixed M, the arguments of h remain in the compact interval $[-2M, 2M]$. Dominated convergence as $N \to \infty$ yields the same inequality for $q_{\alpha,M}$ and $q_{\beta,M}$.

Finally, κ_M is nondecreasing and 1-Lipschitz. Hence, for every $x, y \in \mathbb{R}$, there exists $t_M(x, y) \in [0, 1]$ such that $\kappa_M(x) - \kappa_M(y) = t_M(x, y)(x - y)$. Convexity and nonnegativity imply

$$h(t_M(x, y)(x - y)) \leqslant (1 - t_M(x, y))h(0) + t_M(x, y)h(x - y) \leqslant h(0) + h(x - y).$$

The assumed integrability controls the diagonal term. If the cost of π is finite, the same bound controls the right-hand side; if it is infinite, the desired comparison is immediate. Dominated convergence as $M \to \infty$ therefore gives

$$\int_0^1 h(q_\alpha(r) - q_\beta(r))\mathrm{d}r \leqslant \int h(x - y)\mathrm{d}\pi(x, y)$$

for every $\pi \in \mathcal{U}(\alpha, \beta)$. □

This result is strictly more flexible than the Monge formula. If α has an atom, a map can only send that whole atom to one target point, whereas the quantile interval associated with the atom can be coupled with a nontrivial portion of β. The one-dimensional Kantorovich solution therefore handles mass splitting without changing the monotone geometry.

3.4 Cone of Convex Functions and Monge Gap

For quadratic transport, Brenier's theorem 2.23 identifies optimal maps with gradients, or more generally subgradients, of convex functions. Convex functions form a cone under addition and nonnegative scaling, but imposing this global structure directly on a learned map can be difficult. The Monge gap turns transport optimality into a variational certificate: for suitable costs it is itself convex, on the line it becomes an isotonicity penalty, and in learning problems it can be added to an ordinary regression objective.

Definition and optimality certificate. Matching an output law does not identify an optimal map: many maps can have the same push-forward while incurring different transport costs. The Monge gap of Uscidda and Cuturi [519] separates distributional fitting from transport optimality by comparing the graph coupling generated by a map with the best coupling having exactly the same marginals.

Definition 3.40 (Monge gap)**.** Let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, let $T : \mathcal{X} \to \mathcal{Y}$ be measurable, and assume that the following costs are finite. The c-Monge gap of T relative to α is

$$\mathcal{M}_\alpha^c(T) := \int c(x, T(x))\mathrm{d}\alpha(x) - \mathcal{L}_c(\alpha, T_\sharp\alpha). \tag{3.8}$$

The second marginal is deliberately $T_\sharp\alpha$: the gap tests optimality of T, not agreement with an externally prescribed target.

Proposition 3.41 (Elementary properties of the Monge gap)**.** *Under the standing Polish-space assumptions, suppose that the relevant integrals are finite. Then $\mathcal{M}_\alpha^c(T) \geqslant 0$, with equality if and only if $(\mathrm{Id}, T)_\sharp\alpha$ is an optimal coupling between α and $T_\sharp\alpha$. Moreover,*

$$\mathcal{M}_\alpha^c(T) = \sup_{\eta\in\mathcal{U}(\alpha,\alpha)} \int \big[c(x, T(x)) - c(x, T(z))\big]\mathrm{d}\eta(x, z). \tag{3.9}$$

Finally, the gap is unchanged when $c(x, y)$ is replaced by $c(x, y) + r(x) + s(y)$, provided the modified cost remains admissible and all marginal terms are integrable.

Proof. The graph plan $(\mathrm{Id}, T)_\sharp\alpha$ is feasible for $\mathcal{L}_c(\alpha, T_\sharp\alpha)$, which proves nonnegativity and the equality criterion. Every $\eta \in \mathcal{U}(\alpha, \alpha)$ induces the coupling $(x, z) \mapsto (x, T(z))$. Conversely, disintegrating α along T lifts every coupling in $\mathcal{U}(\alpha, T_\sharp\alpha)$ to such a self-coupling. Subtracting the resulting infimum from the graph cost gives (3.9). Adding $r(x) + s(y)$ shifts both terms in (3.8) by $\int r\mathrm{d}\alpha + \int s\mathrm{d}(T_\sharp\alpha)$. □

Convex gaps and convex potentials. Formula (3.9) need not be convex in T: both occurrences of T lie inside a generic nonlinear cost. Convexity becomes automatic when cost differences are affine in the image variable.

Proposition 3.42 (Convex Monge gaps for affine-difference costs)**.** *Let $\mathcal{Y} \subset \mathbb{R}^m$ be convex and let the nonnegative cost c have the form*

$$c(x, y) = a(x) + b(y) - \langle A(x), y\rangle, \tag{3.10}$$

where $A, T \in L^2(\alpha; \mathbb{R}^m)$ and all displayed terms are finite. Then

$$\mathcal{M}_\alpha^c(T) = \sup_{\eta\in\mathcal{U}(\alpha,\alpha)} \int \langle A(x), T(z) - T(x)\rangle\mathrm{d}\eta(x, z). \tag{3.11}$$

Consequently, $T \mapsto \mathcal{M}_\alpha^c(T)$ is convex, and it is positively homogeneous when the admissible maps take values in $\mathbb{R}^m$. Moreover, $\mathcal{M}_\alpha^c(T) = 0$ if and only if there is a proper lower-semicontinuous convex function φ such that

$$T(x) \in \partial\varphi(A(x)) \qquad \text{for } \alpha\text{-almost every } x. \tag{3.12}$$

Proof. The marginal terms a and b cancel in (3.9), giving (3.11); this is a supremum of linear functionals of T. Next, disintegrating α along A lifts every coupling between $A_\sharp\alpha$ and $T_\sharp\alpha$. Hence zero gap is equivalent to optimality of $(A, T)_\sharp\alpha$ for the bilinear cost $(u, y) \mapsto -\langle u, y\rangle$. Up to marginal terms, this is the quadratic cost $\frac{1}{2}|u - y|^2$. By Rockafellar's characterization [447], recalled in Section 3.5, its optimal plans are exactly those concentrated on the subdifferential of a proper lower-semicontinuous convex function. □

The class (3.10) contains $c(x,y) = \frac{1}{2}|x-y|^2$, for which $A(x) = x$. It also contains the Bregman divergence of Definition 9.1 with x as base point,

$$B_\Phi(y|x) = \Phi(y) - \Phi(x) - \langle \nabla\Phi(x),\, y-x\rangle,$$

where Φ is differentiable and convex: here $a(x) = \langle \nabla\Phi(x),\, x\rangle - \Phi(x)$, $b(y) = \Phi(y)$ and $A(x) = \nabla\Phi(x)$. The argument order is essential: on a full-dimensional domain, the reversed divergence $B_\Phi(x|y)$ is generally not of the form (3.10) unless Φ is quadratic. Thus the zero-gap condition generalizes Brenier's characterization: in the quadratic case it reduces to $T(x) \in \partial\varphi(x)$ α-almost everywhere, and to $T = \nabla\varphi$ when α has a density.

This cost class is also exhaustive under mild smoothness. If $\mathcal{Y}$ is open and $c(x,\cdot) \in C^2(\mathcal{Y})$, then the gap is convex for every source α for which it is finite—and it is enough to test finitely supported sources—if and only if c has the form (3.10). Indeed, for $\alpha = \frac{1}{2}(\delta_{x_0} + \delta_{x_1})$, $u = T(x_0)$, $v = T(x_1)$ and $g(y) = c(x_0,y) - c(x_1,y)$, one has $\mathcal{M}^c_\alpha(T) = \frac{1}{2}(g(u) - g(v))_+$. Convexity in (u,v) forces $\nabla^2 g = 0$, so every such difference is affine.

One-dimensional order and isotonicity. On the line, the order structure gives a more explicit formula, even for convex displacement costs whose Monge gap is not itself convex.

Definition 3.43 (Increasing rearrangement relative to a source)**.** Let $\alpha \in \mathcal{M}^1_+(\mathbb{R})$ be atomless and let $T : \mathbb{R} \to \mathbb{R}$ be measurable. The increasing rearrangement of T relative to α is

$$T^{\uparrow,\alpha} := C^{-1}_{T_\sharp\alpha} \circ C_\alpha. \tag{3.13}$$

It is the unique α-almost-everywhere nondecreasing map satisfying $(T^{\uparrow,\alpha})_\sharp\alpha = T_\sharp\alpha$.

This rearrangement is not intrinsic to T alone: it depends on α through both the source rank C_α and the output law $T_\sharp\alpha$. If α has atoms, a deterministic increasing rearrangement need not exist because an atom may have to split; the quantile coupling of Theorem 3.39 is then the correct replacement.

Proposition 3.44 (One-dimensional Monge-gap formulas)**.** *Let $\alpha \in \mathcal{M}^1_+(\mathbb{R})$ be atomless, let $c(x,y) = h(x-y)$ with h convex, and assume the relevant terms are integrable. Then*

$$\mathcal{M}^c_\alpha(T) = \int \left[h(x - T(x)) - h(x - T^{\uparrow,\alpha}(x))\right] \mathrm{d}\alpha(x). \tag{3.14}$$

If h is strictly convex, this gap vanishes if and only if T has a nondecreasing representative α-almost everywhere.

For the equal-weight atomic measure $\alpha_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ on the uniform grid $x_i = x_1 + (i-1)\Delta$, let $t_i = T(x_i)$ and let $t_{(1)} \leqslant \cdots \leqslant t_{(n)}$ be their order statistics. For $c(x,y) = |x-y|^2$,

$$\mathcal{M}^{|\cdot|^2}_{\alpha_n}(T) = \frac{1}{n}\sum_{i=1}^n \left[(x_i - t_i)^2 - (x_i - t_{(i)})^2\right] = \frac{2\Delta}{n} \sum_{1\leqslant i<j\leqslant n} (t_i - t_j)_+. \tag{3.15}$$

Proof. The one-dimensional rearrangement result of Theorem 2.41, together with its coupling extension in Theorem 3.39, shows that $T^{\uparrow,\alpha}$ is optimal between α and $T_\sharp\alpha$. This gives (3.14). If h is strictly convex, atomlessness makes this optimal map unique α-almost everywhere, so zero gap is equivalent to $T = T^{\uparrow,\alpha}$ α-almost everywhere. For the empirical formula, sorting $(t_i)_i$ gives the increasing optimal assignment and hence the first equality. An adjacent exchange of an inversion $u > v$ decreases the average squared cost by $2\Delta(u-v)/n$; bubble sort exchanges every inverted pair once, which gives the second equality. □

For quadratic cost, formula (3.15) is a convex pairwise hinge penalty whose zero set is the isotonic cone $\{t_1 \leqslant \cdots \leqslant t_n\}$. It is not the usual least-squares isotonic projection: rearrangement preserves the empirical output law, whereas isotonic regression moves and pools values. In higher dimension, Proposition 3.42 replaces scalar order by cyclic monotonicity and convex subgradients. Monge-gap regularization therefore extends the role of isotonic constraints beyond a total order, while its defining comparison keeps $T_\sharp\alpha$ fixed.

Monge-gap-regularized kernel regression. The hinge representation above turns transport optimality into a tractable soft monotonicity prior. Let $x_i = x_1 + (i-1)\Delta$ and let $y_i \in \mathbb{R}$ be noisy observations. For a positive-semidefinite kernel k with reproducing-kernel Hilbert space $\mathcal{H}_k$, consider

$$\min_{T\in\mathcal{H}_k} \left\{ \frac{1}{n}\sum_{i=1}^{n} |T(x_i) - y_i|^2 + \lambda |T|^2_{\mathcal{H}_k} + \gamma \mathcal{M}^{|\cdot|^2}_{\alpha_n}(T) \right\}, \qquad \lambda > 0, \quad \gamma \geqslant 0, \tag{3.16}$$

where $\alpha_n = n^{-1}\sum_i \delta_{x_i}$. The usual orthogonal-projection argument shows that a minimizer has the representer form $T_{\mathrm{q}}(\cdot) = \sum_{j=1}^{n} \mathrm{q}_j k(x_j, \cdot)$. Writing $\mathrm{K}_{i,j} = k(x_i, x_j)$ and $\mathrm{y} = (y_i)_i$, formula (3.15) reduces (3.16) to

$$\min_{\mathrm{q}\in\mathbb{R}^n} \left\{ \frac{1}{n}|\mathrm{Kq} - \mathrm{y}|^2 + \lambda\langle \mathrm{q}, \mathrm{Kq}\rangle + \frac{2\gamma\Delta}{n} \sum_{1\leqslant i<j\leqslant n} \left[(\mathrm{Kq})_i - (\mathrm{Kq})_j\right]_+ \right\}. \tag{3.17}$$

This is a finite-dimensional convex program: the first two terms are convex quadratics because $\mathrm{K} \succeq 0$, and the last term is a sum of hinges of affine functionals. Thus every point along the regularization path can be computed globally, without alternating between regression and transport.

Figure 3.9 shows this path for a Gaussian kernel. The left experiment reproduces a smooth oscillatory example with $n = 55$ observations. The right experiment uses $n = 95$ noisier observations from a signal containing several spatial scales. Increasing γ suppresses order inversions rather than merely smoothing the fit; the black dashed curve is the solution of the hard constraint $T(x_1) \leqslant \cdots \leqslant T(x_n)$.

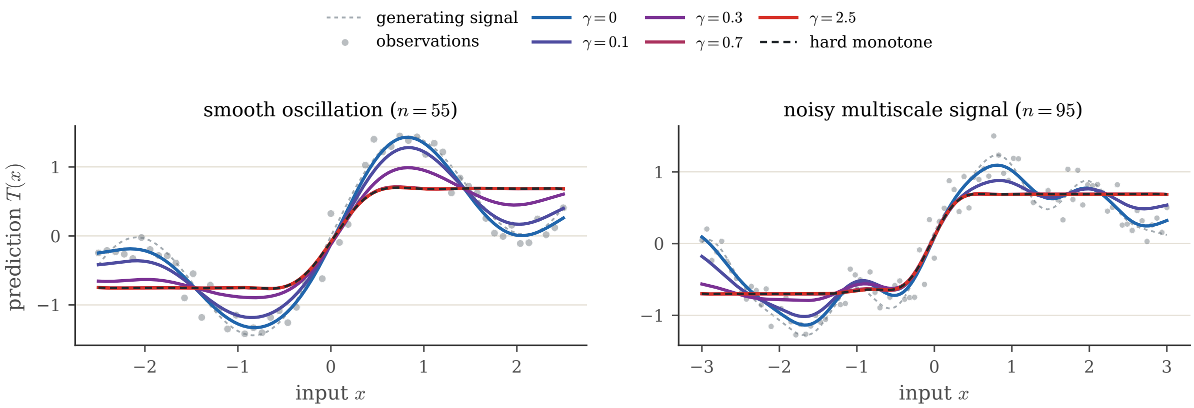


Figure 3.9: A quadratic Monge-gap penalty traces a convex kernel-regression path toward monotonicity. Gray dots are the observations and the light dashed curve is the generating signal. Colored curves solve (3.16) for increasing γ, from ordinary kernel ridge regression in blue to an almost monotone fit in red; the black dashed curve solves the corresponding hard monotonicity-constrained problem. The second experiment uses more points, stronger noise and a more oscillatory signal.

3.5 c-Cyclical Monotonicity

Cyclical monotonicity is the local geometric fingerprint of optimality. It converts a global minimization problem into finite exchange inequalities and is the bridge from Kantorovich plans to convex potentials.

Optimal transport plans behave well when one looks at any finite sub-collection of points in their support: the restriction is still an optimal matching for those points alone. For finitely supported marginals this is immediate, and it leads to the notion of c-cyclical monotonicity.

Support and c-cyclical monotonicity. The support of a coupling is the topological support introduced in Definition 2.4, now applied to a Radon measure on $\mathcal{X} \times \mathcal{Y}$. Thus $(x, y) \in \operatorname{supp}(\pi)$ exactly when every open neighborhood of (x, y) has positive π-mass.

Definition 3.45 (c-cyclical monotonicity)**.** A set $\Gamma \subset \mathcal{X} \times \mathcal{Y}$ is c-cyclically monotone if, for every $k \geqslant 2$, every finite family $(x_i, y_i)_{i=1}^k \subset \Gamma$ and every permutation σ of $\{1, \dots, k\}$,

$$\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

Any permutation is a product of cycles, so it suffices to verify the inequality for cyclic permutations,

$$\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{i+1}), \qquad y_{k+1} = y_1.$$

Optimal matching to optimal transport. Let the marginals be uniform on n points, $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n}\sum_{i=1}^n \delta_{y_i}$. By Corollary 3.18, there exists an optimal plan induced by a permutation. Its support $\Gamma = \{(x_i, y_{\sigma(i)})\}_i$ is c-cyclically monotone: otherwise exchanging the finitely many targets along a violating cycle would lower the matching cost. The following theorem, in the spirit of Rockafellar's cyclic-monotonicity theorem [447], says that the same finite-exchange condition holds for any optimal coupling, not only for finite uniform matchings.

Theorem 3.46 (Optimal plans are c-cyclically monotone)**.** *Assume that $c : \mathcal{X} \times \mathcal{Y} \to [0, +\infty)$ is continuous and that the Kantorovich value is finite. For any optimal plan π solving (3.7), $\mathrm{supp}(\pi)$ is c-cyclically monotone.*

Proof. We prove the contrapositive. Suppose that $\mathrm{supp}(\pi)$ is not c-cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation σ such that $\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)})$. By continuity of c, after shrinking neighborhoods $U_i \ni x_i$ and $V_i \ni y_i$, the same strict inequality holds uniformly for every choice of points in these neighborhoods:

$$\sum_i c(u_i, v_i) > \sum_i c(u_i, \tilde{v}_{\sigma(i)}) \qquad (u_i \in U_i,\ v_i \in V_i,\ \tilde{v}_{\sigma(i)} \in V_{\sigma(i)}).$$

Choose the sets so that $m_i := \pi(U_i \times V_i) > 0$. Taking $0 < \lambda \leqslant \left(\sum_{i=1}^k \frac{1}{m_i}\right)^{-1}$ ensures that the scaled restrictions $\pi_i = \lambda \frac{\pi|_{U_i \times V_i}}{m_i}$ have common mass λ and satisfy $\sum_i \pi_i \leqslant \pi$, even if some rectangles overlap. Let $\alpha_i = (P_{\mathcal{X}})_\sharp \pi_i$ and $\beta_i = (P_{\mathcal{Y}})_\sharp \pi_i$, and define $\tilde{\pi} = \pi - \sum_i \pi_i + \sum_i (\alpha_i \otimes \beta_{\sigma(i)})/\lambda$. The removed and reinserted first marginals are both $\sum_i \alpha_i$, and the removed and reinserted second marginals are both $\sum_i \beta_i$ because σ is a permutation. Hence $\tilde{\pi} \in \mathcal{U}(\alpha, \beta)$. Integrating the uniform strict inequality against the product probability $\otimes_i (\pi_i/\lambda)$ shows that the reinserted crossed terms have strictly smaller cost than the removed diagonal terms. This contradicts the optimality of π. □

Monotonicity. Assume the optimal plan is induced by a measurable map $T : \mathcal{X} \to \mathcal{Y}$, meaning that $\pi = (\mathrm{Id}, T)_\sharp \alpha$. There is a set $G \subset \mathcal{X}$ of full α-measure such that $(x, T(x)) \in \mathrm{supp}(\pi)$ for every $x \in G$. For any $x_1, \dots, x_k \in G$, cyclical monotonicity reads

$$\sum_{i=1}^k c(x_i, T(x_i)) \leqslant \sum_{i=1}^k c(x_i, T(x_{i+1})), \qquad x_{k+1} = x_1.$$

For $c(x, y) = \frac{1}{2}|x - y|^2$, taking $k = 2$ gives $\langle T(x) - T(y), x - y\rangle \geqslant 0$ for every $x, y \in G$, so the optimal representative of T is monotone on G. Brenier's theorem adds that, when α is absolutely continuous, $T = \nabla\varphi$ almost everywhere for a convex potential φ. The converse fails in dimension $d \geqslant 2$: as shown by the rotation example in the Monge section, a small rotation is monotone yet not a gradient.

One dimension. In one space dimension, with cost $c(x, y) = |x - y|^p$, the two-point inequality becomes

$$|x - T(x)|^p + |y - T(y)|^p \leqslant |x - T(y)|^p + |y - T(x)|^p,$$

For $p > 1$, strict convexity implies that a reversed pair $x < y$ and $T(x) > T(y)$ violates this inequality. Hence every optimal map is nondecreasing on the full-measure transport set, recovering the classical monotone rearrangement. For $p = 1$, uncrossing is non-strict: the monotone rearrangement remains optimal, but other optimal maps and non-deterministic plans can occur, as in the book-shifting example of Remark 3.30.

The coupling formalism now provides a well-posed convex relaxation together with structural certificates for its optimizers. The next chapter specializes the cost to powers of a metric and turns these optimal values into a geometry on probability measures.

CHAPTER 14

Dynamic Optimal Transport

Optimal transport becomes especially powerful once distances between measures are seen as actions of moving mass. This chapter first develops the dynamic language: continuity equations describe admissible measure evolutions, while the Benamou–Brenier formula identifies $\mathcal{W}_2$ with a least-action principle. The path-space Schrödinger problem then provides its stochastic, entropy-regularized counterpart. After extending dynamic actions to other transport geometries, the chapter closes with variational mean field games, where congestion and a terminal cost turn the Benamou–Brenier action into a population-planning problem. These ideas prepare the gradient-flow and generative-model chapters that follow.

14.1 Evolutions over the Space of Measures

We start with the continuity equation because it is the common language for particles, densities and weak measure evolutions. It also makes precise which velocity fields actually move mass.

Lagrangian and Eulerian descriptions. We consider the evolution $t \mapsto \alpha_t \in \mathcal{P}(\mathbb{R}^d)$. Such an evolution can be described in a "Lagrangian" way as the advection of particles along a (time-dependent) vector field $v_t(x)$ in $\mathbb{R}^d$. At the particle level, this advection is governed by

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \tag{14.1}$$

and we write T_t for the associated flow map, so that $T_t(x(0)) = x(t)$. The advected measure is then $\alpha_t = (T_t)_\sharp \alpha_0$. For discrete measures, $\alpha_t = \frac{1}{n}\sum_{i=1}^n \delta_{x_i(t)}$, meaning each $x_i(t)$ solves (14.1).

In the Eulerian description, the same motion is written directly on the evolving measure: the particle ODE becomes the PDE

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(v_t \alpha_t) = 0. \tag{14.2}$$

This PDE is often referred to as the advection equation, the continuity equation, or Liouville's equation when operating over a phase space. It is only a classical PDE when α_t has a smooth density. For general measures, and in particular for empirical measures, it is understood in the distributional sense: for every $\varphi \in C_c^1((0,1) \times \mathbb{R}^d)$,

$$\int_0^1 \int_{\mathbb{R}^d} \big(\partial_t \varphi(t,x) + \langle v_t(x), \nabla_x \varphi(t,x)\rangle\big)\, \mathrm{d}\alpha_t(x)\mathrm{d}t = 0. \tag{14.3}$$

This equation is obtained from (14.2) by integration by parts. Hence, for smooth positive densities, the classical and weak formulations are equivalent; the weak formulation is useful because it still makes sense for discrete measures whose particles evolve according to (14.1).

This weak identity provides the measure-theoretic notion of an admissible evolution.

Definition 14.1 (Admissible continuity-equation evolution)**.** A narrowly continuous curve $(\alpha_t)_{t\in[0,1]}$ and a Borel velocity field v are admissible between α_0 and α_1 if

$$\int_0^1 \int |v_t(x)|^2 \mathrm{d}\alpha_t(x)\mathrm{d}t < +\infty$$

and

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0$$

in distributions, with the prescribed endpoints and no flux through the boundary when the domain is bounded.

Proposition 14.2 (Lagrangian flows solve the continuity equation)**.** *Let $(T_t)_{t\in[0,1]}$ be a C^1 family of diffeomorphisms of $\mathbb{R}^d$ and define $\alpha_t = (T_t)_\sharp\alpha_0$. Assume that the derivatives below are integrable with respect to α_0, and define the Eulerian velocity field by*

$$v_t(T_t(y)) = \partial_t T_t(y).$$

Then (α_t, v_t) solves the continuity equation in the weak sense (14.3). In particular, if $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i(0)}$ is empirical, then $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$ is empirical as well, with particle velocities $\dot{x}_i(t) = v_t(x_i(t))$.

Proof. Let $\varphi \in C_c^1((0,1)\times\mathbb{R}^d)$. Since $\alpha_t = (T_t)_\sharp\alpha_0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,x)\mathrm{d}\alpha_t(x) = \frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,T_t(y))\mathrm{d}\alpha_0(y).$$

The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,T_t(y))\mathrm{d}\alpha_0(y) = \int \big(\partial_t\varphi(t,T_t(y)) + \langle\nabla_x\varphi(t,T_t(y)),\, \partial_t T_t(y)\rangle\big)\,\mathrm{d}\alpha_0(y).$$

Using the definition of v_t and the push-forward relation, this equals

$$\int \big(\partial_t\varphi(t,x) + \langle\nabla_x\varphi(t,x),\, v_t(x)\rangle\big)\,\mathrm{d}\alpha_t(x).$$

Integrating in time and using the boundary values of φ gives (14.3). □

From measure evolutions to vector fields. For a given evolution $(\alpha_t)_t$, there are typically infinitely many velocity fields v_t satisfying

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{14.4}$$

This non-uniqueness comes from the kernel of the weighted divergence. The linear space of vector fields that leave a measure α invariant is

$$\mathcal{H}_\alpha = \left\{v \in L^2(\alpha;\mathbb{R}^d)\ ;\ \mathrm{div}(\alpha v) = 0 \text{ in distributions}\right\}.$$

It is usually non-trivial: for instance, if α is an isotropic Gaussian, $\mathcal{H}_\alpha$ contains rotational vector fields generated by anti-symmetric matrices.

Dacorogna–Moser inversion. Reconstructing particles from an observed density evolution is therefore ill-posed. For a smooth positive density $\alpha_t = \rho_t\,\mathrm{d}x$, a simple choice, introduced by Dacorogna and Moser [174], is to impose that the flux $\rho_t v_t$ is a gradient field. With a fixed convention for the inverse Laplacian, this gives formally

$$v_t = -\frac{1}{\rho_t}\nabla\Delta^{-1}(\partial_t\rho_t), \tag{14.5}$$

with suitable boundary conditions, for instance vanishing at infinity. This formula is useful conceptually but delicate when ρ_t vanishes, and it does not generally produce a gradient velocity field.

The classical Dacorogna–Moser construction uses the linear density path. If $\alpha_i = \rho_i\,\mathrm{d}x$ are smooth positive densities with the same total mass on a bounded domain Ω, set

$$\alpha_t = (1-t)\alpha_0 + t\alpha_1 = \rho_t\,\mathrm{d}x, \qquad \rho_t = (1-t)\rho_0 + t\rho_1.$$

Choose a time-independent flux w satisfying

$$\mathrm{div}\, w = \rho_0 - \rho_1, \qquad w\cdot n = 0 \quad \text{on } \partial\Omega,$$

for instance $w = -\nabla\varphi$ with $\Delta\varphi = \rho_1 - \rho_0$ and Neumann boundary condition. Then

$$v_t = \frac{w}{\rho_t}$$

satisfies $\partial_t\rho_t + \mathrm{div}(\rho_t v_t) = 0$. The flow $\partial_t T_t = v_t \circ T_t$, $T_0 = \mathrm{Id}$, therefore transports $\rho_0\mathrm{d}x$ onto $\rho_t\mathrm{d}x$, and T_1 solves the prescribed-Jacobian problem $\rho_1(T_1(x))\det(\nabla T_1(x)) = \rho_0(x)$.

Least-square inversion and gradient structure. A more robust choice, used implicitly in flow matching, optimal transport and Wasserstein gradient flows, is to select among all admissible velocities the one with smallest kinetic energy:

$$\min_v \frac{1}{2}\int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \mathrm{d}\alpha_t(x)\mathrm{d}t \quad \text{subject to} \quad \partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{14.6}$$

Proposition 14.3 (Least-square velocities are gradients)**.** *Assume that $\alpha_t = \rho_t\,\mathrm{d}x$ is a smooth positive density curve, that $\partial_t\rho_t$ has zero integral, and that boundary terms vanish. The minimizer of (14.6), if it exists, is a gradient field*

$$v_t = \nabla\varphi_t,$$

where φ_t, unique up to an additive constant on each connected component, solves the weighted Poisson equation

$$-\mathrm{div}(\rho_t\nabla\varphi_t) = \partial_t\rho_t, \qquad v_t = -\nabla\Delta_{\alpha_t}^{-1}(\partial_t\alpha_t), \quad \Delta_{\alpha_t}\varphi = \mathrm{div}(\alpha_t\nabla\varphi). \tag{14.7}$$

Proof. Introduce a Lagrange multiplier φ_t for the continuity equation. The constrained problem has the formal saddle formulation

$$\min_v \max_\varphi \int_0^1 \left[\frac{1}{2}\int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \mathrm{d}\alpha_t(x) + \int_{\mathbb{R}^d} \varphi_t(x)\big(\mathrm{div}(\alpha_t v_t)(x) + \partial_t\alpha_t(x)\big)\mathrm{d}x\right] \mathrm{d}t.$$

Integrating by parts in the divergence term gives, for each t,

$$\int \left(\frac{1}{2}\|v_t\|^2 - \langle\nabla\varphi_t, v_t\rangle\right) \mathrm{d}\alpha_t + \int \varphi_t\,\partial_t\alpha_t.$$

The pointwise minimizer in v_t is therefore $v_t = \nabla\varphi_t$. Substituting this into the constraint $\partial_t\rho_t + \mathrm{div}(\rho_t v_t) = 0$ gives the weighted Poisson equation in (14.7). The inverse notation is just a shorthand for solving this equation on zero-mean right-hand sides, modulo additive constants. □

In general this inversion is still computationally demanding, but special choices of $(\alpha_t)_t$ lead to simpler formulas; this is the mechanism exploited later by flow matching in Section 16.1.

14.2 Benamou–Brenier Dynamic Formulation of OT

The dynamic formulation identifies $\mathcal{W}_2$ with the kinetic energy of the cheapest continuity-equation path. It is the point where OT becomes a least-action principle.

Benamou–Brenier formulation. Instead of assuming that a whole curve $(\alpha_t)_{t\in[0,1]}$ is prescribed, one only fixes its endpoints α_0 and α_1 and minimizes the least-square energy (14.6). The theorem of Benamou and Brenier states that this geodesic energy is exactly the squared Wasserstein distance [35].

The kinetic energy of an admissible evolution is the basic dynamic action.

Definition 14.4 (Benamou–Brenier action)**.** For an admissible pair (α_t, v_t), define

$$\mathrm{Act}_2(\alpha, v) := \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \mathrm{d}\alpha_t(x)\mathrm{d}t.$$

Theorem 14.5 (Benamou–Brenier)**.** *For probability measures $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{(\alpha_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \mathrm{d}\alpha_t(x)\mathrm{d}t, \tag{14.8}$$

where the infimum is over (α_t, v_t) solving $\partial_t\alpha_t + \nabla\cdot(\alpha_t v_t) = 0$ with $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. If α_0 has a density and T is the optimal Monge map $T_\sharp\alpha_0 = \alpha_1$, a minimizing curve is

$$\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp\alpha_0, \qquad v_t\big((1-t)x + tT(x)\big) = T(x) - x \quad \textit{for } \alpha_0\textit{-a.e. } x \textit{ and a.e. } t \in (0,1). \tag{14.9}$$

Proof. For the inequality "dynamic ⩽ static", assume first that a Monge map T exists and define (α_t, v_t) by (14.9). Since the Lagrangian velocity $T(x) - x$ is independent of t,

$$\int_0^1 \int |v_t|^2 \, \mathrm{d}\alpha_t \mathrm{d}t = \int |T(x) - x|^2 \, \mathrm{d}\alpha_0(x),$$

so the dynamic cost is no larger than the static Monge cost. Without a Monge map, the same construction is made with an optimal coupling π: sample $(X, Y) \sim \pi$ and move along the straight path $\gamma_{X,Y}(t) = (1-t)X + tY$. This path measure has action $\int |x - y|^2 \mathrm{d}\pi(x, y)$; projecting path velocities onto their conditional mean at time t gives an admissible Eulerian velocity with no larger action, so the dynamic value is no larger than the Kantorovich value.

Conversely, for a smooth deterministic path, take the flow T_t defined by $\dot{T}_t = v_t \circ T_t$ and $T_0 = \mathrm{Id}$. Then $\alpha_t = (T_t)_\sharp \alpha_0$ and $(T_1)_\sharp \alpha_0 = \alpha_1$. Jensen's inequality gives

$$|T_1(x) - x|^2 \leqslant \int_0^1 |v_t(T_t(x))|^2 \mathrm{d}t.$$

After integration with respect to α_0, the Monge cost is bounded above by the dynamic action. For general finite-energy solutions of the continuity equation, the superposition principle lifts the curve to a probability measure on absolutely continuous paths; applying Jensen's inequality pathwise gives a coupling of the endpoints whose quadratic cost is no larger than the action. Thus the Kantorovich value is bounded above by the dynamic value. □

Convex moment-based reformulation. Although (14.8) is not jointly convex in (α_t, v_t), it becomes convex after replacing velocities by momenta. Given a velocity $v \in L^2(\alpha; \mathbb{R}^d)$, define the momentum

$$\omega := \alpha v, \qquad \omega(A) = \int_A v(x) \, \mathrm{d}\alpha(x),$$

which is a finite $\mathbb{R}^d$-valued measure. The nonlinear relation $\omega = \alpha v$ is eliminated by the following measure perspective.

Definition 14.6 (Quadratic perspective action)**.**

$$J(a, m) := \begin{cases} |m|^2/a, & a > 0, \\ 0, & a = 0 \text{ and } m = 0, \\ +\infty, & a = 0 \text{ and } m \neq 0, \end{cases} \qquad (a, m) \in [0, +\infty) \times \mathbb{R}^d. \tag{14.10}$$

If λ dominates α and $|\omega|$, define

$$\mathbb{J}(\alpha, \omega) := \int J\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}(x), \frac{\mathrm{d}\omega}{\mathrm{d}\lambda}(x)\right) \mathrm{d}\lambda(x). \tag{14.11}$$

The pointwise function is lower-semicontinuous, convex and positively 1-homogeneous: $J(\eta a, \eta m) = \eta J(a, m)$ for $\eta \geqslant 0$. The definition is independent of the dominating measure λ: after replacing λ by another dominating measure, both Radon–Nikodym densities are rescaled by the same factor, which cancels with the change of reference measure by the 1-homogeneity of J. Equivalently, this is the standard integral functional associated with a convex normal integrand, used in the measure-valued relaxation of dynamic OT [12]; see also Rockafellar's perspective construction [447]. In particular,

$$\mathbb{J}(\alpha, \omega) < +\infty \quad \Longleftrightarrow \quad \omega = v\alpha \text{ with } v \in L^2(\alpha; \mathbb{R}^d), \qquad \mathbb{J}(\alpha, \omega) = \int |v|^2 \, \mathrm{d}\alpha.$$

The Benamou–Brenier problem can now be written with the linear continuity equation in the pair (α_t, ω_t):

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \mathrm{div}\, \omega_t = 0 \\ \alpha_{t=0} = \alpha_0,\ \alpha_{t=1} = \alpha_1}} \int_0^1 \mathbb{J}(\alpha_t, \omega_t) \, \mathrm{d}t. \tag{14.12}$$

In the absolutely continuous case $\alpha_t = \rho_t\,\mathrm{d}x$ and $\omega_t = m_t\,\mathrm{d}x$, this becomes

$$\inf_{\substack{\partial_t\rho_t+\operatorname{div} m_t=0\\ \rho_{t=0}\mathrm{d}x=\alpha_0,\ \rho_{t=1}\mathrm{d}x=\alpha_1}} \int_0^1\int_{\mathbb{R}^d} J(\rho_t(x), m_t(x))\,\mathrm{d}x\,\mathrm{d}t = \inf_{\substack{\partial_t\rho_t+\operatorname{div} m_t=0\\ \rho_{t=0}\mathrm{d}x=\alpha_0,\ \rho_{t=1}\mathrm{d}x=\alpha_1}} \int_0^1\int_{\mathbb{R}^d} \frac{|m_t(x)|^2}{\rho_t(x)}\,\mathrm{d}x\,\mathrm{d}t,$$

with the conventions encoded in (14.10). This convex reformulation enables geodesic interpolation by convex optimization once the domain is discretized.

Dual Hamilton–Jacobi formulation. The momentum formulation also has a useful dual. It turns the least-action problem into a Hamilton–Jacobi subsolution inequality for a scalar potential, with equality on the part of space-time actually visited by the optimal curve. This is the dual dynamic formulation already present in the original Benamou–Brenier perspective [35]; see also the standard accounts [530, 463]. The constants below follow the convention of (14.12), where the kinetic energy has no factor $1/2$.

Proposition 14.7 (Dual Benamou–Brenier problem)**.** *Assume, for simplicity, that the densities are smooth, compactly supported, and that boundary terms vanish. Then the convex dynamic value* (14.12) *has the dual formulation*

$$\mathcal{W}_2^2(\alpha_0,\alpha_1) = \sup_{\varphi}\left\{\int_{\mathbb{R}^d}\varphi_1\,\mathrm{d}\alpha_1 - \int_{\mathbb{R}^d}\varphi_0\,\mathrm{d}\alpha_0\ ;\ \partial_t\varphi_t + \frac{1}{4}|\nabla\varphi_t|^2 \leqslant 0\right\}, \tag{14.13}$$

where the supremum is over smooth test potentials $\varphi : [0,1]\times\mathbb{R}^d \to \mathbb{R}$. In the general metric-measure statement, the same formula is understood after relaxation: φ is a Hamilton–Jacobi subsolution, for instance in the viscosity sense, and finite-dimensional discretizations follow directly from Fenchel–Rockafellar duality.

If (ρ, m) and φ are smooth primal and dual optimizers, then

$$m_t = \frac{\rho_t}{2}\nabla\varphi_t, \qquad \partial_t\varphi_t + \frac{1}{4}|\nabla\varphi_t|^2 = 0 \quad \text{on } \{\rho_t > 0\}. \tag{14.14}$$

Equivalently, the optimal Eulerian velocity is $v_t = m_t/\rho_t = \nabla\varphi_t/2$.

Proof. Let (ρ, m) satisfy the continuity equation and let φ be smooth. Multiplying the constraint by φ and integrating by parts gives

$$\int_{\mathbb{R}^d}\varphi_1\,\mathrm{d}\alpha_1 - \int_{\mathbb{R}^d}\varphi_0\,\mathrm{d}\alpha_0 = \int_0^1\int_{\mathbb{R}^d}\left(\rho_t\,\partial_t\varphi_t + \langle m_t, \nabla\varphi_t\rangle\right)\mathrm{d}x\mathrm{d}t.$$

If $\partial_t\varphi_t + |\nabla\varphi_t|^2/4 \leqslant 0$, Young's inequality yields, pointwise,

$$\rho\,\partial_t\varphi + \langle m, \nabla\varphi\rangle \leqslant -\frac{\rho}{4}|\nabla\varphi|^2 + \langle m, \nabla\varphi\rangle \leqslant J(\rho, m),$$

with the perspective convention (14.10). Hence the dual objective of every feasible φ is no larger than the action of every feasible primal pair.

Conversely, introduce φ as a Lagrange multiplier for $\partial_t\rho + \operatorname{div} m = 0$. After integration by parts, and up to the fixed endpoint contribution in (14.13), the pointwise minimization over (ρ, m) contains

$$J(\rho, m) - \rho\,\partial_t\varphi - \langle m, \nabla\varphi\rangle.$$

For fixed $\rho > 0$, its minimum over m is attained at $m = \rho\nabla\varphi/2$ and equals $-\rho(\partial_t\varphi + |\nabla\varphi|^2/4)$. The recession cases in (14.10) give the same conclusion when $\rho = 0$. Thus the infimum over $\rho \geqslant 0$ is finite exactly when the Hamilton–Jacobi inequality holds, in which case the pointwise infimum is 0. Fenchel–Rockafellar duality then gives no duality gap. Equality in the two pointwise inequalities gives (14.14). □

This also recovers the static Kantorovich inequality from a dynamic principle. If γ is any smooth curve with $\gamma(0) = x$ and $\gamma(1) = y$, then

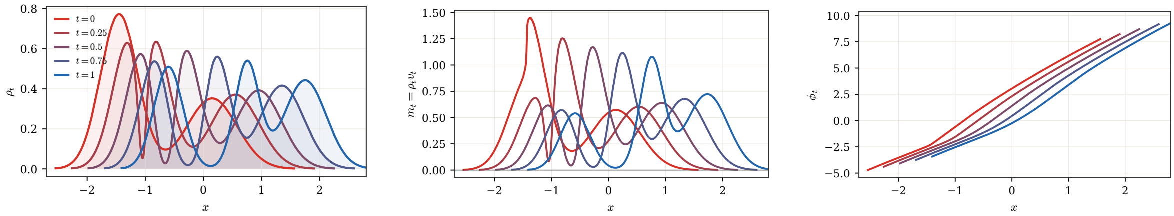
$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\gamma(t)) = \partial_t\varphi_t(\gamma(t)) + \langle\nabla\varphi_t(\gamma(t)), \dot\gamma(t)\rangle \leqslant |\dot\gamma(t)|^2,$$

where the last inequality uses $\partial_t\varphi + |\nabla\varphi|^2/4 \leqslant 0$ and Young's inequality. Integrating and minimizing over curves gives

$$\varphi_1(y) - \varphi_0(x) \leqslant |x - y|^2.$$

Thus $(-\varphi_0, \varphi_1)$ is a feasible static Kantorovich dual pair for the quadratic cost. At optimality the inequality is saturated on the endpoint pairs connected by the primal characteristics, which is the Eulerian version of complementary slackness.

Figure 14.1 displays these primal–dual relations for a one-dimensional mixture transport, including the Hamilton–Jacobi contact identity along the active mass.



density ρ_t momentum density m_t dual potential φ_t

Figure 14.1: One-dimensional Benamou–Brenier primal and dual solutions. The endpoints are Gaussian mixtures and the solution is computed from monotone quantile interpolation. The left and middle panels show the primal density and the momentum density $m_t = \rho_t v_t$. The right panel shows the dual Hamilton–Jacobi potential, normalized up to an additive constant. Along the active transported mass, the notebook checks the primal–dual identities $m_t = \rho_t\partial_x\varphi_t/2$ and $\partial_t\varphi_t + |\partial_x\varphi_t|^2/4 = 0$.

Proximal splitting. The convex momentum formulation also explains the original Benamou–Brenier solver. Papadakis, Peyré and Oudet [412] showed that, after discretization, the ALG2 scheme is a Douglas–Rachford splitting, equivalently ADMM on the Fenchel–Rockafellar dual. Suppressing discretization indices, write $U = (\rho, m)$, where m is the density of the momentum measure. Using the perspective (14.10), set

$$\mathcal{F}(U) = \int_0^1\int_{\mathbb{R}^d} J(\rho_t(x), m_t(x))\,\mathrm{d}x\,\mathrm{d}t, \qquad C = \left\{(\rho, m)\ ;\ \partial_t\rho + \operatorname{div} m = 0,\ \rho_0\mathrm{d}x = \alpha_0,\ \rho_1\mathrm{d}x = \alpha_1\right\},$$

and let $\mathcal{G} = \iota_C$ be the indicator of this affine continuity constraint. The convex Benamou–Brenier problem is therefore

$$\min_U\ \mathcal{F}(U) + \mathcal{G}(U).$$

On the resulting Hilbert space of unknowns, or formally for an L^2-type product structure, the two proximal operators are

$$\operatorname{prox}_{\tau\mathcal{F}}(\bar{U}) = \operatorname{argmin}_U \frac{1}{2}\|U - \bar{U}\|_{\mathrm{H}}^2 + \tau\mathcal{F}(U), \qquad \operatorname{prox}_{\tau\mathcal{G}}(\bar{U}) = \operatorname{argmin}_{U\in C} \frac{1}{2}\|U - \bar{U}\|_{\mathrm{H}}^2.$$

For the standard product metric, the first map is local in (t, x): it is the proximal operator of the convex perspective J. The second map is the orthogonal projection onto the affine set defined by the divergence equation and endpoint constraints. Douglas–Rachford then alternates these two simple operations:

$$\begin{aligned}
U^{(\ell+1)} &= \operatorname{prox}_{\tau\mathcal{F}}(Z^{(\ell)}),\\
\widetilde{U}^{(\ell+1)} &= \operatorname{prox}_{\tau\mathcal{G}}(2U^{(\ell+1)} - Z^{(\ell)}),\\
Z^{(\ell+1)} &= Z^{(\ell)} + \widetilde{U}^{(\ell+1)} - U^{(\ell+1)}.
\end{aligned}$$

Equivalently one may swap the roles of $\mathcal{F}$ and $\mathcal{G}$. At convergence, the two shadow sequences $U^{(\ell)}$ and $\widetilde{U}^{(\ell)}$ agree and give a minimizer of the convex dynamic problem. This viewpoint is useful because it separates the nonlinear but pointwise perspective proximal step from the global but linear projection onto the continuity equation [412, 203].

Figure 14.2 complements the Eulerian optimization viewpoint with the Lagrangian picture: matched particles travel along the straight characteristics of the minimizing curve.

Algorithm 14.1: Douglas–Rachford for dynamic Benamou–Brenier

Input: Functionals $\mathcal{F}, \mathcal{G} = \iota_C$, proximal parameter $\tau > 0$, initial field $Z^{(0)}$, tolerance tol > 0, iteration budget $L \geqslant 1$.
Output: Discrete density-momentum field $U^\star$.
For $\ell = 0, \dots, L-1$ **do**:
  $U^{(\ell+1)} = \operatorname{prox}_{\tau\mathcal{F}}(Z^{(\ell)})$.
  Project reflected point: $\widetilde{U}^{(\ell+1)} = \operatorname{prox}_{\tau\mathcal{G}}(2U^{(\ell+1)} - Z^{(\ell)}) = \operatorname{Proj}_C(2U^{(\ell+1)} - Z^{(\ell)})$.
  Update $Z^{(\ell+1)} = Z^{(\ell)} + \widetilde{U}^{(\ell+1)} - U^{(\ell+1)}$.
  If $\|U^{(\ell+1)} - \widetilde{U}^{(\ell+1)}\| \leqslant \text{tol}$ **then**:
    Return $U^{(\ell+1)}$.
Return $U^{(L)}$.

density sequence midpoint velocities

Figure 14.2: Benamou–Brenier geodesic between two sampled silhouettes. A discrete quadratic OT plan between finely subsampled cat and two-disks point clouds induces the McCann interpolation $Z_t = (1-t)X + tY$, which is the Lagrangian realization of the least-action solution. The left panel renders local color images of the smaller-bandwidth kernel-smoothed densities with enough padding to include the full silhouettes. The right panel overlays shortened velocity arrows centered at evenly subsampled midpoint particles $Z_{1/2}$; each displayed arrow runs in data coordinates from a source-side tail to a target-side head along the matched characteristic direction $Y - X$, but is not drawn as the full endpoint segment from X to Y.

Path-space formulation. Let $\mathcal{S} = C([0,1];\mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0,1]$ define the evaluation map

$$e_t : \mathcal{S} \to \mathbb{R}^d, \qquad e_t(\gamma) = \gamma(t).$$

The Benamou–Brenier cost admits the equivalent formulation

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M \in \mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}} \int_0^1 |\dot{\gamma}(t)|^2 \mathrm{d}t \, \mathrm{d}M(\gamma) \; ; \; (e_0)_\sharp M = \alpha_0, \; (e_1)_\sharp M = \alpha_1 \right\}.$$

The inner kinetic energy is understood as $+\infty$ when γ is not absolutely continuous. If α_0 has a density, the minimizer M^* is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (e_t)_\sharp M^*$ for all t. Furthermore, for a.e. t, denoting $\dot{e}_t(\gamma) = \dot{\gamma}(t)$ on absolutely continuous paths, the conditional law of the velocity is deterministic:

$$(e_t, \dot{e}_t)_\sharp M^*(\mathrm{d}x, \mathrm{d}q) = \alpha_t(\mathrm{d}x)\delta_{v_t^*(x)}(\mathrm{d}q),$$

where v_t^* is the optimal velocity field in the Benamou–Brenier formulation. Hence M^* concentrates on straight-line geodesics and, for a.e. t, assigns exactly one direction at α_t-a.e. spatial point.

14.3 Path-Space Schrödinger Problem

The path-space formulation at the end of Section 14.2 recasts unregularized quadratic transport as a least-action problem over laws of trajectories. Schrödinger's reciprocal problem is its stochastic counterpart: a noisy reference dynamics replaces deterministic least-action paths. After optimizing out the conditional law of a trajectory given its endpoints, only the endpoint coupling remains, and the static entropic problem (8.13) reappears. This section therefore links the Benamou–Brenier dynamics developed above with the Sinkhorn geometry of Chapter 8.

From least-action paths to random bridges. Throughout this section, both endpoint measures live on the same state space $\mathcal{X}$; this is the setting relevant to dynamic transport and Brownian bridges, whereas the static coupling problem (8.13) can also compare measures on different spaces. We first generalize the preceding Euclidean construction by allowing an arbitrary action on paths. The corresponding path-space formulation optimizes directly over laws of trajectories.

Definition 14.8 (Unregularized path-space transport)**.** Let $\Omega = C([0,1];\mathcal{X})$, let $e_t(\omega) = \omega_t$, and let $\mathcal{A} : \Omega \to [0,+\infty]$ be a path action. Define

$$\mathrm{PT}_{\mathcal{A}}(\alpha,\beta) := \inf_{M\in\mathcal{P}(\Omega)} \left\{ \int_\Omega \mathcal{A}(\omega)\,\mathrm{d}M(\omega)\;;\;(e_0)_\sharp M = \alpha,\ (e_1)_\sharp M = \beta \right\}. \tag{14.15}$$

A feasible M is a law of random trajectories whose initial and final laws are α and β. For the quadratic Wasserstein geometry on $\mathbb{R}^d$, one takes

$$\mathcal{A}(\omega) = \begin{cases} \displaystyle\int_0^1 |\dot\omega_t|^2\,\mathrm{d}t, & \text{if } \omega \text{ is absolutely continuous,} \\ +\infty, & \text{otherwise.} \end{cases}$$

For this quadratic action, (14.15) is precisely the unregularized path-space formulation of Section 14.2; its optimizer concentrates on the straight trajectories described there.

The endpoint cost induced by the action is

$$c_{\mathcal{A}}(x,y) := \inf_{\omega\in\Omega} \left\{ \mathcal{A}(\omega)\;;\;e_0(\omega) = x,\ e_1(\omega) = y \right\}. \tag{14.16}$$

For the quadratic action above, the minimizing path is the straight segment $\omega_t = (1-t)x + ty$, and $c_{\mathcal{A}}(x,y) = |x-y|^2$.

Proposition 14.9 (Endpoint reduction of path-space transport)**.** *Assume that minimizing paths in (14.16) can be selected measurably, or more generally that the infimum can be approximated by measurable selections. Then (14.15) has the same value as the Kantorovich problem*

$$\inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y)\,\mathrm{d}\pi(x,y).$$

Moreover, if $\pi^\star$ is an optimal endpoint coupling and $\omega^{x,y}$ is an optimal path from x to y, then

$$M^\star = \int_{\mathcal{X}\times\mathcal{X}} \delta_{\omega^{x,y}}\,\mathrm{d}\pi^\star(x,y)$$

is an optimal path law.

Proof. Let M be any feasible path law and set $\pi = (e_0,e_1)_\sharp M$. Then $\pi \in \mathcal{U}(\alpha,\beta)$ and, by the definition of $c_{\mathcal{A}}$,

$$\int_\Omega \mathcal{A}(\omega)\,\mathrm{d}M(\omega) \geqslant \int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y)\,\mathrm{d}\pi(x,y).$$

This proves that the path-space value is at least the Kantorovich value. Conversely, given a coupling π and a measurable selection $(x,y)\mapsto\omega^{x,y}$ with action arbitrarily close to $c_{\mathcal{A}}(x,y)$, the mixture of Dirac path laws $M = \int \delta_{\omega^{x,y}}\,\mathrm{d}\pi(x,y)$ has endpoints α and β and action equal, up to the selected approximation error, to $\int c_{\mathcal{A}}\,\mathrm{d}\pi$. Optimizing over π and letting the approximation error vanish proves equality. If exact minimizing paths are selected for an optimal $\pi^\star$, the displayed $M^\star$ is optimal. □

Thus the classical coupling problem can be read as a path problem where the endpoints are chosen first and the connecting path is then selected with minimal action. The Schrödinger problem changes exactly this last step: between two endpoints, it keeps the random fluctuations of a reference dynamics instead of collapsing onto a deterministic least-action path.

Entropic path-space problem. Let $\mathcal{R}^\varepsilon \in \mathcal{P}(\Omega)$ be a reference path law, for instance a Brownian or Langevin dynamics at noise level ε. The corresponding entropy projection defines the dynamic Schrödinger problem.

Definition 14.10 (Schrödinger bridge).

$$\mathrm{SB}_\varepsilon(\alpha,\beta) := \inf_{M\in\mathcal{P}(\Omega)} \left\{\varepsilon\,\mathrm{KL}(M|\mathcal{R}^\varepsilon)\ ;\ (e_0)_\sharp M = \alpha,\ (e_1)_\sharp M = \beta\right\}. \tag{14.17}$$

It asks for the most likely path law, relative to the prior dynamics $\mathcal{R}^\varepsilon$, among all path laws matching the observed endpoint marginals. This viewpoint goes back to Schrödinger's reciprocal problem [475] and is surveyed in modern OT language in [336, 337]; stochastic-control formulations are developed in [131].

Stochastic-control interpretation. For Brownian references, the entropy projection has a direct control meaning. To fix constants, suppose that $\mathcal{R}^\varepsilon$ is the law of

$$\mathrm{d}X_t = \sqrt{\varepsilon}\,\mathrm{d}B_t, \qquad X_0 \sim \alpha,$$

so that the generator is $(\varepsilon/2)\Delta$. Under the standard finite-entropy assumptions, a path law M with the same initial marginal and $M \ll \mathcal{R}^\varepsilon$ can be represented, in the weak sense, by an adapted drift u_t of finite energy,

$$\mathrm{d}X_t = u_t\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}B_t, \qquad X_0 \sim \alpha.$$

Conversely, such a drift defines an absolutely continuous change of law when the usual Girsanov integrability conditions hold. Girsanov's formula then gives

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2\varepsilon}\mathbb{E}_M \int_0^1 |u_t|^2\,\mathrm{d}t, \qquad \varepsilon\,\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2}\mathbb{E}_M \int_0^1 |u_t|^2\,\mathrm{d}t.$$

If the reference initial law is not α, an additional term $\varepsilon\,\mathrm{KL}(\alpha|\mathcal{R}_0^\varepsilon)$ appears, but it is fixed by the endpoint constraint. Hence, up to this fixed endpoint cost, the Schrödinger bridge can be read as

$$\inf_u \left\{\frac{1}{2}\mathbb{E}\int_0^1 |u_t|^2\,\mathrm{d}t\ ;\ \mathrm{d}X_t = u_t\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}B_t, \quad X_0\sim\alpha,\ X_1\sim\beta\right\},$$

namely as the least energetic change of drift that steers the Brownian prior from α to β. The optimizer may be chosen Markovian. If h_t denotes the positive space-time harmonic Schrödinger factor associated with the terminal constraint, then the optimal feedback drift has the Doob-transform form

$$u_t^\star(x) = \varepsilon\nabla\log h_t(x).$$

Equivalently, with the value potential $\varphi_t = \varepsilon\log h_t$, one has $u_t^\star = \nabla\varphi_t$ and φ_t solves the viscous Hamilton–Jacobi equation $\partial_t\varphi_t + \frac{1}{2}|\nabla\varphi_t|^2 + \frac{\varepsilon}{2}\Delta\varphi_t = 0$. This control viewpoint and the endpoint-coupling reduction below are two faces of the same object: the controlled diffusion describes the whole path law, whereas the static Sinkhorn coupling records only its two endpoints.

Viscous Benamou–Brenier formulations. The Schrödinger interpolation also admits dynamic optimal-control descriptions, which are viscous analogues of the Benamou–Brenier formula. In this paragraph, v_t denotes the forward drift called u_t above, while u_t denotes the associated current velocity. Write $\sigma > 0$ for the diffusion intensity; thus the uncontrolled noise is $\sqrt{\sigma}\,\mathrm{d}B_t$ and its generator is $(\sigma/2)\Delta$. Formally, for smooth positive densities the controlled Fokker–Planck equation is

$$\partial_t\rho_t + \mathrm{div}(\rho_t v_t) = \frac{\sigma}{2}\Delta\rho_t,$$

one minimizes, among curves joining the prescribed endpoint densities, the kinetic action

$$\int_0^1\int \frac{1}{2}|v_t(x)|^2\rho_t(x)\mathrm{d}x\mathrm{d}t.$$

Equivalently, one can absorb the diffusion into the velocity by writing

$$u_t = v_t - \frac{\sigma}{2}\nabla\log\rho_t, \qquad \partial_t\rho_t + \mathrm{div}(\rho_t u_t) = 0.$$

Expanding $v_t = u_t + \frac{\sigma}{2}\nabla \log \rho_t$ and using the continuity equation for (ρ_t, u_t) gives

$$\int_0^1 \int \frac{1}{2}|v_t|^2 \rho_t \, dx \, dt = \int_0^1 \int \left(\frac{1}{2}|u_t|^2 + \frac{\sigma^2}{8}|\nabla \log \rho_t|^2\right) \rho_t \, dx \, dt + \frac{\sigma}{2}\left[\int \rho_1 \log \rho_1 \, dx - \int \rho_0 \log \rho_0 \, dx\right].$$

Since the entropy term depends only on the prescribed endpoints, the same minimizers are obtained from the modified Benamou–Brenier action

$$\int_0^1 \int \left(\frac{1}{2}|u_t(x)|^2 + \frac{\sigma^2}{8}|\nabla \log \rho_t(x)|^2\right) \rho_t(x) dx dt.$$

Thus the Schrödinger bridge is a least-action interpolation with both transport kinetic energy and a Fisher-information penalty. If one instead writes the viscous equation with diffusion coefficient $\sigma\Delta\rho_t$, the same formula is obtained after replacing σ above by 2σ, so the Fisher coefficient becomes $\sigma^2/2$.

The reduction to endpoint couplings follows from disintegration. We write

$$\mathcal{R}^\varepsilon(d\omega) = \int \mathcal{R}^{\varepsilon,x,y}(d\omega)\, \mathcal{R}^\varepsilon_{0,1}(dx, dy), \qquad \mathcal{R}^\varepsilon_{0,1} := (e_0, e_1)_\sharp \mathcal{R}^\varepsilon,$$

where $\mathcal{R}^{\varepsilon,x,y}$ is the reference bridge conditioned on endpoints (x, y). Similarly, for any feasible M, write

$$M(d\omega) = \int M^{x,y}(d\omega)\, \pi(dx, dy), \qquad \pi := (e_0, e_1)_\sharp M.$$

Proposition 14.11 (Endpoint reduction of the Schrödinger problem)**.** *Assume that the regular conditional laws above exist and that the relative-entropy chain rule applies, with value $+\infty$ when absolute continuity fails. Then*

$$\mathrm{SB}_\varepsilon(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha,\beta)} \varepsilon\, \mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1}). \tag{14.18}$$

For a fixed endpoint coupling π with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1})$, the minimizing path law is the mixture of reference bridges

$$M^\pi = \int \mathcal{R}^{\varepsilon,x,y}\, d\pi(x, y). \tag{14.19}$$

Consequently, if $\pi^\star$ solves (14.18), then $M^{\pi^\star}$ solves the path-space problem (14.17).

Proof. If M has finite relative entropy with respect to $\mathcal{R}^\varepsilon$, then $\pi = (e_0, e_1)_\sharp M$ is necessarily absolutely continuous with respect to $\mathcal{R}^\varepsilon_{0,1}$. The chain rule for relative entropy gives

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1}) + \int_{\mathcal{X}\times\mathcal{X}} \mathrm{KL}(M^{x,y}|\mathcal{R}^{\varepsilon,x,y})\, d\pi(x, y). \tag{14.20}$$

The second term is nonnegative and vanishes exactly when $M^{x,y} = \mathcal{R}^{\varepsilon,x,y}$ for π-almost every (x, y). Thus, once an endpoint coupling π with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1})$ is fixed, the best path law is (14.19), and the remaining minimization is precisely (14.18). If no such endpoint coupling exists, both sides are $+\infty$. □

This proposition makes the connection between the dynamic and static views precise. The static Schrödinger problem stores only the endpoints of the optimal random trajectories; the full bridge is recovered by filling each transported endpoint pair with the corresponding reference bridge. In the zero-noise limit, these bridges concentrate on least-action paths and one recovers the unregularized path-space transport problem, hence the Monge–Kantorovich problem [336].

Brownian bridges and Sinkhorn couplings. For $\mathcal{X} = \mathbb{R}^d$, take $\mathcal{R}^\varepsilon$ to be a Brownian reference dynamics. With the convention $dX_t = \sqrt{\varepsilon}\, dB_t$ used above, its unit-time transition density is proportional to $e^{-|x-y|^2/(2\varepsilon)}$, so multiplying the endpoint entropy by ε produces the static cost $|x - y|^2/2$. Equivalently, after renaming the static temperature by a factor of two, the heat-kernel density has the form

$$p_\varepsilon(x, y) \propto \exp\left(-\frac{|x - y|^2}{\varepsilon}\right).$$

More generally, suppose that the endpoint prior can be written, up to a normalization constant independent of π, as

$$\mathcal{R}^{\varepsilon}_{0,1}(\mathrm{d}x, \mathrm{d}y) \propto \exp\left(-\frac{c(x,y)}{\varepsilon}\right) \alpha(\mathrm{d}x)\beta(\mathrm{d}y).$$

This includes the usual heat-kernel reference after rewriting it with respect to $\alpha \otimes \beta$, whenever the one-time endpoint densities are fixed and mutually absolutely continuous. The additional one-body density factors only add constants under the marginal constraints. Then, for every $\pi \in \mathcal{U}(\alpha, \beta)$,

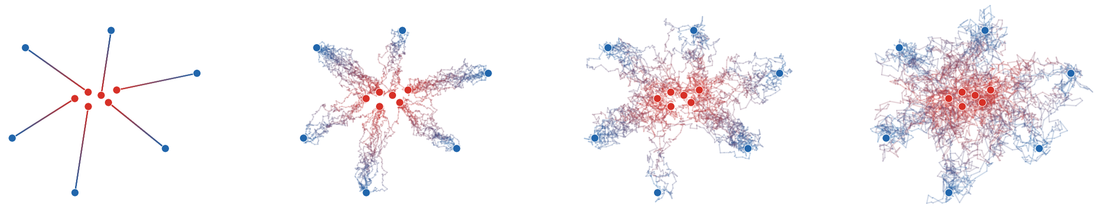
$$\varepsilon\,\mathrm{KL}(\pi|\mathcal{R}^{\varepsilon}_{0,1}) = \int c(x,y)\,\mathrm{d}\pi(x,y) + \varepsilon\,\mathrm{KL}(\pi|\alpha \otimes \beta) + \text{constant},$$

where the constant does not depend on π. Hence (14.18) is exactly the continuous Sinkhorn problem (8.13), up to an additive constant in the value. The optimal static coupling $\pi^{\star}_{\varepsilon}$ is the endpoint law of the most likely controlled noisy dynamics, while the associated path law is

$$M^{\star}_{\varepsilon} = \int \mathcal{R}^{\varepsilon,x,y}\,\mathrm{d}\pi^{\star}_{\varepsilon}(x,y).$$

In words, Sinkhorn computes which endpoints should be paired; the path-space Schrödinger bridge then connects each paired endpoint by a Brownian bridge rather than by a deterministic straight line.

Figure 14.3 illustrates this endpoint-to-path lifting on a small discrete example. The red atoms form a compact source cloud and the blue atoms surround them. The first panel uses the unregularized OT coupling and zero bridge noise; the next panels increase ε, which simultaneously softens the endpoint coupling and amplifies the Brownian fluctuations between paired endpoints.



$\varepsilon = 0$ $\varepsilon = 0.035$ $\varepsilon = 0.10$ $\varepsilon = 0.28$

Figure 14.3: Endpoint couplings lifted to Brownian bridges. Each panel transports six equally weighted red atoms concentrated near the center to six blue atoms distributed around them. For the displayed value of ε, the endpoint coupling is either the exact quadratic OT plan ($\varepsilon = 0$) or the entropic Sinkhorn plan ($\varepsilon > 0$). A total of 60 paths is sampled by a multinomial allocation with probabilities $\pi_{i,j}$, and each selected pair (x_i, y_j) is filled by a Brownian bridge with covariance proportional to $\varepsilon t(1-t)$. Colors encode time from red to blue.

The deterministic and entropic path-space formulations thus differ in how each transported endpoint pair is filled: by a least-action path in the former and by a conditional reference bridge in the latter. We now return to deterministic transport and vary its instantaneous action to generate other dynamic geometries.

14.4 Generalized Dynamic Wasserstein Distances

The quadratic Benamou–Brenier formula is only one instance of a broader fixed-mass dynamic language. The goal of this section is to define a large family of geodesic-like distances on spaces of probability measures by modifying the action minimized in the Benamou–Brenier formula. The objects introduced here are metric: they specify admissible curves, tangent variables and path energies. All descent constructions are postponed to Section 15.5, where these distances are used to generate gradient-flow PDE models.

Path actions. The common starting point is an instantaneous cost for moving a measure with a prescribed Eulerian velocity. Integrating this cost along continuity-equation curves produces the associated endpoint geometry.

In the mass-preserving Euclidean setting, the basic input is an instantaneous action $\mathbb{A}(\alpha, w)$, where α is the current measure and w is an admissible velocity representative. Its endpoint geometry is defined dynamically.

Definition 14.12 (Generalized dynamic action distance)**.** Let $\mathbb{A}$ be a nonnegative extended-valued action on measure–velocity pairs. Define

$$\mathsf{D}^2_{\mathbb{A}}(\alpha_0, \alpha_1) := \inf_{\alpha_t, v_t} \left\{ \int_0^1 \mathbb{A}(\alpha_t, v_t)\,\mathrm{d}t : \partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0,\ \alpha_{t=0} = \alpha_0,\ \alpha_{t=1} = \alpha_1 \right\}. \tag{14.21}$$

Equivalently, one may quotient by velocity fields that induce the same first-order variation of the measure. The formula (14.21) should be read as a dynamic definition of the distance, not as a property automatically satisfied by an arbitrary discrepancy. Some standard distances, such as $\mathcal{W}_p$, are first written with a p-homogeneous action and then squared by taking a constant-speed parametrization; this normalization is made explicit below. Different choices of $\mathbb{A}$ change the resulting geometry; Section 15.5 later reuses these choices when dynamics are introduced.

Quadratic, or Riemannian, tangent actions. A particularly transparent case occurs when $w \mapsto \mathbb{A}(\alpha, w)$ is quadratic. For simplicity, take admissible velocities in $L^2(\alpha; \mathbb{R}^d)$; in some applications this Hilbert space is replaced by a closed subspace encoding additional constraints. If the polarization of $\mathbb{A}$ is represented by a positive self-adjoint operator $Q_\alpha : L^2(\alpha; \mathbb{R}^d) \to L^2(\alpha; \mathbb{R}^d)$,

$$\mathbb{A}(\alpha; w, z) = \langle Q_\alpha w, z \rangle_{L^2(\alpha)}, \qquad \mathbb{A}(\alpha, w) = \langle Q_\alpha w, w \rangle_{L^2(\alpha)},$$

and if the quadratic form is nondegenerate after quotienting velocities that induce the same measure variation, then the least-action distance generated by this tensor is

$$\mathsf{D}^2_Q(\alpha_0, \alpha_1) = \mathsf{D}^2_{\mathbb{A}}(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0 \\ \alpha_{t=0} = \alpha_0,\ \alpha_{t=1} = \alpha_1}} \int_0^1 \langle Q_{\alpha_t} v_t, v_t \rangle_{L^2(\alpha_t)}\,\mathrm{d}t. \tag{14.22}$$

The usual $\mathcal{W}_2$ geometry corresponds to $Q_\alpha = \mathrm{Id}$ in this simplified notation. Thus Q_α records how the chosen geometry deforms the Euclidean $L^2(\alpha)$ tangent norm: no deformation for $\mathcal{W}_2$, and a nontrivial tensor for generalized Riemannian geometries. In Section 15.5, the same tensor is reused as a preconditioner for metric descent.

Local velocity actions. Many dynamic distances are local with respect to a reference measure λ. Write $\alpha = a\lambda$. A velocity action is specified by a pointwise integrand

$$A : [0, +\infty) \times \mathbb{R}^d \to [0, +\infty], \qquad (a, w) \mapsto A(a, w),$$

where $a \in \mathbb{R}_+$ is a density value and $w \in \mathbb{R}^d$ is a velocity value, and defines

$$\mathbb{A}(\alpha, w) = \int A\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}(x), w(x)\right) \mathrm{d}\lambda(x). \tag{14.23}$$

For a fixed reference λ, this covers density-dependent mobilities and congestion constraints. If, in addition, A is positively 1-homogeneous in its first variable, $A(\eta a, w) = \eta A(a, w)$ for $\eta \geqslant 0$, then the same formula is intrinsic: replacing λ by another dominating measure gives the same value. The usual Benamou–Brenier action is the model case

$$A_2(a, w) = a|w|^2, \qquad \mathbb{A}(\alpha, w) = \int |w|^2 \mathrm{d}\alpha. \tag{14.24}$$

Here $a = \mathrm{d}\alpha/\mathrm{d}\lambda$ is the density of transported mass and w is the Eulerian velocity. The factor a says that moving twice as much mass with the same velocity costs twice as much.

Homogeneous momentum actions. The same action can be written in momentum variables, and this is the form in which convexity and metric properties are easiest to read. Set $\omega = \alpha w$, so that ω is a vector-valued measure. When the local description is written with the same reference λ, so that $\alpha = a\lambda$ and $\omega = m\lambda$, the pointwise momentum perspective is

$$J_A(a, m) := \begin{cases} A(a, m/a), & a > 0, \\ 0, & a = 0 \text{ and } m = 0, \\ +\infty, & a = 0 \text{ and } m \neq 0, \end{cases} \tag{14.25}$$

and the measure action relative to λ is

$$\mathbb{J}_{A,\lambda}(\alpha,\omega) := \int J_A\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}, \frac{\mathrm{d}\omega}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda, \tag{14.26}$$

with value $+\infty$ if α or the total variation $|\omega|$ is not absolutely continuous with respect to λ. This zero-density convention is the lower-semicontinuous one for the superlinear actions used below; other growths use the corresponding recession extension. If A is positively 1-homogeneous in a, then J_A is jointly 1-homogeneous: $J_A(\eta a, \eta m) = \eta J_A(a,m)$. In that intrinsic case the value of $\mathbb{J}_{A,\lambda}$ is independent of the dominating reference measure, and we write simply $\mathbb{J}_A$. For $A_2(a,w) = a|w|^2$, one recovers the quadratic perspective

$$J_2(a,m) = \frac{|m|^2}{a},$$

which is the integrand used in the convex Benamou–Brenier formulation (14.12). In this case $\omega = \alpha w$; if $\alpha = \rho\mathrm{d}x$ and $\omega = m\mathrm{d}x$, then $m = \rho w$, and $J_2(\rho, m) = \rho|w|^2$.

Proposition 14.13 (Concave mobilities give convex momentum actions)**.** *Let $I \subset [0,+\infty)$ be a convex interval, let $\theta : I \to [0,+\infty)$ be concave, and let $L : \mathbb{R}^d \to [0,+\infty]$ be convex with $L(0) = 0$. Define, on the set where $\theta(a) > 0$,*

$$J_{\theta,L}(a,m) := \theta(a)L\left(\frac{m}{\theta(a)}\right), \qquad A_{\theta,L}(a,w) := J_{\theta,L}(a,aw) = \theta(a)L\left(\frac{aw}{\theta(a)}\right), \tag{14.27}$$

and extend $J_{\theta,L}$ to the boundary by lower semicontinuity. Then $J_{\theta,L}$ is convex in (a,m). In particular, this single construction contains

$$A(a,w) = aL(w) \quad \textit{by taking } \theta(a) = a,$$

and the concave-mobility quadratic action

$$A(a,w) = \frac{a^2|w|^2}{\theta(a)} \quad \textit{by taking } L(u) = |u|^2.$$

Proof. The perspective $P(s,m) = sL(m/s)$ of a convex function is convex on $s > 0$. Since $L(0) = 0$, it is also nonincreasing in the scalar s for fixed m: if $s_2 \geqslant s_1 > 0$, then

$$L(m/s_2) = L\big((s_1/s_2)(m/s_1)\big) \leqslant (s_1/s_2)L(m/s_1),$$

hence $P(s_2,m) \leqslant P(s_1,m)$. Let $a_\zeta = (1-\zeta)a_0 + \zeta a_1$ and $m_\zeta = (1-\zeta)m_0 + \zeta m_1$. By concavity of θ,

$$\theta(a_\zeta) \geqslant (1-\zeta)\theta(a_0) + \zeta\theta(a_1).$$

Using the monotonicity in s, and then the convexity of P, gives

$$J_{\theta,L}(a_\zeta, m_\zeta) = P(\theta(a_\zeta), m_\zeta) \leqslant P((1-\zeta)\theta(a_0) + \zeta\theta(a_1), m_\zeta) \leqslant (1-\zeta)J_{\theta,L}(a_0,m_0) + \zeta J_{\theta,L}(a_1,m_1).$$

The boundary cases follow by the chosen lower-semicontinuous extension. □

Remark 14.14 (Perspective transform gives convexity). The convexity mechanism in Proposition 14.13 is the same one used earlier for φ-divergences. In both cases one starts from a convex function of a ratio and multiplies by the denominator: the density-ratio perspective $v\varphi(u/v)$ proves the joint convexity and lower-semicontinuity of $\mathcal{D}_\varphi$ in Proposition 7.17, see (7.6) and (7.8); the momentum perspective $J_A(a,m) = A(a,m/a)$ in (14.25) plays the same role for dynamic transport actions. The classical Benamou–Brenier relaxation uses the quadratic perspective $J(a,m) = |m|^2/a$ from (14.10)–(14.11); concave mobilities use the modified perspective (14.27); and unbalanced dynamic transport uses the three-variable analogue $J_\kappa(a,m,r)$ in (14.45)–(14.46). Thus the same perspective transform is the hidden convexification behind density-ratio divergences, balanced dynamic OT, concave-mobility distances and Wasserstein–Fisher–Rao dynamics.

The next proposition isolates the additional assumptions under which the momentum formulation generated by A defines a genuine path metric rather than only a variational principle.

Proposition 14.15 (Homogeneous dynamic actions define distances). *Fix a reference measure λ, omitted from the notation only in the intrinsic case where $\mathbb{J}_{A,\lambda}$ does not depend on λ. Assume that the momentum perspective J_A defined in (14.25) is lower semicontinuous, convex in (a, m), even in m, and satisfies $J_A(a, 0) = 0$. Assume moreover that, for some $r > 1$,*

$$J_A(a, \xi m) = |\xi|^r J_A(a, m) \qquad \text{for all admissible } a, \text{ all } m, \text{ and all } \xi \in \mathbb{R},$$

and that $J_A(a, m) = 0$ if and only if $m = 0$. Equivalently, the evenness, homogeneity and nondegeneracy assumptions translate, away from zero density, into

$$A(a, -w) = A(a, w), \qquad A(a, \xi w) = |\xi|^r A(a, w), \qquad A(a, w) = 0 \iff w = 0 \qquad (a > 0).$$

Define, on every fixed-mass class,

$$\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t \alpha_t + \operatorname{div} \omega_t = 0 \\ \alpha_{t=0} = \alpha_0,\ \alpha_{t=1} = \alpha_1}} \left(\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t, \omega_t)\, \mathrm{d}t \right)^{1/r}.$$

Assume finally that the relaxed dynamic problem is sequentially closed and attains its infimum whenever the value is finite. Then $\mathsf{D}_{A,\lambda}$ is an extended distance on each finite-action component: it is symmetric, satisfies the triangle inequality, and $\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_1) = 0$ only when $\alpha_0 = \alpha_1$. In the intrinsic case, we write D_A.

Proof. The constant curve $\alpha_t = \alpha_0$, $\omega_t = 0$, gives $\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_0) = 0$. Conversely, if $\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_1) = 0$, attainment provides a relaxed zero-action minimizer, which satisfies

$$\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t, \omega_t)\, \mathrm{d}t = 0.$$

Since $\mathbb{J}_{A,\lambda} \geqslant 0$, this implies $\omega_t = 0$ for a.e. t. The continuity equation then gives $\partial_t \alpha_t = 0$ in the sense of distributions, so $\alpha_0 = \alpha_1$.

Symmetry follows by time reversal. If $(\alpha_t, \omega_t)_{t \in [0,1]}$ connects α_0 to α_1, then

$$\widetilde{\alpha}_t = \alpha_{1-t}, \qquad \widetilde{\omega}_t = -\omega_{1-t}$$

connects α_1 to α_0, satisfies the same continuity equation, and has the same action because J_A is even in m.

It remains to prove the triangle inequality. Let (α^1, ω^1) connect α_0 to α_1 with action E_1, and let (α^2, ω^2) connect α_1 to α_2 with action E_2. For $\tau \in (0, 1)$, concatenate the two curves after the time changes $s = t/\tau$ and $s = (t - \tau)/(1 - \tau)$:

$$(\alpha_t, \omega_t) = \begin{cases} \left(\alpha^1_{t/\tau},\ \tau^{-1} \omega^1_{t/\tau}\right), & 0 \leqslant t \leqslant \tau, \\ \left(\alpha^2_{(t-\tau)/(1-\tau)},\ (1 - \tau)^{-1} \omega^2_{(t-\tau)/(1-\tau)}\right), & \tau \leqslant t \leqslant 1. \end{cases}$$

The rescaled curve is admissible from α_0 to α_2. By r-homogeneity in the momentum variable, its action is $\tau^{1-r} E_1 + (1 - \tau)^{1-r} E_2$. Writing $X = E_1^{1/r}$ and $Y = E_2^{1/r}$, the optimal allocation is $\tau = X/(X + Y)$, with the evident limiting convention if $X + Y = 0$, and gives

$$\inf_{\tau \in (0,1)} \left\{ \tau^{1-r} E_1 + (1 - \tau)^{1-r} E_2 \right\} = (X + Y)^r.$$

Hence $\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_2) \leqslant E_1^{1/r} + E_2^{1/r}$. Taking E_1 and E_2 arbitrarily close to the two optimal action values yields

$$\mathsf{D}_{A,\lambda}(\alpha_0, \alpha_2) \leqslant \mathsf{D}_{A,\lambda}(\alpha_0, \alpha_1) + \mathsf{D}_{A,\lambda}(\alpha_1, \alpha_2),$$

with the usual extended-real convention when one of the distances is infinite. □

Without homogeneity or nondegeneracy, the same momentum action remains useful as a variational principle, but its r-th root need not define a distance.

Example 14.16 (Wasserstein-p action). The usual $\mathcal{W}_p$ distances correspond to changing only the homogeneity of the Benamou–Brenier action. The case $p = 2$ is the original formula [35]; for

$1 < p < +\infty$, the same dynamic viewpoint gives the standard description of absolutely continuous curves in Wasserstein spaces [12, 530, 463]. The specific objects to insert in the general framework above are

$$A_p(a,w) = a|w|^p, \qquad J_p(a,m) = \begin{cases} |m|^p/a^{p-1}, & a > 0, \\ 0, & (a,m) = (0,0), \\ +\infty, & a = 0,\ m \neq 0, \end{cases}$$

With $A = A_p$, Proposition 14.15 gives the usual identity $\mathsf{D}_{A_p} = \mathcal{W}_p$. The corresponding squared length-space normalization is

$$\mathbb{A}_p(\alpha, w) = \left(\int |w|^p \, d\alpha \right)^{2/p}.$$

This is the squared version of the p-homogeneous action: minimizing $\int_0^1 \mathbb{A}_p(\alpha_t, v_t) dt$ gives $\mathcal{W}_p^2$ after constant-speed reparametrization. Thus A_p, J_p denote the local p-homogeneous velocity and momentum densities, whereas $\mathbb{A}_p$ denotes the squared tangent action used in the length formulation. The endpoint $p = 1$ can be treated separately: the perspective becomes $J_1(a,m) = |m|$, and the dynamic problem collapses to Beckmann's formulation of $\mathcal{W}_1$ [31].

Concave-mobility actions. One can instead keep a quadratic momentum action and change the mobility. Dolbeault, Nazaret and Savaré introduced this construction as a class of generalized transport distances adapted to nonlinear diffusion [192]. The defining objects are collected below.

Definition 14.17 (Concave-mobility dynamic distance)**.** Let $I \subset [0, +\infty)$ be a convex interval and let $\theta : I \longrightarrow [0, +\infty)$ be concave. Define

$$J_\theta(a,m) := \begin{cases} |m|^2/\theta(a), & \theta(a) > 0, \\ 0, & \theta(a) = 0 \text{ and } m = 0, \\ +\infty, & \theta(a) = 0 \text{ and } m \neq 0. \end{cases}$$

The corresponding velocity action is

$$A_\theta(a,w) = J_\theta(a, aw) = \frac{a^2|w|^2}{\theta(a)}.$$

For a fixed reference measure λ, set

$$\mathsf{W}_{\theta,\lambda}(\alpha_0, \alpha_1) := \mathsf{D}_{A_\theta,\lambda}(\alpha_0, \alpha_1).$$

The convexity of J_θ is the special case $L(u) = |u|^2$ of Proposition 14.13. This is why concavity of the mobility, rather than convexity, is the structural condition that makes the continuity-equation formulation convex.

Fix now a reference measure λ. If $\alpha = a\lambda$, the induced squared tangent action is

$$\mathbb{A}_{\theta,\lambda}(\alpha, w) = \int A_\theta(a(x), w(x)) \, d\lambda(x) = \int \frac{a(x)^2}{\theta(a(x))} |w(x)|^2 \, d\lambda(x),$$

and it is set to $+\infty$ when $\alpha \not\ll \lambda$. Equivalently, on the set where $\theta(a(x)) > 0$,

$$\mathbb{A}_{\theta,\lambda}(\alpha, w) = \int \frac{a(x)}{\theta(a(x))} |w(x)|^2 \, d\alpha(x).$$

Hence this is a local Riemannian case, in the sense of (14.22), whenever the multiplier $a(x)/\theta(a(x))$ is finite and positive. The associated tensor is the multiplication operator

$$(Q_{\theta,\lambda,\alpha} w)(x) = \frac{a(x)}{\theta(a(x))} w(x), \qquad \alpha = a\lambda, \tag{14.28}$$

defined α-a.e. on the set where $\theta(a) > 0$. Except for linear mobilities $\theta(a) = ca$, and in particular the normalized case $\theta(a) = a$ which recovers $\mathcal{W}_2$, the pointwise velocity action $A_\theta(a,w)$ is not positively

1-homogeneous in the density variable a. Consequently the construction is not intrinsic under a change of λ: the resulting distance depends on the chosen reference measure and is finite only between endpoints that can be joined by a finite-action curve with $\alpha_t \ll \lambda$.

The subscript λ recalls that the action is measured through the density $a = \mathrm{d}\alpha/\mathrm{d}\lambda$. Equivalently, because this action is quadratic in the momentum, $\mathsf{W}^2_{\theta,\lambda}$ is the path value (14.21) with $\mathbb{A} = \mathbb{A}_{\theta,\lambda}$.

Proposition 14.18 (Concave-mobility dynamic distances)**.** *For a fixed reference measure λ, and under the standard compactness and lower-semicontinuity hypotheses ensuring existence of relaxed minimizers [192], $\mathsf{W}_{\theta,\lambda} = \mathsf{D}_{A_{\theta},\lambda}$ is an extended distance on each finite-action component contained in $\{\alpha : \alpha \ll \lambda\}$. In particular, whenever the relevant component has finite pairwise distances, $\mathsf{W}_{\theta,\lambda}$ is a genuine metric there.*

Proof. Proposition 14.13, applied with $L(u) = |u|^2$, gives the lower-semicontinuous convex momentum density J_θ. It is even in m, satisfies $J_\theta(a, 0) = 0$, and obeys $J_\theta(a, \xi m) = |\xi|^2 J_\theta(a, m)$. Moreover $J_\theta(a, m) = 0$ if and only if $m = 0$, with the boundary convention used in its definition. For the fixed reference λ, the hypotheses of Proposition 14.15 therefore hold with $r = 2$; the existence assumption is supplied by the compactness hypothesis in the statement. That proposition gives symmetry, separation and the triangle inequality for $\mathsf{D}_{A_\theta,\lambda}$, hence for $\mathsf{W}_{\theta,\lambda}$. □

The choice $\theta(a) = a$ recovers $\mathcal{W}_2$. Other choices encode different geometry: $\theta(a) = a^\gamma$ with $0 < \gamma \leqslant 1$ changes the cost of moving dilute mass, while $\theta(a) = a(1 - a/M)$ on $[0, M]$ models a volume-filling or exclusion effect, since mobility vanishes both at vacuum and at the maximal density. The distance is comparable with $\mathcal{W}_2$ on classes where $\theta(a)$ is bounded above and below by positive multiples of a; otherwise zero-mobility barriers can make some pairs infinitely far apart. Its use as a geometry for nonlinear diffusion is discussed in Section 15.5.

Dynamic spectral Wasserstein distances. The static spectral distances of Section 11.5 penalize a coupling through the covariance of its displacement. A dynamic version keeps the continuity equation but replaces the pointwise kinetic energy by a gauge of the whole velocity covariance. The resulting action is nonlocal in space: velocity directions are charged globally through their covariance, rather than independently at each point.

The spectral geometry charges the covariance of the velocity field through a monotone gauge. The corresponding action and length distance are defined together.

Definition 14.19 (Dynamic spectral Wasserstein distance)**.** Let γ be a monotone spectral gauge on $\mathbb{S}^d_+$. Define

$$\mathbb{A}_\gamma(\alpha, v) := \gamma\left(\int v(x)v(x)^\top \mathrm{d}\alpha(x)\right). \tag{14.29}$$

The associated distance is

$$\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0, \alpha_1) := \mathsf{D}^2_{\mathbb{A}_\gamma}(\alpha_0, \alpha_1). \tag{14.30}$$

The trace gauge gives the usual Wasserstein tangent action, while the operator gauge $\gamma(M) = \lambda_{\max}(M)$ charges only the largest directional velocity variance. In density–momentum variables, this corresponds to the measure action

$$\mathbb{J}_\gamma(\alpha, \omega) = \gamma\left(\int \left(\frac{\mathrm{d}\omega}{\mathrm{d}\alpha}\right)\left(\frac{\mathrm{d}\omega}{\mathrm{d}\alpha}\right)^\top \mathrm{d}\alpha\right),$$

or, when $\alpha = \rho\,\mathrm{d}x$ and $\omega = m\,\mathrm{d}x$,

$$\mathbb{J}_\gamma(\rho, m) = \gamma\left(\int \frac{m(x)m(x)^\top}{\rho(x)}\,\mathrm{d}x\right).$$

This functional is convex in the density–momentum fields (ρ, m) by the matrix perspective, together with the monotonicity and convexity of γ. It is nevertheless not, in general, obtained by integrating a pointwise action density, because the covariance is computed globally before applying γ. It becomes local only for linear spectral gauges. For instance, if $\gamma(M) = \mathrm{tr}(GM)$ with $G \succeq 0$, then the velocity and momentum densities are

$$A_{\mathrm{lin}}(a, w) = a\,w^\top G w, \qquad J_{\mathrm{lin}}(a, m) = \frac{m^\top G m}{a},$$

and the trace gauge, $G = \mathrm{Id}$ in $\gamma(M) = \mathrm{tr}(GM)$, recovers the usual Benamou–Brenier action.

The following result, in the form used for normalized spectral flows in [425], shows that this dynamic construction is not merely infinitesimal: it exactly recovers the static displacement-covariance formulation.

Proposition 14.20 (Static/dynamic equality for spectral Wasserstein)**.** *Let γ be a monotone spectral gauge and let $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$. Then*

$$\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0, \alpha_1) = \mathcal{W}^2_\gamma(\alpha_0, \alpha_1).$$

The common value defines a distance on $\mathcal{P}_2(\mathbb{R}^d)$; the triangle inequality is established by the robust representation in Proposition 11.47.

Proof. We first prove $\mathsf{W}^2_{\gamma,\mathrm{dyn}} \leqslant \mathcal{W}^2_\gamma$. Let $\pi \in \mathcal{U}(\alpha_0, \alpha_1)$ and let $(X, Y) \sim \pi$. Set $Z_t = (1-t)X + tY$ and $\alpha_t = (Z_t)_\sharp \pi$. Define the Eulerian velocity as the conditional mean $v_t(z) := \mathbb{E}[Y - X \mid Z_t = z]$. Then (α_t, v_t) solves the continuity equation. If

$$M_\pi := \int (x-y)(x-y)^\top \mathrm{d}\pi(x,y), \qquad C_t := \int v_t(z)v_t(z)^\top \mathrm{d}\alpha_t(z),$$

conditional Jensen gives $C_t \preceq M_\pi$. Indeed, for every direction $u \in \mathbb{R}^d$,

$$u^\top C_t u = \mathbb{E}\big[\mathbb{E}[\langle u, Y - X\rangle \mid Z_t]^2\big] \leqslant \mathbb{E}[(\langle u, Y - X\rangle)^2] = u^\top M_\pi u.$$

By the Loewner monotonicity of γ, $\int_0^1 \gamma(C_t)\mathrm{d}t \leqslant \gamma(M_\pi)$. Taking the infimum over π gives the first inequality.

Conversely, let (α_t, v_t) be an admissible finite-action competitor. Since γ is a finite positive gauge on the finite-dimensional cone $\mathbb{S}^d_+$, it is equivalent to the trace on this cone; hence finite spectral action implies the usual finite kinetic energy needed for the superposition principle. Thus there exists a probability law η on absolutely continuous paths $\omega.$ such that $(e_t)_\sharp\eta = \alpha_t$ and $\dot\omega_t = v_t(\omega_t)$ for a.e. t. Let $\pi = (e_0, e_1)_\sharp\eta$ be the induced endpoint coupling. With $C_t = \int v_t v_t^\top \mathrm{d}\alpha_t$, Jensen's inequality along each path gives

$$(\omega_1 - \omega_0)(\omega_1 - \omega_0)^\top = \left(\int_0^1 \dot\omega_t \mathrm{d}t\right)\left(\int_0^1 \dot\omega_t \mathrm{d}t\right)^\top \preceq \int_0^1 \dot\omega_t \dot\omega_t^\top \mathrm{d}t.$$

After integration over η, $M_\pi \preceq \int_0^1 C_t \mathrm{d}t$. Monotonicity of γ, followed by convexity, yields

$$\gamma(M_\pi) \leqslant \gamma\left(\int_0^1 C_t \mathrm{d}t\right) \leqslant \int_0^1 \gamma(C_t)\mathrm{d}t.$$

Since π is admissible for the static problem, $\mathcal{W}^2_\gamma(\alpha_0, \alpha_1)$ is bounded by the action of the dynamic competitor. Taking the infimum over all competitors gives the reverse inequality.

The crucial assumption is the monotonicity of γ: both parts of the proof produce Loewner-order comparisons between covariance matrices, and these comparisons imply inequalities between actions only for monotone gauges. □

The use of this geometry for normalized flows, including the operator-gauge connection with Muon-type normalization, is developed in Section 15.5.

Kernelized Benamou–Brenier distances. A different way to deform the Benamou–Brenier geometry is to keep the local continuity equation but to measure velocities in a reproducing-kernel Hilbert space rather than in $L^2(\alpha)$. This construction is motivated by Stein variational gradient descent, studied later in Section 16.2: the kernel makes the velocity field smooth and computable from particles, at the price of defining a much more restrictive transport geometry.

The kernelized distance uses a vector-valued RKHS as its restricted velocity class.

Definition 14.21 (Kernelized Benamou–Brenier distance)**.** Let k be a positive definite kernel on $\mathbb{R}^d$ with scalar RKHS $\mathcal{H}_k$. Set

$$\mathcal{H}_k^d := \mathcal{H}_k \times \cdots \times \mathcal{H}_k, \qquad \|v\|^2_{\mathcal{H}_k^d} := \sum_{\ell=1}^d \|v_\ell\|^2_{\mathcal{H}_k}.$$

$$\mathbb{A}_k(\alpha, v) := \|v\|^2_{\mathcal{H}_k^d}, \qquad \mathcal{W}_k^2(\alpha_0, \alpha_1) := \mathrm{D}^2_{\mathbb{A}_k}(\alpha_0, \alpha_1). \tag{14.31}$$

This is the vector-valued analogue of the scalar RKHS norm used for MMDs in Section 7.2. Formula (14.21) is restricted to velocities $v_t \in \mathcal{H}_k^d$. The action is independent of α; the measure enters only through the continuity equation. This type of Stein geometry was introduced in the analysis of SVGD by Liu and Wang [346, 345] and later developed geometrically in [197, 403]. The important caveat is that the admissible tangent space is the smooth RKHS class, not the whole Wasserstein tangent space.

Proposition 14.22 (Kernelized dynamic distance). *Assume that the kernel has uniformly bounded diagonal,*

$$\kappa_k^2 := \sup_{x \in \mathbb{R}^d} k(x,x) < +\infty.$$

Then the quantity $\mathcal{W}_k$ defined by (14.31) is an extended distance on probability measures. More precisely, it is symmetric, satisfies the triangle inequality, and $\mathcal{W}_k(\alpha_0, \alpha_1) = 0$ implies $\alpha_0 = \alpha_1$. It can take the value $+\infty$ between different finite-action components.

Proof. The constant curve gives zero self-distance. To prove separation without assuming existence of a minimizing curve, use the RKHS evaluation bound $|v(x)| \leqslant \kappa_k \|v\|_{\mathcal{H}_k^d}$. For every $\varphi \in C_c^1(\mathbb{R}^d)$ and every admissible curve of action E, the weak continuity equation and Cauchy–Schwarz give

$$\begin{aligned}\left|\int \varphi \, \mathrm{d}(\alpha_1 - \alpha_0)\right| &\leqslant \int_0^1 \int |\nabla\varphi(x)| \, |v_t(x)| \, \mathrm{d}\alpha_t(x) \mathrm{d}t \\ &\leqslant \kappa_k \|\nabla\varphi\|_\infty \left(\int_0^1 \|v_t\|^2_{\mathcal{H}_k^d} \mathrm{d}t\right)^{1/2} = \kappa_k \|\nabla\varphi\|_\infty \sqrt{E}.\end{aligned}$$

Taking the infimum over admissible curves shows that $\mathcal{W}_k(\alpha_0, \alpha_1) = 0$ forces equality of the integrals of every compactly supported smooth test function, hence $\alpha_0 = \alpha_1$.

Symmetry follows by time reversal and by replacing v_t with $-v_{1-t}$. For the triangle inequality, concatenate an almost optimal curve from α_0 to α_1 of action E_1 with one from α_1 to α_2 of action E_2. Compressing the first curve into a time interval of length τ multiplies its action by $1/\tau$, and compressing the second into length $1-\tau$ multiplies its action by $1/(1-\tau)$. Optimizing over $\tau \in (0,1)$ gives $(\sqrt{E_1} + \sqrt{E_2})^2$, and taking infima yields the triangle inequality. □

One should read $\mathcal{W}_k$ as an extended distance on finite-action components, not as a replacement for $\mathcal{W}_2$ on all of $\mathcal{P}_2(\mathbb{R}^d)$. A useful sufficient condition for finiteness is that the endpoints lie on the same RKHS-flow orbit: if there exists $v \in L^2([0,1]; \mathcal{H}_k^d)$ whose flow map Φ_t solves $\dot{\Phi}_t = v_t \circ \Phi_t$ and satisfies $\alpha_1 = (\Phi_1)_\sharp \alpha_0$, then $\mathcal{W}_k^2(\alpha_0, \alpha_1) \leqslant \int_0^1 \|v_t\|^2_{\mathcal{H}_k^d} \mathrm{d}t$. In particular, for strictly positive definite kernels, two discrete measures with the same weights and distinct moving support points are at finite distance whenever their atoms can be connected by noncolliding smooth paths, because RKHS interpolation constructs vector fields realizing the prescribed atom velocities along the paths.

The same condition also explains the limitation. If k is smooth enough that $\mathcal{H}_k^d$ embeds into Lipschitz vector fields, finite-action curves are induced by flows of regular maps. Atomic measures therefore remain atomic with the same number of atoms along such curves, so a Dirac mass cannot be transported at finite kernelized action to a measure with a density. This lack of splitting is precisely what makes the geometry useful for deterministic particle methods, and also what makes it a nontrivial extended object rather than a full probability metric.

14.5 Nonlocal Wasserstein Distances

The local dynamic distances above transport mass through a vector field on the base space and a classical continuity equation. Nonlocal geometries use a different tangent model: the elementary motion is an exchange across an edge or a jump from x to y. The common data are a reversible kernel K, a symmetric edge or jump measure J, a pairwise increment $\bar{\nabla}$, and a pair-space action $\mathbb{A}_K$. The tangent variable is therefore attached to pairs of points, not to a single point x, so these constructions are not simply obtained by choosing another pointwise local cost $A(a, w)$.

There are two complementary versions. On a finite state space, the goal is to put a Wasserstein-like geometry on the probability simplex so that the entropy gradient flow is exactly a prescribed reversible

Markov chain [360, 383, 145]. On a continuum space, the same edge calculus becomes a jump calculus over $\mathcal{X} \times \mathcal{X}$, which models nonlocal motion, heavy-tailed jumps, and fractional-type diffusion; this is the construction of Erbar [207], building on nonlinear mobilities [192], with subsequent metric and asymptotic refinements [487].

In both settings the canonical mobility for entropy is the logarithmic mean.

Definition 14.23 (Logarithmic mean)**.**

$$\theta(a,b) := \begin{cases} \dfrac{a-b}{\log a - \log b}, & a,b>0 \text{ and } a \neq b, \\ a, & a=b, \\ 0, & ab=0, \end{cases} \tag{14.32}$$

for $a, b \geqslant 0$.

It appears because $\theta(a,b)(\log a - \log b) = a - b$, which is the edge-wise chain rule identifying entropy-driven flows with the underlying reversible Markov or jump dynamics.

Continuum jump kernels. The continuum version replaces graph edges by a symmetric measure on pairs. Its action is still quadratic, but the tangent variable is an antisymmetric jump velocity $v(x,y)$, and the mobility depends simultaneously on the two endpoint density values. It is best viewed as a convex action on the pair space $\mathcal{X} \times \mathcal{X}$, rather than as an integral of independent costs attached to single base points x.

Let $(\mathcal{X}, \mathfrak{m})$ be a reference measure space, and let $K(x,\cdot)$ be a nonnegative measure on $\mathcal{X}$ for each $x \in \mathcal{X}$, possibly of infinite total mass. We write this kernel as $K(x, \mathrm{d}y)$ to emphasize that the integration variable is y. The pair measure J on $\mathcal{X} \times \mathcal{X}$ is defined by testing against nonnegative measurable functions Φ:

$$\int_{\mathcal{X}\times\mathcal{X}} \Phi(x,y)\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y) := \int_{\mathcal{X}} \left(\int_{\mathcal{X}} \Phi(x,y)\,K(x,\mathrm{d}y) \right) \mathfrak{m}(\mathrm{d}x). \tag{14.33}$$

The reversibility assumption is precisely that this measure J is symmetric, i.e. invariant under $(x,y) \mapsto (y,x)$. For a density $\rho = \mathrm{d}\alpha/\mathrm{d}\mathfrak{m}$, write

$$\bar{\nabla}\varphi(x,y) := \varphi(y) - \varphi(x)$$

for the nonlocal gradient, and use the logarithmic mean θ defined in (14.32). These objects define the nonlocal continuity equation and its least-action distance.

Definition 14.24 (Continuum nonlocal Wasserstein distance)**.** For $\alpha_t = \rho_t \mathfrak{m}$ and antisymmetric $v_t(x,y)$, require, for every test function φ,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi\,\mathrm{d}\alpha_t = \frac{1}{2} \iint \bar{\nabla}\varphi(x,y)\,v_t(x,y)\,\theta(\rho_t(x),\rho_t(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y). \tag{14.34}$$

Define

$$\mathbb{A}_K(\alpha, v) := \frac{1}{2} \iint |v(x,y)|^2 \theta(\rho(x),\rho(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y),$$

and

$$\mathcal{W}_K^2(\alpha_0,\alpha_1) := \inf_{\rho_t, v_t} \int_0^1 \mathbb{A}_K(\alpha_t, v_t)\,\mathrm{d}t, \tag{14.35}$$

where the infimum is over all curves solving (14.34) with endpoints α_0, α_1.

Here v is not a vector field on $\mathcal{X}$ but an antisymmetric velocity on pairs (x,y); the action is therefore genuinely nonlocal.

Proposition 14.25 (Metric properties of nonlocal transport)**.** *Under the regularity and irreducibility assumptions of [207], $\mathcal{W}_K$ is an extended distance on the set of probability measures absolutely continuous with respect to $\mathfrak{m}$, and finite-distance pairs are connected by constant-speed geodesics.*

Proof. We use the analytic compactness and lower-semicontinuity theorem of [207] for the logarithmic-mean action. In particular, action-bounded sequences of admissible curves are compact for the narrow topology, the weak nonlocal continuity equation is closed under this convergence, and the action is lower semicontinuous. These facts are the nonlocal analogues of the compactness theorem used in the Benamou–Brenier formulation.

Nonnegativity is immediate from the definition of $\mathbb{A}_K(\alpha, v)$. If $\alpha_0 = \alpha_1$, the constant curve $\rho_t = \rho_0$, $v_t = 0$, is admissible and has zero action, hence $\mathcal{W}_K(\alpha_0, \alpha_0) = 0$.

Symmetry follows by time reversal. If (ρ_t, v_t) transports α_0 to α_1, set $\tilde{\rho}_t = \rho_{1-t}$, $\tilde{v}_t = -v_{1-t}$. For every test function φ, differentiating $\int \varphi \tilde{\rho}_t \, \mathrm{d}\mathfrak{m}$ and using (14.34) shows that $(\tilde{\rho}_t, \tilde{v}_t)$ solves the same continuity equation from α_1 to α_0. Since the action is quadratic in v, $\mathbb{A}_K(\tilde{\alpha}_t, \tilde{v}_t) = \mathbb{A}_K(\alpha_{1-t}, v_{1-t})$, so the action is unchanged.

For the triangle inequality, let (ρ_t^0, v_t^0) connect α_0 to α_1 with action E_0, and let (ρ_t^1, v_t^1) connect α_1 to α_2 with action E_1. For $0 < \zeta < 1$, concatenate the two curves after the rescaling

$$(\rho_t, v_t) = \begin{cases} \left(\rho^0_{t/\zeta}, \ \zeta^{-1} v^0_{t/\zeta}\right), & 0 \leqslant t \leqslant \zeta, \\ \left(\rho^1_{(t-\zeta)/(1-\zeta)}, \ (1-\zeta)^{-1} v^1_{(t-\zeta)/(1-\zeta)}\right), & \zeta < t \leqslant 1. \end{cases}$$

The factors ζ^{-1} and $(1-\zeta)^{-1}$ are exactly those needed for the weak continuity equation after changing time. Since $v \mapsto \mathbb{A}_K(\alpha, v)$ is quadratic,

$$\int_0^1 \mathbb{A}_K(\alpha_t, v_t) \, \mathrm{d}t = \frac{E_0}{\zeta} + \frac{E_1}{1-\zeta}.$$

Optimizing in ζ, or taking $\zeta = \sqrt{E_0}/(\sqrt{E_0} + \sqrt{E_1})$ when both actions are positive, gives a concatenated action $(\sqrt{E_0} + \sqrt{E_1})^2$. Taking infima over the two curves yields

$$\mathcal{W}_K(\alpha_0, \alpha_2) \leqslant \mathcal{W}_K(\alpha_0, \alpha_1) + \mathcal{W}_K(\alpha_1, \alpha_2).$$

It remains to justify definiteness. If $\mathcal{W}_K(\alpha_0, \alpha_1) = 0$, choose admissible curves with actions tending to zero. By the compactness and lower-semicontinuity theorem quoted above, a subsequence converges to an admissible curve (ρ_t, v_t) of zero action. Hence $v_t = 0$ for $\theta(\rho_t(x), \rho_t(y))\mathrm{J}(\mathrm{d}x, \mathrm{d}y)\mathrm{d}t$-a.e. (t, x, y). Substituting in the weak continuity equation gives $\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi \, \mathrm{d}\alpha_t = 0$ for every admissible test function φ. The irreducibility/separation assumption in [207] ensures that these test functions determine the measure, so α_t is constant and $\alpha_0 = \alpha_1$.

Finally, if $\mathcal{W}_K(\alpha_0, \alpha_1) < +\infty$, the same direct-method compactness applied to a minimizing sequence gives a minimizer of (14.35). Reparametrizing this minimizing curve by metric arclength gives a constant-speed curve. Equivalently, for every $0 \leqslant s < t \leqslant 1$, the restriction of the minimizer to $[s, t]$, rescaled to unit time, is admissible between α_s and α_t; the triangle inequality and equality of the total minimal action force

$$\mathcal{W}_K(\alpha_s, \alpha_t) = (t-s)\mathcal{W}_K(\alpha_0, \alpha_1)$$

after this constant-speed parametrization. Thus finite-distance pairs are joined by constant-speed geodesics. The consequences for entropy dynamics and fractional PDE examples are developed in Section 15.6. ◻

For a fixed jump kernel, this geometry is genuinely nonlocal and does not coincide with ordinary $\mathcal{W}_2$. The local metric is nevertheless recovered in a small-jump limit. More explicitly, on $\mathcal{X} = \mathbb{R}^d$, let $\eta(z) = \bar{\eta}(|z|)$ be a nonnegative radial profile with

$$M_2(\eta) := \int_{\mathbb{R}^d} |z|^2 \eta(z) \, \mathrm{d}z \in (0, +\infty),$$

and define
$$K_\varepsilon(x, \mathrm{d}y) := \eta_\varepsilon(y - x) \, \mathrm{d}y, \qquad \eta_\varepsilon(z) := \varepsilon^{-d} \eta(z/\varepsilon). \tag{14.36}$$

Radiality implies symmetry and isotropy, and a change of variables gives

$$\int |y - x|^2 K_\varepsilon(x, \mathrm{d}y) = \varepsilon^2 M_2(\eta), \qquad \int (y - x)(y - x)^\top K_\varepsilon(x, \mathrm{d}y) = \varepsilon^2 \frac{M_2(\eta)}{d} \mathrm{Id}. \tag{14.37}$$

Equation (14.37) is the precise sense in which the second-moment jump scale is ε. To obtain a nontrivial local limit, one simultaneously accelerates the jump rate by ε^{-2} and sets

$$\widehat{K}_\varepsilon := \frac{2d}{\varepsilon^2 M_2(\eta)} K_\varepsilon, \qquad \int (y-x)(y-x)^\top \widehat{K}_\varepsilon(x, \mathrm{d}y) = 2\mathrm{Id}.$$

Multiplying a jump kernel by a constant $c > 0$ divides the associated distance by $\sqrt{c}$. Hence, under the regularity and irreducibility hypotheses of [487], for endpoints supported in a fixed compact set,

$$\mathcal{W}_{\widehat{K}_\varepsilon} = \varepsilon \sqrt{\frac{M_2(\eta)}{2d}}\, \mathcal{W}_{K_\varepsilon} \underset{\varepsilon \to 0}{\longrightarrow} \mathcal{W}_2 .$$

This makes precise how sharply concentrated isotropic jumps recover the local Benamou–Brenier geometry. Without isotropy, the covariance matrix in (14.37) need not be proportional to Id, and the limit is instead an anisotropic Wasserstein geometry.

Discrete Wasserstein distances on Markov chains. The finite-state version keeps the same pair-space philosophy, but with a finite graph of admissible exchanges. It is not the naive Euclidean metric on the simplex. The key idea, introduced by Maas and independently developed in related forms by Mielke and by Chow–Huang–Li–Zhou, is to use the transition graph of a reversible Markov chain to define both the admissible directions and the mobility of the mass [360, 383, 145]. The resulting distance is the finite-dimensional counterpart of the Wasserstein geometry behind the heat equation, and it is also the geometric structure used in numerical work on discrete metric-measure spaces [208, 209]. Its entropy interpretation is stated later in Proposition 15.61.

Let $\mathcal{X} = \{1, \ldots, n\}$ and let $K = (K_{i,j})$ denote the off-diagonal transition rates of an irreducible continuous-time Markov chain which is reversible with respect to a probability vector π, so that $\pi_i K_{i,j} = \pi_j K_{j,i}$ for $i \neq j$. We write

$$\mathsf{J}_{i,j} := \pi_i K_{i,j} = \pi_j K_{j,i}$$

for the symmetric edge measure, the finite counterpart of the jump measure used above. The transported object is a mass histogram

$$\Sigma_n := \left\{ a \in \mathbb{R}^n_+ : \sum_i a_i = 1 \right\}.$$

Relative densities only enter as auxiliary variables with respect to the invariant law:

$$\rho_i(a) := \frac{a_i}{\pi_i}, \qquad a_i = \pi_i \rho_i(a).$$

The logarithmic mean θ defined in (14.32) is the mobility selected so that the entropy calculus later recovers exactly the Markov evolution; see Proposition 15.61. The identity $\theta(a,b)(\log a - \log b) = a - b$ converts entropy gradients into density differences along graph edges. For a potential $\psi \in \mathbb{R}^n$, set

$$\bar{\nabla}\psi(i,j) := \psi_j - \psi_i.$$

The finite nonlocal divergence, tangent action, and distance are the discrete counterparts of Definition 14.24.

Definition 14.26 (Discrete Markov-chain Wasserstein distance)**.**

$$(\mathcal{K}_\rho\psi)_i := \sum_j K_{i,j}\theta(\rho_i,\rho_j)(\psi_i-\psi_j), \qquad (\mathcal{L}_a\psi)_i := \pi_i(\mathcal{K}_{\rho(a)}\psi)_i. \tag{14.38}$$

Its associated tangent action is

$$\mathbb{A}_K(a,\psi) := \frac{1}{2}\sum_{i,j} \mathsf{J}_{i,j}\theta(\rho_i(a),\rho_j(a))(\bar{\nabla}\psi(i,j))^2. \tag{14.39}$$

$$\mathcal{W}_K^2(a_0,a_1) := \inf_{a_t,\psi_t}\int_0^1 \mathbb{A}_K(a_t,\psi_t)\,\mathrm{d}t, \qquad \dot{a}_t+\mathcal{L}_{a_t}\psi_t=0, \tag{14.40}$$

with endpoint conditions $a_{t=0}=a_0$ and $a_{t=1}=a_1$.

The tangent variable is the potential ψ, or equivalently the induced edge flux, rather than an ambient Euclidean vector field. In edge-flux variables, flux is allowed only where $K_{i,j}>0$, and its kinetic denominator is the logarithmic mean of the two relative endpoint densities.

The first nontrivial finite Markov geometries already show how the logarithmic mean bends the simplex. In both examples below, take the uniform random walk on the complete neighbor graph, i.e. every pair of distinct points is declared neighboring. Thus $\pi_i=1/n$, and $K_{i,j}=1/(n-1)$ for $i\neq j$.

Example 14.27 (Two-point complete graph). On Σ_2, write $a_0=(r,1-r)$ and $a_1=(s,1-s)$. Since there is only one edge, the discrete dynamic problem reduces to the scalar Riemannian length

$$\mathcal{W}_K(a_0,a_1)=\left|\int_s^r \frac{\mathrm{d}u}{\sqrt{\theta(u,1-u)}}\right|, \qquad 0<r,s<1. \tag{14.41}$$

This formula is closed but not Euclidean: the logarithmic mean changes the cost of moving mass depending on the current split between the two points.

Example 14.28 (Three-point complete graph). On Σ_3, the complete-neighbor graph is a triangle. For $a\in\mathrm{int}(\Sigma_3)$, set

$$\Theta_{i,j}(a) := \frac{1}{2}\theta(a_i,a_j), \qquad 1\leqslant i<j\leqslant 3.$$

For a fixed a, write $\Theta_{i,j}=\Theta_{i,j}(a)$. For a tangent vector $h\in\mathbb{R}^3$ with $h_1+h_2+h_3=0$, orient the edges as $1\to 2$, $1\to 3$, $2\to 3$. The squared norm induced by the discrete Wasserstein metric is

$$\|h\|_a^2 = \min_{m_{1,2},m_{1,3},m_{2,3}}\left\{\frac{m_{1,2}^2}{\Theta_{1,2}}+\frac{m_{1,3}^2}{\Theta_{1,3}}+\frac{m_{2,3}^2}{\Theta_{2,3}}\right\}, \tag{14.42}$$

subject to

$$h_1+m_{1,2}+m_{1,3}=0, \qquad h_2-m_{1,2}+m_{2,3}=0, \qquad h_3-m_{1,3}-m_{2,3}=0.$$

Eliminating the three edge fluxes gives an explicit formula. With $D=\Theta_{1,2}^{-1}+\Theta_{1,3}^{-1}+\Theta_{2,3}^{-1}$,

$$m_{1,2}^*=\frac{h_2/\Theta_{2,3}-h_1/\Theta_{1,3}}{D}, \qquad m_{1,3}^*=-h_1-m_{1,2}^*, \qquad m_{2,3}^*=m_{1,2}^*-h_2,$$

and $\|h\|_a^2$ is obtained by inserting these values in (14.42). Therefore

$$\mathcal{W}_K^2(a_0,a_1)=\inf_{a_t\in\mathrm{int}(\Sigma_3)}\int_0^1 \|\dot{a}_t\|_{a_t}^2\,\mathrm{d}t, \qquad a_{t=0}=a_0, \quad a_{t=1}=a_1. \tag{14.43}$$

Thus the three-state distance is an explicit two-dimensional Riemannian geodesic problem on the open triangle. The formula is simple enough to compute directly, but it already shows the main difference with Euclidean geometry on the simplex: the local metric depends nonlinearly on the current density through logarithmic edge mobilities.

Figure 14.4 visualizes these small-dimensional geometries and compares them with the ordinary Wasserstein distance associated with the 0/1 ground metric, for which $\mathcal{W}_2^2$ is exactly the total variation distance.

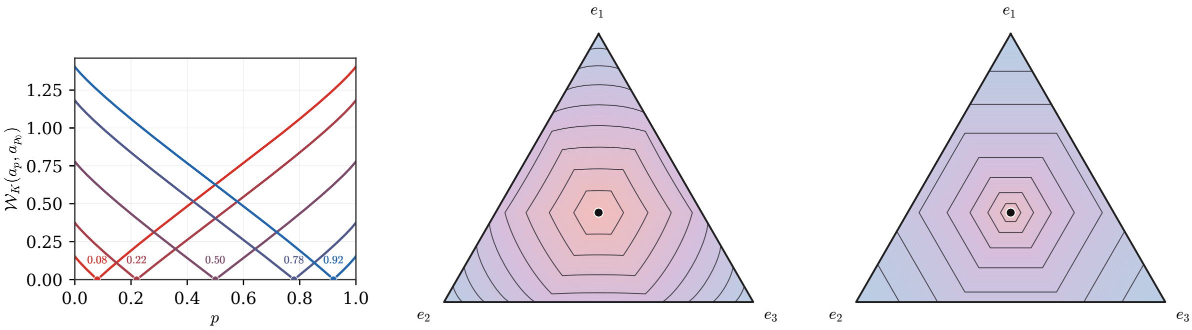


Figure 14.4: Discrete Wasserstein distances on small Markov-chain simplices. Left: the closed-form profiles $r \mapsto \mathcal{W}_K(a_r, a_{r_0})$, with $a_r = (r, 1-r)$, for several anchors r_0 on Σ_2. Middle: numerical level sets of $\mathcal{W}_K(a, \bar{a})$ on Σ_3, where $\bar{a} = (1/3, 1/3, 1/3)$, computed from the local Riemannian norm induced by the complete-neighbor Markov chain. Right: the corresponding level sets for the ordinary $\mathcal{W}_2$ distance with $d(i,j) = 1$ for $i \neq j$, i.e. $\mathcal{W}_2^2(a, \bar{a}) = |a - \bar{a}|_{\mathrm{TV}}$.

14.6 Dynamic Unbalanced Wasserstein Distances

Balanced dynamic distances keep the total mass fixed: their tangent vectors are transport velocities or fluxes satisfying a continuity equation. Unbalanced distances use a different tangent model. A tangent vector now has both a spatial component and a reaction component, so mass can move, disappear, and reappear. This section isolates this reaction–transport geometry before its use for gradient flows in Section 15.7.

Balance equation and tangent variables. Unbalanced dynamic transport is obtained by allowing mass to be created and destroyed along the path. The continuity equation is replaced by a balance equation, and, at the density level, an admissible tangent direction is a pair (m, s): the flux density m transports mass in space, while the source density s changes the amount of mass locally. This dynamic formulation underlies the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics [339, 143]; its equivalence with static entropy-transport and cone formulations is developed in [340, 141]. A representative quadratic balance equation is

$$\partial_t \rho_t + \nabla \cdot m_t = s_t, \qquad \int_0^1 \int \left(\frac{|m_t|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) dx\, dt,$$

with the usual perspective convention: zero flux and zero source through zero density cost nothing, whereas nonzero flux or source through zero density has infinite cost. At the measure level these densities become the flux measure $\omega_t = m_t\, dx$ and the signed source measure $\sigma_t = s_t\, dx$.

Reaction–transport action. The action attaches one price to displacement and another to local growth. Its velocity and momentum forms are collected in one definition.

Definition 14.29 (Dynamic Wasserstein–Fisher–Rao distance)**.**

For $\kappa > 0$, define
$$A_\kappa(a, w, g) := a\left(|w|^2 + \kappa^2 g^2\right), \tag{14.44}$$

and
$$J_\kappa(a, m, r) := \begin{cases} (|m|^2 + \kappa^2 r^2)/a, & a > 0, \\ 0, & (a, m, r) = (0, 0, 0), \\ +\infty, & a = 0,\ (m, r) \neq (0, 0). \end{cases} \tag{14.45}$$

If λ dominates α, $|\omega|$, and $|\sigma|$, set

$$\mathbb{J}_\kappa(\alpha, \omega, \sigma) := \int J_\kappa\left(\frac{d\alpha}{d\lambda}, \frac{d\omega}{d\lambda}, \frac{d\sigma}{d\lambda} \right) d\lambda. \tag{14.46}$$

The induced dynamic Wasserstein–Fisher–Rao distance is

$$\mathrm{WFR}^2_\kappa(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t \alpha_t + \nabla\cdot\omega_t = \sigma_t \\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \int_0^1 \mathbb{J}_\kappa(\alpha_t, \omega_t, \sigma_t)\mathrm{d}t. \tag{14.47}$$

Here w is velocity, g is relative growth, $m = aw$ is flux, and $r = ag$ is source density. The parameter κ fixes the relative cost of reaction and transport. Since J_κ is convex and positively one-homogeneous, the measure action is independent of λ; finite action forces $\omega, \sigma \ll \alpha$.

Static and dynamic viewpoints. The dynamic balance-equation formula is not a separate object from static unbalanced OT: it is the least-action representation of the same cone distance. To make the normalization explicit, define on the cone $\mathfrak{C}[\mathbb{R}^d]$ the squared cost

$$\Delta_\kappa\big((x,r),(y,s)\big)^2 := 4\kappa^2 \left[r^2 + s^2 - 2rs\cos\left(\frac{|x-y|}{2\kappa} \wedge \frac{\pi}{2}\right)\right]. \tag{14.48}$$

The radii encode masses through the weighted projection P_2 defined in Section 11.1. Accordingly, set

$$\mathrm{CW}_\kappa(\alpha_0, \alpha_1) := \inf_{\substack{\gamma\in\mathcal{M}_+(\mathfrak{C}[\mathbb{R}^d]^2) \\ \mathsf{P}_2\gamma_1=\alpha_0,\ \mathsf{P}_2\gamma_2=\alpha_1}} \int \Delta_\kappa\big((x,r),(y,s)\big)^2 \mathrm{d}\gamma(x,r,y,s). \tag{14.49}$$

For $\kappa = 1/2$, this is exactly the normalization of the Hellinger–Kantorovich cone cost used in Section 11.1. The next proposition identifies its general rescaling with the dynamic action above.

Thus Definition 14.29 gives both the reaction–transport action and the induced dynamic WFR metric.

Proposition 14.30 (Static/dynamic equivalence for unbalanced OT)**.** *For nonnegative finite measures α_0, α_1 on $\mathbb{R}^d$, the dynamic value (14.47) equals the static cone value (14.49). Hence $\mathrm{WFR}_\kappa = \mathrm{CW}_\kappa^{1/2}$ is the geodesic distance generated by the balance-equation least-action problem.*

Proof. The cone construction turns variation of mass into radial motion and spatial transport into angular motion on $\mathfrak{C}[\mathbb{R}^d]$. The normalization can be checked on a smooth cone path. If its base position is x_t, its cone radius is r_t, and the projected mass is $a_t = r_t^2$, then its relative growth rate is $g_t = \dot{a}_t/a_t = 2\dot{r}_t/r_t$. The infinitesimal energy induced by (14.48) is therefore

$$4\kappa^2\dot{r}_t^2 + r_t^2|\dot{x}_t|^2 = a_t\left(|\dot{x}_t|^2 + \kappa^2 g_t^2\right),$$

which is precisely the velocity action (14.44).

Applying the Benamou–Brenier theorem on the cone to the lifted endpoint measures now gives a dynamic least-action problem whose static value is CW_κ. Projecting a cone curve back to the base space with the squared cone radius as weight produces a measure curve α_t, a spatial flux measure ω_t, and a source measure σ_t satisfying the balance equation. The cone kinetic energy decomposes exactly into the three-variable perspective action $\mathbb{J}_\kappa$ in (14.47). Conversely, any finite-action triple $(\alpha_t, \omega_t, \sigma_t)$ can be lifted, after relaxation, to a cone curve whose radial and spatial velocities realize the source and flux with the same action. This is the standard static/dynamic identification for the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics [339, 340, 143, 141].

The two infima are therefore equal; lower semicontinuity gives the general finite-measure statement from the smooth positive case. □

Balanced versus unbalanced interpolations. The distinction is visible already for one-dimensional mixtures with mismatched modal masses. Balanced transport must physically move the excess mass, whereas unbalanced transport can trade transport against reaction. Figure 14.5 uses entropic balanced and KL-relaxed barycenters as a qualitative numerical surrogate; its unbalanced row illustrates the reaction–transport mechanism but is not asserted to be an exact WFR_κ geodesic.

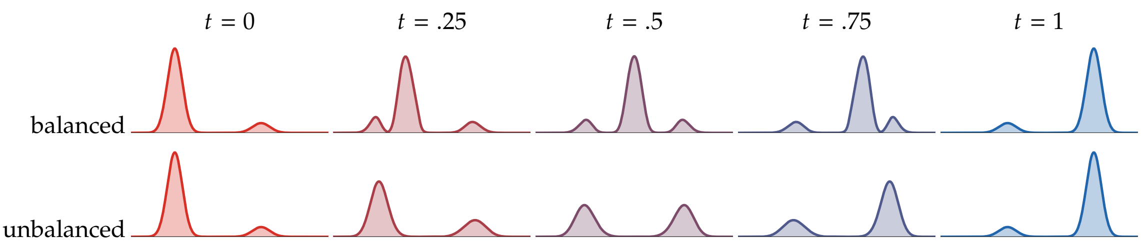


Figure 14.5: Balanced and unbalanced Sinkhorn-barycenter interpolations between two one-dimensional Gaussian mixtures with swapped modal masses. The balanced row conserves total mass, so excess mass from the dominant left mode must move along the line toward the dominant right target mode, producing transient mass in the middle. The unbalanced row uses KL-relaxed marginal constraints; mass can be attenuated near overrepresented modes and recreated near underrepresented modes, giving a reaction–transport interpolation closer to the Wasserstein–Fisher–Rao intuition.

14.7 Variational Mean Field Games

Mean field games use a population law to summarize strategic interactions among a very large number of individually negligible agents. This section only considers the special *potential* subclass in which the equilibrium system is the optimality condition of one convex planning problem. In that setting, the relation with dynamic OT is particularly transparent: one augments the Benamou–Brenier kinetic action by congestion and may replace a prescribed final measure by a terminal penalty.

From individual control to a population equilibrium. Mean field games were introduced by Lasry and Lions [323] and, in a parallel large-population stochastic-control framework, by Huang, Malhamé and Caines [283]. Given a candidate population density $(\rho_t)_{t\in[0,1]}$, a representative deterministic agent starting from x chooses a path γ by minimizing

$$\inf_{\gamma(0)=x} \left\{ \int_0^1 \left(\frac{1}{2}|\dot\gamma(t)|^2 + g\big(\rho_t(\gamma(t))\big) \right) \mathrm{d}t + \Psi(\gamma(1)) \right\}. \tag{14.50}$$

Here the kinetic term prices individual effort, $g(\rho)$ is the running cost created by the local population density, and Ψ assigns a cost to the final state. A mean field Nash equilibrium requires consistency: when all agents use an optimal feedback for (14.50), the resulting law of their positions must be precisely $\alpha_t = \rho_t\,\mathrm{d}x$.

A Benamou–Brenier planning problem. The consistency condition is usually coupled to an individual Hamilton–Jacobi–Bellman equation. A global variational reduction is available when the local coupling is a first variation: assume

$$g(r) = G'(r) \tag{14.51}$$

for a proper convex congestion potential $G : [0,+\infty) \to \mathbb{R}\cup\{+\infty\}$. Starting from $\alpha_0 = \rho_0\,\mathrm{d}x$, the associated potential MFG is the following planning problem.

Definition 14.31 (Variational mean-field-game planning problem)**.**

$$\inf_{(\rho_t,v_t)} \left\{ \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2}\rho_t(x)|v_t(x)|^2 + G(\rho_t(x)) \right) \mathrm{d}x\,\mathrm{d}t + \int_{\mathbb{R}^d} \Psi(x)\rho_1(x)\mathrm{d}x \right\}, \tag{14.52}$$

subject to $$\partial_t\rho_t + \mathrm{div}(\rho_t v_t) = 0, \qquad \rho_{t=0}\,\mathrm{d}x = \alpha_0. \tag{14.53}$$

Unlike in (14.8), the final measure is not prescribed. The terminal potential Ψ selects desirable final states softly, while the integral of $G(\rho_t)$ penalizes crowded intermediate configurations. It is important that the planner pays $G(\rho)$ rather than $\rho g(\rho)$: the marginal cost perceived by one infinitesimal agent is the first variation $G'(\rho) = g(\rho)$. For the usual congestion models, g is nondecreasing, equivalently G is convex; superlinear choices of G increasingly penalize concentration.

This potential formulation and its deterministic and diffusive variants are developed by Benamou, Carlier and Santambrogio [40]; first-order MFG systems with local couplings are analyzed, in particular, by Cardaliaguet and Graber [102].

Convex momentum formulation. As in the Benamou–Brenier problem, the density–velocity integrand and the constraint in (14.52)–(14.53) are not jointly convex. Set $\omega_t = \alpha_t v_t$, or $m_t = \rho_t v_t$ when densities exist, and define the congestion functional

$$C_G(\alpha) := \begin{cases} \displaystyle\int_{\mathbb{R}^d} G(\rho(x))\mathrm{d}x, & \alpha = \rho\,\mathrm{d}x, \\ +\infty, & \text{otherwise.} \end{cases} \tag{14.54}$$

Using the measure perspective action (14.11), the same problem becomes the generalized Benamou–Brenier formula

$$\inf_{(\alpha_t,\omega_t)} \left\{ \int_0^1 \left(\frac{1}{2}\mathbb{J}(\alpha_t, \omega_t) + C_G(\alpha_t) \right) \mathrm{d}t + \int_{\mathbb{R}^d} \Psi\,\mathrm{d}\alpha_1 \right\}, \qquad \partial_t \alpha_t + \operatorname{div}\omega_t = 0, \quad \alpha_{t=0} = \alpha_0. \tag{14.55}$$

Equivalently, in density–momentum variables, the running integrand is

$$\frac{1}{2}J(\rho_t, m_t) + G(\rho_t) = \frac{|m_t|^2}{2\rho_t} + G(\rho_t), \tag{14.56}$$

with the vacuum convention of (14.10), under the affine equation $\partial_t \rho_t + \operatorname{div} m_t = 0$.

The perspective term $(\alpha, \omega) \mapsto \mathbb{J}(\alpha, \omega)$ is convex by (14.10)–(14.11), while C_G is convex whenever G is proper and convex. Since the terminal term is linear and the constraints are affine, problem (14.55) is then convex in (α, ω). If G is nonconvex, the momentum substitution still convexifies the kinetic term and linearizes the constraint, but it does not convexify the congestion term.

Optimality system and game interpretation. The potential structure matters because the optimality conditions of the planner reproduce the best-response and consistency equations of the game. The following smooth calculation makes this identification explicit.

Proposition 14.32 (Potential MFG system)**.** *Assume that an optimizer of* (14.55) *has a smooth positive density* ρ*, that* G *is differentiable, and set* $g = G'$*. Then there is a value function* u *such that*

$$\begin{cases} -\partial_t u + \dfrac{1}{2}|\nabla u|^2 = g(\rho), & u_{t=1} = \Psi, \\ \partial_t \rho - \operatorname{div}(\rho \nabla u) = 0, & \rho_{t=0}\,\mathrm{d}x = \alpha_0, \end{cases} \qquad m = -\rho\nabla u. \tag{14.57}$$

Proof. Use $-u$ as the multiplier for $\partial_t \rho + \operatorname{div} m = 0$. After integration by parts, the space–time Lagrangian density is

$$\frac{|m|^2}{2\rho} + G(\rho) + \rho\,\partial_t u + m \cdot \nabla u.$$

Stationarity in m gives $m = -\rho\nabla u$. Stationarity in ρ, followed by this substitution, gives the first equation in (14.57). The free terminal density contributes $(\Psi - u_1)\rho_1$, hence $u_1 = \Psi$ under the positivity assumption. Primal feasibility gives the second equation. □

The backward Hamilton–Jacobi–Bellman equation describes the representative agent's optimal response to the density-dependent cost $g(\rho)$; the forward continuity equation transports the population under the resulting feedback $v = -\nabla u$. Their coupling enforces the Nash consistency condition. For nonsmooth G, $g(\rho)$ is replaced by an element of $\partial G(\rho)$; hard density caps then produce a pressure supported on the saturated region.

Hard congestion through a bottleneck. A global hard crowd-capacity constraint is obtained from

$$G_\kappa(r) = \iota_{[0,\kappa]}(r) = \begin{cases} 0, & 0 \leqslant r \leqslant \kappa, \\ +\infty, & \text{otherwise.} \end{cases} \tag{14.58}$$

In order to make the cap active while retaining a prescribed terminal preference, one may replace the linear terminal term in (14.55) by the convex penalty

$$\Gamma_\eta(\rho_1) = \frac{\eta}{2} \int_\Omega \left|\rho_1(x) - \rho_\star(x)\right|^2 \mathrm{d}x, \tag{14.59}$$

where $\rho_\star$ is a desired terminal density. Such L^1 and L^2 endpoint penalties are part of the augmented-Lagrangian experiments of Benamou and Carlier [36]; the two-room hard-congestion geometry follows Benamou, Carlier and Santambrogio [40].

The role of endpoint relaxation depends on the geometry. If two fixed endpoint densities on $\mathbb{R}^d$ satisfy a spatially constant cap, their ordinary $\mathcal{W}_2$ geodesic satisfies the same cap by Proposition 15.22, so the cap does not alter that geodesic. This argument no longer applies inside a nonconvex domain, where straight displacement segments may leave the domain and a narrow passage creates a genuine capacity bottleneck. The terminal penalty (14.59) additionally lets the planner trade terminal mismatch against kinetic cost while enforcing the running cap everywhere.

Figure 14.6 illustrates this mechanism in two rooms connected by a one-cell doorway. Here we specialize the terminal term to the hard constraint $\rho_{t=1} = \rho_\star$, equivalently the indicator terminal cost $\Psi = \iota_{\{\rho_\star\}}$. Both endpoints are uniform on mirrored disks. Writing $M_0 = \|\rho_0\|_\infty$, the constrained experiment uses the tight feasible capacity $\kappa = M_0$: the crowd starts at saturation and cannot compress above its initial packing density. The uncapped hard-endpoint path forms a thin, dense jet through the doorway, whereas the capped path builds a broad saturated queue upstream. The convex density–momentum problem is discretized on a 48×31 spatial grid with eight time slabs. Douglas–Rachford splitting computes the variational solution, and a final Dykstra projection onto the intersection of the affine continuity/end-point constraints and the global density box removes the small feasibility error of the finite splitting iterate.

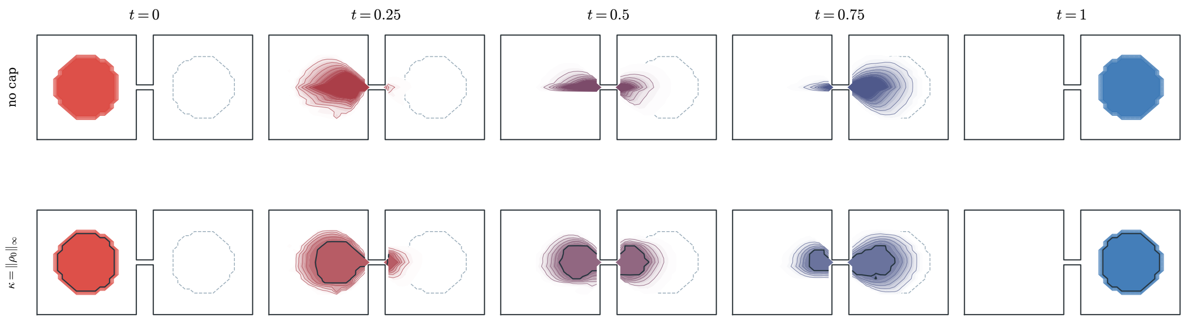


Figure 14.6: Hard-congestion variational MFG through two communicating rooms. The initial and hard terminal profiles are uniform on mirrored disks; the dashed outlines show the latter, and the five columns advance from red to blue. The top row has no cap and concentrates into a narrow doorway jet. The bottom row imposes $\kappa = \|\rho_0\|_\infty$ everywhere, forcing a broader saturated queue. Dark contours mark regions within 1% of the cap, and both rows use the same density levels.

Remark 14.33 (Scope of the variational reduction). A general MFG need not be the optimality system of a global convex functional. The reduction above applies because the coupling is the first variation of the potential C_G and the terminal cost is linear, or more generally convex, in the final law. More general nonlocal, non-potential or non-monotone couplings retain the coupled best-response/consistency interpretation but generally fall outside (14.55).

Computational consequence. The convexification is not merely formal. After conservative space–time discretization, the objective is a sum of local perspective and congestion terms subject to one affine divergence constraint. Augmented-Lagrangian and primal–dual methods can therefore alternate local proximal updates with the solution of a linear space–time equation, closely paralleling the Benamou–Brenier solver. This route was developed for transport and variational MFGs by Benamou and Carlier [36] and is reviewed with numerical examples in [40].

The chapter has separated three dynamic transport mechanisms—local advection, nonlocal jumps, and coupled displacement–reaction—and then shown how the local Benamou–Brenier action becomes a convex population game after adding a potential congestion cost. The next chapter uses dynamic transport geometries to turn energies on measures into evolution equations.